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HOSPITAL SUPPORT SERVICES

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INTRODUCTION

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Healthcare has become one of India's largest sectors—both in terms of revenue and employment. Healthcare comprises hospitals, medical devices, clinical trials, outsourcing, telemedicine, medical tourism, health insurance and medical equipment. The Indian healthcare sector is growing at a brisk pace due to its strengthening coverage, services and increasing expenditure by public as well private players. Indian healthcare delivery system is categorized into two major components—public and private. The Government, i.e. public healthcare system, comprises limited secondary and tertiary care institutions in key cities and focuses on providing basic healthcare facilities in the form of primary healthcare centres (PHCs) in rural areas. The private sector provides majority of secondary, tertiary and quaternary care institutions with a major concentration in metros, tier I and tier II cities.

Hospitals which utilize most of the health budget play a very important role in the country's health system. Hospitals are among the most complex organizations in modern society. The modern hospital itself is a universe, with a variety of objectives, and a scalar division of labour to achieve those objectives. Medical, para-medical and support services of a hospital must be well aware of the scientific methods to run and evaluate the hospital functions and services in an objective fashion. They should have a good understanding of hospital organization and management for better care of their patients. Moreover, they should also have enough knowledge for management of human, material and financial resources in a cost effective way with optimum time approach.

This book, *Hospital Support Services*, is divided into four blocks, which, in turn, are sub-divided into various units. There are in all fourteen units in the book, dealing with topics that include principles and methods of organizing; clinical and support services for hospitals; role of support services/departments in the hospital management; laboratory administration and management; safety in laboratories and laundry arrangements; dietary services, store management; and general kitchen facility. Each unit contains an Introduction and a list of Objectives to prepare the student for what to expect in the text. At the end of each unit are a Summary and a list of Key Words, to aid in recollection of concepts learnt. All units contain Self-Assessment Questions and Exercises, and strategically placed Check Your Progress questions so the student can keep track of what has been discussed.

BLOCK - I
BASICS OF HOSPITAL SUPPORT SERVICES

*Principles and Methods
of Organising Hospital
Support Services*

**UNIT 1 PRINCIPLES AND
METHODS OF
ORGANISING HOSPITAL
SUPPORT SERVICES**

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Structure

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1.0 INTRODUCTION

The image with respect to medicinal consideration in some nations including India can be portrayed as clamorous. Hospital beds are deficient, the healing facilities are situated far from the locations which require them the most, and mostly concentrated in towns, vigorously among the one-sided urban populace. Thus, in spite of the fact that the normal hospital bed to patient ratio should be under 0.8

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beds per 1000 populace, it is as high as 2.5 beds per 1000 populace in metropolitan areas like Delhi and Mumbai and as low as 1 bed for every 3700 to 14000 populace in provincial and native territories in India. Where there is insufficient medicinal consideration, there is also a surfeit of consideration in a few spots. In any case, what has progressively become critical is that they are expensive to work, with their running expenses adding up to roughly 33% along with the underlying development costs during every activity. Heedless planning at the initial stages by the unpractised and uncommitted workforce has created problems. It has resulted in changes at constructional organisation bringing about avoidable seepage of assets. Thus, in this unit we will discuss the importance of organisation, when it comes to hospital support services, along with the methods to do so.

1.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the importance of organisation in Hospital Support Services
- Analyse the principles of organisation in Hospital Support Services
- Discuss the methods of organisation in Hospital Support Services

1.2 ORGANISATION IN HOSPITAL SUPPORT SERVICES

There is a serious lack of doctors due to insufficient funds in most of the developing nations, despite the fact that there is a huge need for restorative care. This is barring some private healing facilities, which are working for the special classes of individuals. Basic services required for the hospital network can be met most efficiently, just with a little thought given to planning, development, and task of social insurance offices. Indeed, even where money is not an issue, best utilisation of assets can be made with proper organisation.

The word 'executives' has been defined from numerous points of view by numerous specialists. According to the first definition by Henri Fayol, thought to be the father of present day theory of management, the role entails to oversee, to gauge, plan, to compose, to order, to facilitate and to control. This encapsulates the true meaning of being an executive. The task concerning any venture constitutes:

- Deciding the objective and targets of the endeavour
- Procurement and use of assets
- Establishing correspondence frameworks
- Deciding control techniques
- Assessing the execution of the endeavour

The job, capacity, and undertakings of general administration have been classified as 'standards of the management'. These standards include rules for the role of an administrator. These devise guidelines, which are normal for the exercise of overseeing organisations as various as a departmental store, an inn, a shoe industrial facility, or a political gathering. However, they are excessively objective—almost machine-like and forget the imperative human component. With the human component, the executives are all about choosing what anyone can do, doing it, planning the material, and afterward completing things through individuals. For instance, HRs work with this approach. The standards of general administration originally created by the French industrialist Henry Fayol in the mid- twenties, became popular in the forties. The division of the work should be as mentioned in the points below:

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- I. Division of work:** The work allotted to every worker ought to be characterised and clearly defined to ensure its proper performance. For financial experts, this specialisation is the most important to effectively utilise human asset.
- II. Specialist and responsibility:** Specialist and duty are indistinguishable, with the latter emerging from the former. Without power one can't free themselves from the obligation. Notwithstanding, specialist ought to be synonymous with duty.
- III. Control:** Considering discipline as a tool for inciting compliance, it requires great managers at all levels. Managers and overseers should set good precedent through their activities and conduct.
- IV. Solidarity of command:** This implies executives ought to get orders from one of their seniors. Every executive in the organisation should be aware of their respective manager and be answerable to them.
- V. Solidarity of direction:** Each group with a similar target must have one head and one arrangement. Rather than solidarity of order, it is solidarity of course, which identifies with the overall organisation. There ought to be cooperation and solidarity in the organisation. On the off chance that the individuals at different levels disagree about the goals, there will be wastage of authoritative assets.
- VI. Centralisation of authority:** This refers to the degree to which expertise is focused or scattered. It ought to be clear in the organisation as to who is supposed to issue orders. Moreover, contradicting requests will create confusion in the organisation.
- VII. Scalar chain:** The chain of administrators from the highest level of superiority to the lowest positions in the organisation is called the scalar chain. While this should be left alone, it could be tweaked now and then, when following it would be hindering in a particular circumstance.

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VIII. Request: As a rule of organisation, in the course of action of things and individuals, this will result in the streamlining of the assets.

IX. Compensation: Compensation should be equivalent to the work. Every individual ought to be paid according to their work, the compensation and methods of hiring should be reasonable.

X. Dependability of tenure: Unnecessary turnover of workers is both the reason and the impact of terrible administration. The worker requires acknowledgement about the long-term relationship, bringing about a sentiment of security and organisation in work.

XI. Designation of authority: Since managers oversee work done by others, the responsibility should be assigned. Through designation, subordinates get prepared for higher obligation. The requirements, preparation, and inspiration of the delegator and delegate must match.

XII. Work: Work is the reasoning out and execution of an arrangement. Worker ought to be given space for the utilisation of creativity in their work. It is a duty towards administrative goals.

XIII. Subordination of individual interest to organisational interest: When these two contrast, the management must accommodate them. As the organisation is set up to address the issues of society, the individual must forfeit some narrow-minded interests in the general interest of the organisation and society.

XIV. Value: Reliability and commitment ought to be inspired in the workforce by a blend of fairness, generosity, and equity, when managing subordinates. Subordinates must be treated with no predisposition for race, religion, sex, or class.

The above guidelines propounded by the traditional school of the management are still widely use today.

1.2.1 Management

The terms, 'organisation' and 'management', have frequently been interchangeably used. A few people have endeavoured to characterise management and the organisation as two different elements. To them, organisation means something superior and more extensive than just overseeing. In any case, the management isn't just a scholastic order. It is a pragmatic craftsmanship and a science, calling for advancement of learning, abilities and dispositions. Overseeing and organisation make utilisation of information, for example the administration science. Craftsmanship is putting science to the best use. The science and speciality of the management are not fundamentally unrelated, but rather reciprocal. For this reason, organisation and the management are taken to refer to a similar concept.

Check Your Progress

1. Who is known as the father of present day theory of management?
2. Why does the dependability of tenure important?

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1.3 ADMINISTRATIVE ACTIVITIES OF A HOSPITAL

The following activities are common to the administration of all hospitals:

- **Determination of objectives and goals:** This essentially deals with the organisation of the hospital involving redesigning existing management, sorting out new offices, and projects.
- **Financial management:** This refers to the budgetary undertakings of the hospitals. It incorporates planning and costing.
- **Managing staff:** This division refers to the determination, inspiration, and aim of representatives. It incorporates wage and compensation management.
- **Planning:** This incorporates activities related to managing the inward working of all hospitals divisions. A meeting with departmental heads would be one example.
- **Program survey and assessment:** The audit and assessment refers to the working of the clinical administrations and programs, and is a constant procedure.
- **Open and network exercises:** This refers to the improvement and support of connection with other wellbeing administration organizations incorporating shared-benefit courses of action with different healing facilities.
- **Wellbeing industry exercises:** This grouping alludes to exercises that are outside to the clinic. It incorporates support in hospitals affiliations, outsider payers (insurance agencies, businesses), and so forth.
- **Government-related exercises:** This activity is concerned with the legal issues of the hospitals and managing local, state, and central government offices.
- **Educational improvement:** It incorporates all education related training and training of hospital's workforce and cooperation in meetings and proceeding with remedial instruction.

NOTES

1.3.1 The Purpose of Planning

The purpose of planning is to guarantee the most efficient usage of assets and finance. The requirement for planning is felt on the grounds that a specific circumstance shows a chance to accomplish something positive or a specific circumstance represents an issue and attracts consideration regarding that issue. All supervisors take part in planning work, according to their position. Given the shortage of assets, hospital administrators are required to make long and short term plans so as to avoid indiscriminate use of assets. While an administrator can be aided by the different staff members in the planning process, it ought to be noted by every manager that a definitive duty regarding planning rests only with him. The responsibility of planning can't be designated. Since the management of an organisation chooses, where it needs to reach (or what it needs to accomplish), sooner or later, planning will include the goal alongside approaches, methodologies, projects, systems, and principles which are vital for their achievement.

1.3.2 Key and Operational Planning

Planning in the human resources can be comprehensively separated into two classes.

- (a) Key planning
- (b) Operational planning

Key Planning

Key or technical planning conveys a more extensive point of view for the entire organisation at the most important planning level. It concerns the formation of the primary mission of the organisation, creating expansive goals, followed by deciding the organisation required, and deciding the methods for satisfying the same. This likewise includes methodologies for gathering, growing, and assorting variety.

Operational Planning

Operational planning, which is performed at the lower authoritative level, by and large focuses on plan's particulars and execution. It is all about actualising the essential plan in the entirety of its components at the operational level. This piece of planning can likewise be independent of key planning, since it concerns the structural planning at all levels of the foundation—when another company or office is included, when there is a requirement for updating of a current company or office, or when another plan of organisation is being presented.

1.3.3 Planning Committees

It has just been said in the beginning of this section that planning is the obligation of supervisors. Be that as it may, planning committees are of extraordinary help in numerous issues from the social point of view, because they can adjust the

perspectives of the overseeing management, medicinal staff, organisation and other key gatherings both inside and outside the establishment. This gives freedom from the traditional choices and execution of the techniques becomes clearer. A single administrator in charge of planning, provides suitable foundation data to the advisory group. With obviously focussed issues, data, and informed sources of details, this individual fills in as a staff consultant to the advisory group and channelises their collective reasoning. All conceivable effort must be put into inciting valuable ideas from every one with different interests on the issue of management.

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Check Your Progress

3. How many categories can planning be divided?
4. Mention any two activities in the administration of hospitals.

1.4 BASICS OF HOSPITALS SUPPORT SERVICES

There are some basic services without which a hospital won't be called a hospital. These essential services makes it a centre of health and care.

1.4.1 Outpatient Services

Walk-in health care is given to patients who are most certainly not confined to bed and can be treated at a general hospital, a specialised hospital, or a welfare hospital. When such care is rendered at premises which are a piece of a hospital (outpatient division), then they are called outpatient care. Furthermore, the management stemming from it is named outpatient services. Outpatient division is characterised as a section of the hospital with designated physical offices, medicinal staff, regular planned hours, to give care to patients, who are not enlisted as inpatients.

Size and importance of outpatient services

A survey of the rank of outpatient services provided by medical facilities in India requires attention. The level of the services is high, but the issues of sorting them out are numerous. There are still huge chunks of populace, who do not have any healthcare facility, even the walk-in one. This is based on the following results:

- 2.7 to 3.0 crore inpatients are served with 8, 70, 160 beds in the nation every year.
- Around 25-35 inpatients are served per bed in a year.
- Then again, for every hospital bed, around 600 outpatients every year are provided with services.
- This implies more than 52 crore outpatients in a year are treated in the outpatient bureau of medical centres.

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1.4.2 Nursing Services

Nursing services stand out amongst the most critical services of medical facility. Nursing services in a more extensive setting is that part of the complete healthcare management, which intends to fulfil the nursing needs of the network, the significant targets of which are:

- Nursing care required for the prevention of sickness and improvement of well-being.
- The nursing care of debilitated patients—in the interest of his or her psychological and physical solace and by reason of the malady from which the person in question is suffering. Medical attendants shape an essential group—the biggest single specialised group—of staff occupied with patient care in hospitals along with specialists, taking up around one third of hospital costs.

Responsibility of the nursing services

The elements of nursing services are:

- A basic responsibility of helping the individual patients in the execution of those activities adding to their wellbeing or recuperation (or to peaceful demise).
- As an expansion of the above basic responsibility, to help and urge the patient to receive the remedial treatment advised by the doctor.
- As an individual from the healthcare group, to help different people, to plan, and perform the complete program of healthcare.

1.4.3 Radiology and Imaging Services

It goes without saying that the technical aspects of hospitals are of immense value. Without its aid, the whole healthcare system will be on the road to ruin. Thus, services like Radiology and Imaging holds especial place in the scheme of healthcare. The modern drug can't be prescribed without certain insights. Radiology or Radiodiagnosis is one such vital division of the hospital, which clearly contributes to the patient's care. It gives, alongside pathology, an essential demonstrative reinforcement to every one of the specialities which can't work successfully without its help. Consequently, the organisation turns into a necessary part with everything being the same except for little hospitals and nursing homes. Radiology has immensely expanded in the recent decades. The term imaging includes:

- X-Ray
- Ultrasound
- CT Scan
- Magnetic Resonance Imaging (MRI)
- Digital Subtraction Angiography (DSA)

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Both radiodiagnosis and radiotherapy have progressed to such an extent, that it isn't possible for one individual to ace both these specialities, which share little, practically speaking, with one another. Radiotherapy has turned into a highly specialised, expensive, and modern gear. Such a duty has some expert responsibilities, since they require extraordinarily built structures, costly hardware and profoundly prepared radiotherapists, specialists, physicists, and experts who are involved in cancer treatment and medical procedure. Oncology has turned into a separate order requiring collaboration of pathologists, cell cytologists, specialists, physicians, along with chemotherapy, requiring joint conversation between all of these specialists.

Types of radiological functions

The responsibilities, which are common to radiodiagnosis department of all hospitals are as follows:

- Routine X-Rays of organs like chest, stomach, intestines, and so on.
- Unique techniques like barium test, IVPs, Myelograms, etc.
- Ultrasound
- Angiographies
- Training and research

1.4.4 Laboratory Services

Similar to the radiology and imaging department, laboratory services too are indispensable part of the hospitals. The whole healthcare system rely on it to diagnose the patients' ailments correctly and treat them successfully.

The significance of medical laboratories can't be overestimated in light of the fact that the effect of medicine, today, requires more laboratory tests, and doctors have the requirement for such space, regardless of whether they are private specialists or hospital-based specialists. Without a doubt, one explanation behind the concentration of doctors in urban areas and towns is the accessibility of such services in comparison to the rural and less densely populated areas. A medical laboratory can be a source of high revenue for the hospital. An effective laboratory for outpatients can diminish the quantity of patients admitted exclusively for laboratory tests, hence, decreasing burden on hospital beds. A proficient lab additionally helps to diminish the normal period of rest of the admitted patients.

The essential responsibilities of a laboratory are:

- To help specialists in touching base at or affirm an analysis and to aid the treatment and follow-up of patients.
- To generate expeditious and dependable reports, as well as works as a storage facility of reports for future references.
- To complete dire tests at any time of day or night, subsequently give benefit 24 hours per day.

- To help showing programs for specialists, medical attendants and research facility technologists.

Important departments

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A hospital's laboratory is a combination of numerous developments and strategies. The hospital's lab work by and large falls under these five departments:

- Haematology
- Microbiology
- Clinical science/Biochemistry
- Histopathology and cytology
- Urine and stool test

Organisation

The planning and the structure of the laboratories in the hospital have failed to such a degree that numerous hospitals today are confronting cramped working conditions and an ineffective working laboratory. Growing demand for these services can be a motivating force for due consideration of the responsibility and necessity of this quickly developing service in the hospitals. As in planning of other particular services, people with aptitude and involvement in the laboratory field are expected to fill in as a group with the manager and the engineer, to build up a moderate program for the laboratory. Advancement of the practical programs depends, for the most part, on the present number of tests required to be completed, which in turn, will rely upon the number and sorts of patients served. Different variables, are to be considered including the tests and the style of working, as given in the manual and the level of mechanical or computerised structure. This knowledge is converted into successful floor designs, territories, spaces, hardware, organisation, and staffing.

1.4.5 Pharmacy

Purchasing medicines and keeping up an excellent stock of medications, manufactured substances, and chemicals render a considerable amount of cash, next to compensations and wages. Around 20 percent of the hospital costs are represented by drugs and pharmaceutical supplies. Accessibility to the correct medication at the required place at the critical moment is the means to the hospital's existence. Delays are deplorable in this regard, as the impact of no availability of the right medication at the right time can be frightening. It can be a question of life and death. A high level of dependence of the hospitals on pharmacies highlights the need for exceptional care when it comes to the impact of drug store on medical services in the hospitals. There ought to be a legitimately build drug store department under the supervision of an expertly skilled and qualified pharmacist. Every large hospital will require a pharmacy, with a well-qualified pharmacist in-charge of providing all the services included under the components of pharmacy. Small

hospitals may not be able to manage the cost of a pharmacy with a completely qualified specialist due to insufficient funds. Be that as it may, the benefits of such a service for the skilful task of operating the hospitals have been acknowledged by staff and managers.

Components of the pharmacy

The components of the pharmacy are as follows:

- Provisioning, acquiring, putting away and distributing drugs, chemicals, and pharmaceutical sundry things
- Ensuring the effectiveness and quality of medications, along with prescribing medicines to inpatients and outpatient
- Maintaining data with respect to quality, cost, and sources of supply, everything being equal, of manufactured chemicals
- Researching pharmaceutical issues emerging in the utilisation of the drugs
- Guaranteeing adherence to the laws, rules, and control appropriate to pharmacy
- Promoting thriftiness in the utilisation of medications and building up accounting methods for pharmacy's costs and supplies
- Keeping a watch on the adherence of all the rules

1.4.6 Administrative Services

For any organisation to work properly, its administration should be well-oiled machinery. This is especially true for hospitals. It is extremely important for the proper functioning of a hospital.

Medical records

The requirement for the appropriately composed documentation of the facts related to a patient's treatment in the clinics can't be neglected, on the grounds that inability to maintain records implies failure in the duty towards the patient. Through medical records insights into the working of hospital are created for the eyes and the ears of the hospital's chairman. These records are of significance for the hospital for the assessment of its services for better healthcare. They also fill in as an asset for instruction and preparation of doctors as well as information for clinical research. In order to be successful, research requires logically recorded facts as reflected in the medical record. Moreover, the significance of accurate records for legal reasons is well-established. To put it simply, the need for maintaining legal medical records by a hospital can be understood through following points:

- I. Patient's needs:** It fills in as an account of the patient's entry into a hospital, maintaining coherence in that story. From this relevant data is separated for example, a medical prescription of his/her treatment, reports, etc. It saves time by avoiding unnecessary tests, if the patient is re-admitted into the

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hospital. Usually doctors are not assigned to same room resulting in lack of familiarity with the personal life of a patient. Therefore, a composed report of the family ancestry and individual history are essential. From a financial point of view, utilisation of medical record by different offices for protection claims, organisation advantages, joblessness, and modern remuneration is of fundamental significance to the patient. Data contained in the medical records is the deciding factor in supplying the patient with monetary help or healthcare for the rest of his or her life.

II. Doctors' needs: Medical records addresses the doctors' issues as:

- Procedure with regards to drug prescription on the basis of recorded facts
- Consistency in medical care
- Assessment of their own abilities as well as weaknesses
- Viable correspondence for the medical group. Records, when appropriately compiled, can be immediately recovered for study and research.

III. The medical the records help hospitals in:

- Generating hospital insights
- Teaching and helping in research and planning
- Admission control
- Planning of services
- Improving nature of healthcare
- Saving from any false law suits

1.4.7 Marketing of Health Services

Marketing is thought of by the vast majority as just advertising. But marketing is so much more. It can cover a wide scope of basic business exercises that bring items, ideas, or services to the clients. What's more, these items, ideas, or services are supplied to the clients at the time of their need, at the place, where they need, and at the costs that they can manage, providing all the facts the clients would need to make educated and satisfying decisions. Thus, with numerous characterisation there no commonly agreed upon meaning of marketing. Two of the broadly acknowledged definitions, which represent the subject are:

- Marketing is a procedure of planning and executing the creation, estimation, advancement, and conveyance of thoughts, merchandise, and ventures to make trades that fulfil individual or authoritative objectives (American Marketing Affiliation).
- Marketing is a social and administrative process by which people acquire what they need by offering and trading with others.

- Marketing is the service of distinguishing, fulfilling, and meeting client's requirements commercially.

Some basic features resulting from these definitions are:

- Marketing is an administration procedure
- Marketing offers and trades ideas, products, or services
- Marketing is a distinguishing service and envisioning client's necessities
- Marketing is providing clients with what they need
- Marketing includes valuing, advancement and appropriation of thoughts, merchandise, or services.

Consequently, marketing would cover:

- Scanning of the environment (demographic, creative, lawful and socio-social condition)
- Finding out clients' needs
- Deciding which of these needs or whose needs the organisation can fulfil (this is the subdivision called Target Market)
- Developing a 'Marketing Mix' (these are the four Ps of Marketing: Product, Price, Place, and Promotion) to fulfil the requirements and needs of the target group.
- Periodically, assessing the marketing efforts. Promoting is by definition a framework in the case, we acknowledge the recommendation factor of the said framework. Unquestionably, the connection between organisation of such factors as item, value, and marketing mix fits the definition.

Services given by a hospital include components, which can be analysed dispassionately, emotionally, or both. Each endeavour is concerned with quality affirmation by deciding the nature of the item it produces and staying in contact with buyers to ensure its fulfilment. Due to advances in medical innovation, demonstration of complex symptomatic and remedial techniques, and highly experimental and other advanced components, some crucial issues are being raised, for example, what is the measure of profit and level of accuracy of hospital service? What is the expense of operating the hospitals? Is the hospital spending more due to wastefulness of hospital labour? Could the equal quality of medical care be made accessible at lesser costs? What is the degree of patients' satisfaction? What is the ultimate effect or the final results when it comes to recuperation rate, partial recuperation rate, demise rate, etc.

Need for evaluation

Considered from all angles, there are three fundamental reasons which warrant target assessment of hospitals:

- It is to protect interests of the beneficiaries of hospitals. A person can't in any way, decide for themselves whether the care he/she is accepting is

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reasonable and logical. He has insufficient insurance against misbehaviours, misuse, and wasteful aspects of hospitals' medical staff and structure. Consequently, it is the moral and legal duty of the specialists to guarantee that hospitals render protected and proficient medical services to its patients.

- It is to find shortcomings and inadequacies of the hospital staff, its establishment, and system and the most essentially, its business structure. Obviously, hospitals' final outcome can't be great if there are no appropriate offices or proper condition in which the doctors can work.
- It is to provide the experts, viz., administering body, board of trustees, or owners a sound evaluation system for the performance of administrative staff at different levels, like hospital chairmen and individual doctors, by providing legitimate information to direct their future advancement.

Meaning of evaluation

Evaluation has been characterised as the way toward deciding the level of achievement in accomplishing predetermined goals. It is additionally characterised as 'Estimation of activity against acknowledged criteria and interpretation of relationship among them'. Examination, appraisal, promotion, appraisal, and survey are a section of the terms, which have been utilised synonymously with assessment. Assessment is one of the last task during the time spent in the management. In hospitals and public insurance, there are five indicators through which the nature of healthcare and services can be evaluated:

- i. The organisation
- ii. The procedure
- iii. The importance
- iv. The result
- v. The effect.

1.4.8 Hospital Linen and Laundry Service

Quality bed material, arrangement of good, clean cloth for patients and continuous supply of required material to wards and departments positively affect the image of a hospital. However, because of insufficient availability of Operation Theatre material and complaints of inadequately washed and rarely changed bed cloth by patients can be often heard in hospitals. The significance of providing spotless, sufficient bed material and patient's cloth at regular intervals according to the different requirement of different departments in the hospitals needs no emphasis.

Hospital laundry service

The services rendered by a hospital laundry are drastically different from a commercial laundry. A commercial laundry is responsible for household laundry, which include massive amount of material that must be assorted, pressed by hand,

and arranged. Then, again a hospital also has a pile of clothes, for example, sheets, covers, clothes, and towels, which require no work or planning. Commercial laundry practices should not be followed in planning a hospital laundry. Hospital laundries are quite dependable, cause less harm to material, are increasingly stable and decrease the peril of infections. Automatic laundry is safe, trustworthy, and affordable over the long haul. It guarantees required stock, safe treatment of grimy and contaminated cloth, and complete control of the washing procedure.

Linen classification

Linen utilised in the hospitals can be divided into General purpose linen, Patient linen, Ward linen, and Operation Theatre, Labour room, Procedure room linen:

- I. General purpose linen: This incorporates curtains, drapes, table clothes, and similar clothes generally utilised in all parts of the hospital. This is the material which isn't utilised for patient care.
- II. Patient linen: This comprises patient apparel, for example, night wear, shirts, gowns, coats, and so on worn by the patients.
- III. Ward linen: This comprises patient's bed linen, for example, bed-sheets, cushions, covers, and so on utilised by the patients.
- IV. Operation Theatre, Labour room, and Procedure room linen: This incorporates attire, for example, night wear, shirts, gowns and, coats worn by surgeons, anaesthetists, and OT staff along with surgical gowns, caps, masks, trolley covers, OT towels, etc. required in labour rooms, procedure room, and OT.

Check Your Progress

5. What is the benefit of an efficient laboratory for outpatients?
6. What does the term imaging include?
7. How many inpatients are served with respect to a number of beds every year on an average?
8. State any one reason for the need for evaluation of hospital administration.
9. Mention any three benefits of medical records for hospitals.
10. What is General purpose linen?

1.5 ETHICAL AND LEGAL ASPECT OF HOSPITAL SERVICES

The expanding number of medico legal cases has made it necessary for hospitals and their chairmen to be mindful of the legal angles in order to limit public criminal suits and guarantee nature of medical care. In India, legal advisors, till now, have

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almost no preparation in the puzzling issues related to medical care. Thus, public policy department must comprehend the fundamental medico-legal issues associated with individual damage, carelessness, negligence, medical fraud, incompetent doctors, and ethical issues inherent in human experimentation, organ transplants, and controlling genetics. In such situations, they are the only ones who can do the correct things, and shield themselves from silly suits.

1.5.1 General Acts/Legislations for Hospitals

In order to secure hospitals', hospital staff's, as well as patients' rights, certain acts were imposed. These acts were meant to bring justice to the victims and avoid wrong conviction. Some of these acts are:

- I. Industrial Disputes Act, 1948:** Under this act, clinics were incorporated under the term 'business'. The Industrial Disputes Act is relevant, where the number of workers is at least 50. It provides a method to deal with disputes emerging in a modern organisations. A 1982 alteration to the act exempted hospitals and dispensaries, among others, from the purview of the act, nevertheless, this arrangement was revoked by a revision in 1984. The administration has now an extensive enactment under its effective idea to manage modern disputes.
- II. Minimum Wages Act, 1948:** Under this act, minimum wages are fixed for various classes of labourers.
- III. Employees' Provident Fund Act, 1952:** The clinics as businesses are required to contribute an equivalent sum as the employees' salary and acknowledge the reserve consistently for the government. It is relevant to hospitals if the number of employees is more than twenty. Violation of this act is can lead to imprisonment.
- IV. The Payment of Bonus Act, 1956:** This act is pertinent to hospitals with at least forty workers. Welfare or non-profit hospitals are exempted from this act.
- V. The Payment of Gratuity Act, 1972:** Gratuity is payable to workers for each finished year of service at the time of his or her retirement or demise. This act is applicable, where there are at least 10 surgeons and applied on those who have finished five years of service (or at least one year if there they died).
- VI. The Payment of Wages Act, 1936:** This act ensures that the employees are paid on time without any illegal deductions from their salaries.
- VII. Indian Medical Council Act, 1933:** Indian Medical Council Act, 1933 sets out the code of morals for medical surgeons and directs medical education. State Medical Councils built under the act have the same responsibility in their respective states. Apart from that supporting hospitals is partly administered by the IMC act.

VIII. The Indian Nursing Council Act, 1947: Indian Nursing Council Act, 1947 sets out the instructive models and necessities for enrolment of medical caretakers.

IX. The Pharmacy Act, 1948: The Pharmacy Act, 1948 manages the foundation of drug stores and medication stores. A hospital needs to procure a medication permit if it sells drugs over the counter, where only a pharmacist can be utilised for distributing medications.

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Conclusion

Good organisation is an indispensable component of any health sector. Organisation is an essential part of people's wellbeing, along with different factors, including social determinants of wellbeing. The exact organisation and features of health services will vary from one nation to another, however in any efficient health system, the system of management ought to have the accompanying key attributes:

- I. Extensiveness:** A far reaching scope of health organisation is provided according to the requirements of the target population, including preventive, healing, sedative, and rehabilitative services and health boosting exercises.
- II. Openness:** Services are clearly available at all times with no undue obstructions of expense, dialect, culture, or topography. Health services should be available near the general population. Services might be provided at the home, the network, the work environment, or health departments as suitable.
- III. Inclusion:** Health services are designed with the goal that all individuals in a characterised target populace are secured, for example people from each income group.
- IV. Progression:** Health services are composed to furnish a person with the consistent care through various health conditions, levels of care, and the life-cycle.
- V. Quality:** Health services should be of high quality, for example they should be practical, stable, focused on the patient's needs and provided in a timely manner.
- VI. Individual-centered:** Services should be organised according to the individual, not the ailment or the money. Patients should consider health services to be responsive and adequate. There is support from the individuals in general populace in the management structure and evaluation. Individuals are accomplices in their own social insurance.
- VII. Coordination:** Local health administration system should be effectively organised, across different sorts of suppliers, care, levels of services, and for both daily as well as sudden problems. The patient's main care provider should encourage the method through the required services, and work as a team within different levels and types of providers. Coordination should

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likewise happen within different parts (for example, public services) and partners (for example, network services).

VIII. Responsibility and productivity: Health services should be supervised everywhere in order to accomplish the main tasks little to no wastage of assets. Managers are provided with the significant specialist to accomplish targets and are considered responsible for general execution and results.

Check Your Progress

11. What is Employees' Provident Fund Act, 1952?
12. List the elements needed for the efficient health system.

1.6 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. Henri Fayol is known as the father of present day theory of management.
2. The dependability of tenure is important as unnecessary turnover of workers is both the reason and the impact of terrible administration. The worker requires acknowledgement about the long-term relationship, bringing about a sentiment of security and organisation in work.
3. Planning can be comprehensively separated into two categories:
 - (a) Key planning
 - (b) Operational planning
4. The activities in the administration of hospitals are:
 - (a) **Financial management:** This refers to the budgetary undertakings of the hospitals. It incorporates planning and costing.
 - (b) **Managing staff:** This division refers to the determination, inspiration, and aim of representatives. It incorporates wage and compensation management.
5. An effective laboratory for outpatients can diminish the quantity of patients admitted exclusively for laboratory tests, hence, decreasing burden on hospital beds. A proficient lab additionally helps to diminish the normal period of rest of the admitted patients.
6. The term imaging includes:
 - (a) X-Ray
 - (b) Ultrasound
 - (c) CT Scan

- (d) Magnetic Resonance Imaging (MRI)
- (e) *Digital Subtraction* Angiography (DSA)
- 7. 3.0 crores inpatients are served with 8, 70, 160 beds in the nation every year.
- 8. There is the need for evaluation of hospital administration in order to find shortcomings and inadequacies of the hospital staff, its establishment, and system and the most essentially, its business structure.
- 9. The benefits of keeping medical records for hospitals are:
 - (a) Teaching and helping in research and planning
 - (b) Admission control
 - (c) Planning of services
- 10. General purpose linen incorporates curtains, drapes, table clothes, and similar clothes generally utilised in all parts of the hospital. This is the material which isn't utilised for patient care.
- 11. According to Employees' Provident Fund Act, 1952, the clinics as businesses are required to contribute an equivalent sum as the employees' salary and acknowledge the reserve consistently for the government. It is relevant to hospitals if the number of employees is more than twenty. Violation of this act is can lead to imprisonment.
- 12. The elements needed for the efficient health system are Extensiveness, Openness, Inclusion, Progression, Quality, Individual-centred, Coordination, and Responsibility and productivity.

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1.7 SUMMARY

- Basic services required for the hospital network can be met most efficiently, just with a little thought given to planning, development, and activity of social insurance offices.
- The purpose of planning is to guarantee the most efficient usage of assets and finance. The requirement for planning is felt on the grounds that a specific circumstance shows a chance to accomplish something positive.
- Walk-in health care is given to patients who are most certainly not confined to bed and can be treated at a general hospital, a specialised hospital, or a welfare hospital. When such care is rendered at premises which are a piece of a hospital (outpatient division), then they are called outpatient care.
- Nursing services stand out amongst the most critical services of medical facility. Nursing services in a more extensive setting is that part of the complete healthcare management, which intends to fulfil the nursing needs of the network.

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- The modern drug can't be prescribed without certain insights. Radiology or Radiodiagnosis is one such vital division of the hospital, which clearly contributes to the patient's care.
- A medical laboratory can be a source of high revenue for the hospital. An effective laboratory for outpatients can diminish the quantity of patients admitted exclusively for laboratory tests, hence, decreasing burden on hospital beds.
- There ought to be a legitimately build drug store department under the supervision of an expertly skilled and qualified pharmacist.
- Quality bed material, arrangement of good, clean cloth for patients and continuous supply of required material to wards and departments positively affect the image of a hospital.
- The expanding number of medico legal cases has made it necessary for hospitals and their chairmen to be mindful of the legal angles in order to limit public criminal suits and guarantee nature of medical care.

1.8 KEY WORDS

- **Scalar chain:** The chain of administrators from the highest level of superiority to the lowest positions in the organisation
- **Tenure:** The holding of an office, guaranteed permanent employment

1.9 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. What is Operational planning?
2. Write a short note on the Radiology and Imaging services.
3. State the components of the pharmacy.
4. Mention all the acts imposed on the hospitals.

Long Answer Questions

1. How should the division of the work take place? Explain.
2. Describe the administrative activities of hospitals.
3. Evaluate the significance of laboratories for hospitals.
4. Discuss marketing as applied to hospitals.

1.10 FURTHER READINGS

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*Principles and Methods
of Organising Hospital
Support Services*

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UNIT 2 CLINICAL AND SUPPORT SERVICES IN A HOSPITAL

Structure

- 2.0 Introduction
- 2.1 Objectives
- 2.2 Functions and Classification of the Hospitals
 - 2.2.1 Classification of Hospitals
 - 2.2.2 Staff Classification in Hospitals
- 2.3 Supportive or Associative Medical Service
- 2.4 Answers to Check Your Progress Questions
- 2.5 Summary
- 2.6 Key Words
- 2.7 Self Assessment Questions and Exercises
- 2.8 Further Readings

2.0 INTRODUCTION

The hospital is an intricate organisation, which provides healthcare to people through proper scientific equipment and group of skilled and knowledgeable staff. They are all synchronised with each other to achieve the general aim of restoring and maintaining the physical condition of the patients.

2.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the classification of the hospitals on the basis of different factors
- Analyse the supportive or associative medical services
- Discuss the different services with respect to hospitals

2.2 FUNCTIONS AND CLASSIFICATION OF THE HOSPITALS

Hospital acts as a healthcare centre to analyse and treat the problem of the patients:

- Provides services for hospitalisation
- Operates as a vaccination centre for the prevention diseases
- Acts as a teaching medium for health professionals like nurses, pharmacists, etc. and also helps in educating people

- Coordinates a variety of fields of medicine and improves the principles of medical performance.

Clinical and Support Services in a Hospital

2.2.1 Classification of Hospitals

The hospitals can be classified on the basis of various factors like age, size, system of medicines, ownership, diseases, etc.

Table 2.1 Classification of Hospitals based on Orientation

Clinical-oriented Based on System of Medicines	Non-clinical Oriented Based on Ownership
• Allopathic hospitals • Ayurvedic hospitals • Siddha hospitals • Unani hospitals • Homeopathic hospitals • Physiotherapy hospitals	• Government Controlled • Central government controlled a) Military hospitals b) Railway hospitals c) JIPMER, Pondicherry d) AIIMS, New Delhi • State Controlled a) General hospitals b) ESIC (employee state insurance) • Private trusts controlled Institutes a) Ramachandra hospitals –Chennai b) KMC hospitals-Udipi and Mangalore c) Bombay hospitals–Bombay Municipality corporation • Religion bodies controlled a) Hindu mission hospitals–Chennai b) Ramakrishna mission hospitals–Chennai and Kolkata c) Meenakshi mission hospital–Madurai d) Christian medical college hospitals-Vellore

Based on diseases

- Tuberculosis/Infection related hospitals
- Leprosy hospital
- Cancer hospitals
- Psychiatric/Rehabilitation hospitals
- Heart Institutes

Based on anatomy/physiology

- Ophthalmic hospitals
- Renal hospitals
- Dental hospitals
- Bone disorders related hospitals
- Neurological hospitals

Based on age

- Maternity hospitals
- Child disorders hospitals

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Based on bed number

- Large hospitals—more than 1,000 beds
- Medium hospitals—Between 500 and 1,000 beds
- Small hospitals—Between 100-500 beds
- Very small hospitals—less than 100 beds

Miscellaneous hospitals

- Training hospitals (hospitals attached to medical and dental colleges)

Depending upon the type and extent of the hospitals, the organisation of the institution will be different but major section that is, the clinical services will remain common and all the further functions like nursing, pharmacy, maintenance, and manager will depend upon the size of the hospital.

2.2.2 Staff Classification in Hospitals

The healthcare system is quite useless without the medical staff. The medical staff plays a huge role in the operation of any hospital, whether big or small. The staff can be classified on various factors, which are discussed in this section.

Administrative staff

- Run by panel of trustees or the government.
- Chosen by the management or the firm.
- Maintain the hospitals and report about it to the administrator beside the employees or acquaintances, helpers, managers, and branch leaders.
- Implement strategy as prescribed by governing council.
- Act as a link connecting medical employees and board of trustees.

Medical staff

- Accountable for every medical care that is to be provided to the patients.
- Backbone of the hospitals
- Plays a vital role in restoring and sustaining excellent physical conditions of the patients.
- It is categorised into:
 - a) Active health check-up employees or housing health check personnels
 - b) Supporting health check-up staff
 - c) Consulting medical staff
 - d) Voluntary medical staff

Residential medical staff

- Accessible on 24/7 basis.
- Accessible at any time to examine the patients.
- Also, concerned with associational and managerial duties.

Associated medical staff

- Appointed and assigned for the various services in the same manner as members of active medical staff.
- Considered for advancement to residential medical staff.

Consulting medical staff

- Experts in particular fields of medicine.
- Patients come to hospitals on the specific days and time, according to the availability of concerned doctors depending upon the problems of the patient.

Honorary medical staff

- Amateur seeing medical personnel.
- Possibly retired or retired doctors.
- May have their individual private clinic, but voluntarily provide their services to the hospitals.

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2.3 SUPPORTIVE OR ASSOCIATIVE MEDICAL SERVICE

As already discussed in the previous unit, the clinical support services are the most important services provided by a hospital so as to maintain patient care. Analytical Imaging, Pharmacy, and Laboratory tests, although not openly concerned with patient care, are all significant for the accurate as well as appropriate analysis. Provided by every hospital, clinical support services are the mechanism that keeps the hospitals working. It is necessary to keep these services updated with the latest technology.

I. Pathology and Microbiological/Biochemistry services

- At these places, samples of blood, urine, and faeces are tested for any sign of diseases along with biochemical indicators like blood sugar, BUN, LDL, and TBC.
- Ailments like cancer, malaria, syphilis, AIDS, tuberculosis, etc. can be detected by pathological reports.

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a. Laboratory

- Whether you are inquiring for a blood or urine sample, the majority of patients are familiar with the drill. The following procedure is little less familiar.
- Samples go through apparatus like haematology analysers, or pathology immunohistochemistry auto-stainers to test the effects of a treatment and to ascertain any infection in patients.
- The demand of laboratory services have only increased as apparatus and tests needs have amplified.

b. Radiology

- Radiologists are responsible for conducting various imaging-related tests for the diagnosis and the treatment of the hospitals.
- Apart from X-Rays, it also provides facilities such as Sonography, Computed Aided Tomography (CAT) scanning, and Magnetic Resonance Imaging (MRI).
- By means of apparatus like CT Scanners, MRIs, X-Rays, and Ultrasound, Radiologists assist doctors in making a diagnosis for treating sickness and injuries.
- With knowledge in the field of health care developing all the time, the lifetime of testing equipment is less than ten years. Thus, it should be regularly updated and supplied with new equipment, along with developing new enhanced techniques to conduct tests.
- Funds contributed for Imaging-related tests will aid in restoring older machinery and offer a means to update the equipment to the latest version.

c. Hematology

- Hematology related activities are closely connected with the hospital's laboratory. These doctors treat blood diseases and malignancies related to the blood, according to the latest recommendations.

d. Blood bank

- It is responsible for collecting, processing, and delivering blood and its content like blood plasma and platelets, whenever and wherever it is needed.

II. Medical–Social Service Department

- This department maintains connection among hospitals, patients, and their relatives.
- Usually offers information to patients concerning dangers and advantages of a particular medical and surgical treatment.

III. Anaesthesia Services

- This service includes anaesthetic, which are responsible for the correct administration of the painkillers, prior to and throughout the process of medical treatment or surgery.

IV. Nursing Services

- Improve physical condition by therapy and act as an emotional support for the patients.
- Nursing individuals ought to value individuality, self-respect, and rights of every being despite their race, colour, nationality, or social and financial status.
- Nursing section works under the head of nurse.

V. Dietary Service

- Food managed and supplied in hospitals should in no way be corrupted or contaminated.
- Food has to be prepared under hygienic conditions and should be healthy. It should be provided according to the diseases.
- Dieticians are responsible for recording nutritional history of patients particularly of those with allergic reactions to certain food.
- Dieticians are also concerned with counselling patients and their families with regards to their diet.
- Skilled dieticians and nutritionists offer professional recommendation on diet for hospital wards and outpatient clinics, forming part of a multidisciplinary team.
- This department works across an extensive series of specialties such as:
 - o Diabetes
 - o Cancer
 - o Kidney problems
 - o Paediatrics
 - o Elderly care
 - o Surgery and critical care
 - o Gastroenterology.
- Moreover, they provide group education to patients with diabetes, heart disease and osteoarthritis, and work closely with weight supervision groups.

VI. Medical record services

- It includes crucial information to facilitate the treatment of a patient by one or more doctors.

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- It has information for identifying patients by their name, reference number, location, etc. as well as health insurance plan.
- Medical account page presents information of previous hospital admittance for major injuries, procedure history of transfusion, family history, along with pregnancies with particulars concerning dates, consequences, and problems.

VII. Pharmacy

- A pharmacy in the hospital is responsible for fulfilling the medication requirements of the patients.
- The whole apparatus of a pharmacy mostly reports the safety of the medicines in addition to quality of the supply, transportation and stockpiling.
- Particular dosage, bar code, electronic checking of medicines, cabinets, and freezers are all elements of the pharmacy that significantly impacts the health care.
- A new method will assist in the transportation of medications with added accuracy from one corner to another in every branch, in a small amount of time.

Check Your Progress

1. Classify hospitals on the basis of the system of medicines.
2. How are the medical staff categorised?
3. What is the responsibility of blood bank?
4. Mention any two responsibilities of dieticians.

2.4 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The following is the classification of hospitals on the basis of the system of medicines:
 - a) Allopathic hospitals
 - b) Ayurvedic hospitals
 - c) Siddha hospitals
 - d) Unani hospitals
 - e) Homeopathic hospitals
 - f) Physiotherapy hospitals

2. The medical staff are categorised into:
 - a) Active health check-up employees or housing health check personnel
 - b) Supporting health check-up staff
 - c) Consulting medical staff
 - d) Voluntary medical staff
3. The blood bank is responsible for collecting, processing, and delivering blood and its content like blood plasma and platelets, whenever and wherever it is needed.
4. The responsibilities of dieticians are:
 - a) Dieticians are responsible for recording nutritional history of patients particularly of those with allergic reactions to certain food.
 - b) Dieticians are also concerned with counselling patients and their families with regards to their diet.

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2.5 SUMMARY

- Hospital acts as a healthcare centre to analyse and treat the problem of the patients.
- The hospitals can be classified on the basis of various factors like age, size, system of medicines, ownership, diseases, etc.
- Depending upon the type and extent of the hospitals, the organisation of the institution will be different but major section that is, the clinical services will remain common and all the further functions like nursing, pharmacy, maintenance, and manager will depend upon the size of the hospital.
- The healthcare system is quite useless without the medical staff. The medical staff plays a huge role in the operation of any hospital, whether big or small.
- The staff can be classified on various factors. The clinical Support Services are the most important services provided by a hospital so as to maintain patient care.
- Analytical Imaging, Pharmacy, and Laboratory tests, although not openly concerned with patient care, are all significant for the accurate as well as appropriate analysis.

2.6 KEY WORDS

- **Ayurveda:** Developed more than 3,000 years ago in India, it's based on the belief that health and wellness depend on a delicate balance between the mind, body, and spirit.

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- **Siddha:** Refers to perfected masters who have achieved a high degree of physical as well as spiritual perfection or enlightenment.
- **Unani:** The term for Perso-Arabic traditional medicine as practiced in Mughal India and in Muslim culture in South Asia and modern day Central Asia based on the teachings of the Greek physicians Hippocrates and Galen.

2.7 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. State the functions of hospitals.
2. What are the functions of a laboratory?

Long Answer Questions

1. Describe the different types of medical staff.
2. Discuss, in detail, the dietary services provided by a hospital.

2.8 FURTHER READINGS

Llewelyn, Davies R; Macaulay, H.M.C. 1966. *Hospital Planning and Administration*. Geneva: W.H.O, Jaypee Brothers.

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UNIT 3 **ROLE OF SUPPORTING DEPARTMENTS IN HOSPITAL MANAGEMENT**

*Role of Supporting
Departments in
Hospital Management*

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Structure

- 3.0 Introduction
- 3.1 Objectives
- 3.2 Structure of the Organisation
- 3.3 Governance
 - 3.3.1 Duties and Responsibilities of the Governing Board
 - 3.3.2 Conflict of Interest
- 3.4 Management Structure
 - 3.4.1 Functions of a Committee
 - 3.4.2 Advantages/Disadvantages of a Committee
 - 3.4.3 CEO (Chief Executive Officer)
- 3.5 Answers to Check Your Progress Questions
- 3.6 Summary
- 3.7 Key Words
- 3.8 Self Assessment Questions and Exercises
- 3.9 Further Readings

3.0 **INTRODUCTION**

Organisation of the hospital is vital and plays an important role in its effective operation. It can be defined as a mechanism or the executive body governing business and enabling people to work together effectively without any problem. It also provides structural basis and acts as the final authority in maintaining relationships. As per the particulars of planning processes and various organisational hospital structures, general hospitals act as the central prototype, as it is the primary health care facility, common in any community. These categories, which are structurally different varies according to factors like design, governing style, and management.

Some philosophies and goals specific to the individual hospitals separates them from each other. Various kinds of hospitals which include Government, Municipal, Private and Voluntary hospitals are totally different from each other based on their governance and management structure. A teaching hospital (Medical College), on the other hand requires more facilities, staff, and beds than a normal service hospital or small rural hospitals. Likewise, a speciality hospital, for instance, psychiatric hospital or a heart-diseases hospital have certain structural difference, which make them different from simply designed hospitals providing secondary care.

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Having all these above features is not mandatory as they can vary from place to place, with respect to the basic and important preferences of the place at which the hospital is located. Sometimes general hospitals differ according to their basic needs, facilities, structure, and function. Each and every organisation should be organised based solely on the fundamental needs of the community and should be moulded accordingly. Provision of quality patient care, and maintenance of effective and efficient fee structure, along with the organisation, fuels the proper functioning of the hospitals.

3.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the structure of the organisation
- Analyse the governance of the organisation
- Discuss the management structure of the organisation

3.2 STRUCTURE OF THE ORGANISATION

The administrative centre or the action governing committee, also known as the members of administrative council and governance, or board of directors are the point of intersection of various organisations and functioning committees within a hospital. The duties and responsibilities remain the same everywhere with name being the only differentiating factor.

The board of governance is the primary body in the hospital responsible for all the legal steps and actions. The board delegates have the major authority and power to provide suggestions for performing certain actions. The Director, President, or the Chief Board Member discusses his/her plans with the members, which is then passed on with his/her permission. The Chief Executive Officer (CEO) is the person, who takes all the responsibility for managing and supervising all hospital operations. He/she is generally helped by a group of highly educated associates in all of his/her tasks.

Modern day hospitals are of a very complex nature and thus, it's not possible for a single individual to handle all the departments single-handedly. Every individual is specially trained in the specific areas to look upon the functioning of the various departments of a hospital and its operations, efficiently. Supportive experts in certain departments such as finance, public relations, nursing services, strategic planning, legal matters, human resources management, and material management are very helpful in the functioning of business aspects of the hospital. However, the CEO is the boss and give directions to them in various matters. All associates are directly answerable to him/her. In smaller hospitals or clinics, the various departments of may be concentrated into a small group of associates, who look into all the matters simultaneously and are responsible solely for the functioning of

the organisation. They are headed by few higher officials. The departments in a hospital are broadly divided into two major groups:

- Professional care of patients
- Business Management

On the basis of the service provided by the hospitals, it is divided into five divisions:

- Medical
- Nursing
- Ancillary
- Fiscal
- Support Services

You have learnt in the previous unit that classification of hospital staff may include administrative, medical, associated medical, residential medical, consulting medical and honorary medical staff. Another way, this classification can be done is this:

- Hospital-based staff
- Full-time salaried staff (majority of the hospitals)
- Temporary staff

This is a commonly followed pattern in all over the world and now it is also being followed in our country.

The hospital chief can either be a physician or an administrative officer. Both of these types are present in India. The medical chief is generally known by the name or title of director, medical director, or medical superintendent. The non-medical head on the other hand is known by the name of administrator, director, or executive director. In corporate hospitals, he/she may be called President, CEO, Managing Director, Executive Director, or General Manager.

3.3 GOVERNANCE

The governing board is the primary authority in the organisation of a hospital. Thus, selection of the governing staff is done with utmost care. To be a member of the governing body is of great respect and everyone wants to have their name enlisted for the same. For a person filled with social values and philanthropic enthusiasm, it creates an avenue of great ideas and possibilities. These responsibilities should be given to such people, who are eager to be in hospitality department and are educated enough for their respective posts. Those who think of these posts as business ideas and means to earn profit should never be entertained. Political members should not be included in the governing body as the hospital organisation should be free from the political interference and ups and downs of political events.

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These people can be a great menace to the organisation of the hospital as they only think about self-gain. Some of them love the power provided by the posts so much that they do not even hesitate to book their position in the board through bribery. They are bound to fight for lower positions and be happy with them because they lack the knowledge needed to be in a huge organisation on the basis of their qualification. The institute generally decides whether it wants to include banking and legal professionals in among the board members. Some of the organisations focus on hiring professionals from certain fields to face any problem in any fields. Some others positions can also be filled by government officials and newspaper editors. The selection of members should be solely based on the type of officials needed and their professional achievements, along with their personal skills and type of person he/she is. These steps ensure a dynamic management of the organisation.

Inclusion of people in the board, who have never excelled in their own professional fields is a total loss for the board and should be avoided completely because one cannot expect them to change the destiny of a big and complex organisation such as a hospital, when they were unable to maintain their own business affairs. Some of the businessmen leave all responsibilities in the hands of paid officials and enjoy their peaceful lives. People like them are of no use in maintaining a steady organisation. Membership of the board comes with a huge responsibility, which needs to be taken seriously and performed with utmost care. The members are also prohibited from having any personal gains from their positions.

Sometimes it is also seen that the organisation sticks to a basic boundary and cannot cross it to see the profit. Most of the times this is because of some members, who do not perform their jobs correctly and are prone to excuses, which ultimately results in the loss for the organisation. It is very important to boost them up or chuck out and replace them, with active and capable officials for the sake of fluidity in the functioning of the organisation. Even if the central committee takes decision, without involving each and every member, there must be an efficient system, through which other members are informed. This must be found out and cleared before it could harm the functioning of the organisation and break the unity among the members of the board.

It is often seen that first-timers in the board are as amateurs and are easily manipulated by the senior members and made to follow their decisions. This should not be done and steps should be taken by the governing officer to avoid such incidents. There must be a mechanism to check the performance of the members and to mark their improvement level. This is very important as it brings up competitive mentality among the members and a healthy competition is always good for the betterment of the organisation. It will also help in detecting problems among the members if any and lead to their correction.

A very good and positive relation between the board members and the CEO is a must for the proper and successful operation of the organisation. Thus, this should be maintained at each and every possible level.

3.3.1 Duties and Responsibilities of the Governing Board

The hospital is centrally controlled by the governing body. Usually, it is responsible for important and dynamic decisions. It is helped in its job and functioning by many other small officials and departments of the hospital. Governing board as the name suggests consist of certain members of the same or other departments, who work in unison on an issue and find the solution to the problems after discussions and meetings. This is done by the majority, after going through different strategies required in the process of finding a suitable answer. The governing board has huge responsibilities and need to perform certain duties. They include:

- To nominate and elect the perfect CEO on the basis of his/her knowledge and achievements.
- To formulate and reshuffle the goals, objectives, and mission of the organisation.
- To determine, evaluate, and review all the major problems of the hospital with respect to the community it serves.
- To raise and provide adequate funding for the proper functioning of the hospital.
- To ensure control over expenditure and have detailed business records.
- To increase the total assets of the hospitals in terms of finance, material, space, and personnel.
- To enforce and maintain the proper professional standards in the treatment of the patients.
- To fulfil its legal obligations.
- To select responsible candidates for medical and non-medical fields in a hospital.

3.3.2 Conflict of Interest

Circumstances that can create possible conflict of interests in the governing body are:

- A board member having financial interest with regards to from where a hospital buys its medical and non-medical resources.
- A board member rendering management, professional, or consultative services to another organisation.
- A board member having business with other hospitals, institutions, and organisations.

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- A board member accepting gifts or hospitality from the business-seeking persons or such officials.
- Member disclosing the trade secrets, management style, and other classified information of the hospital.
- Member caring more about his personal relationship with other organisations more than that with the hospital.
- A member who is concerned with a transfer or promotion in any other organisation can be a threat.
- A member treating other members badly in the name of caste and creed.

Check Your Progress

1. What are the names used for the non-medical head?
2. Mention any three duties of a governing board.
3. What could be the problem with first-timers in the governing board?

3.4 MANAGEMENT STRUCTURE

In an organisation, the management structure is frequently baffled with the governance. Maybe there is a lack of understanding about the different functions that constitute the management and governance. There is a person called CEO or Chief Executive Officer, who manages the management team. He/she represents the governing board legally, the board who invested him/her with the authority to manage the hospital. The CEO, on the other side, may give some authority and responsibilities to the associates and department heads to perform their respective functions of every department properly, but ultimately he/she is responsible for everything that happens in the hospital. The motivation of the staff members and the human resource management is very much crucial in hospital management.

There are organisations like the committee organisation, in which work is done by the committees, which has its own advantages and disadvantages. It is basically an association of people to solve various problems. A committee is an important requirement, when it comes to a large complex organisation, which has problems too big and complex to be dealt effectively by one person. There is a participation of several persons, who propose their ideas, which are later developed, resulting in the recommendations with regards to the procedures and policies. Coordination and control become convenient because there is an open discussion in the committee, among the committee members.

3.4.1 Functions of a Committee

The functions of a committee in an organisation are:

- To get the members of the committee to work as a team and not in an individual manner.
- To secure the unity of thought and action, upon which success in any firm depends.
- To serve as an effective communication network.
- To develop mutual cooperation among the members of the committee.
- To provide more realistic decisions through their collective knowledge, skills, and experience.
- To promote cooperation.
- To recognise an executive's ability and enable training for those who will later occupy the top management.

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3.4.2 Advantages/Disadvantages of a Committee

There are various advantages as well as disadvantages in working with the committees. Some of the advantages in working through the committees are as follows:

- The committee members can share the information and easily exchange their thoughts and views.
- There are several people working in a group, so the problem or task can be targeted by a wider range of experience and perspectives. The members of the committee can bring the collective wisdom, judgement and group deliberation to do task.
- As there are more than one head working in the committee so the decisions can be made on collective basis.
- A committee can generate more ideas to work on.
- Objective decisions are made personal by the committee, when there is a need to discuss about the pros and cons of an issue or a topic.
- Members can recommend action as a committee.
- The members of a well-organised and responsible committee perform at a high level of commitment and cooperation.
- Committee constitutes of more varied professionals and specialists from various disciplines like management, public relations, architecture, and finance. They can all contribute to the discussion with respect to their fields or expertise.

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Disadvantages in working with the committees are:

- Due to some aggressive members in the committee, the situation gets out of control as they are dominating and overbearing. They may unduly influence other members in their final decision.
- The manager sometimes makes the other members of the committee to come to a decision that he has already made. He may obtain group sanction for certain actions to make it look like he is not railroading.
- There is a prevalent opinion that committees are a waste of time. They take more time to reach to a conclusion and their cost in ways of time and money is considerable.
- The original ideas of the members of the committee and their plans and project proposals are modified and compromised, before the committees approve them.
- There is a popular saying about the committees that if you want to destroy an idea, you should appoint a committee. It happens when the manager himself does not want to make a decision, so he appoints a committee, who makes decisions on his behalf.
- The inherent danger is that the members of the committee may use their power of position to further their self-interest, which may cause conflicts in the organisation.
- There are some non-performers who do not take part in the affairs of the committee and the organisation. They are absent from the meetings regularly.
- Another threat to the committee is the external members of the committee interfering in the day to day operation.
- The decisions that are made by the committee generally represents a compromised position and do not truly reflect the real feelings of the individual committee members.

3.4.3 CEO (Chief Executive Officer)

The hospital's CEO must be a person, who is remarkable, equipped, and ready to meet the challenges that comes in the professional and technical fields. The job of managing an organisation as broad and complex as the hospital, is too difficult for anyone. Thus, the CEO of the hospital is helped by associates, who are able to assist in various fields. The CEO can also be compared to an automobile driver here, he/she is not an automobile engineer or a mechanic. The driver is not aware about the engine and the various functions of a car, but he is still a good driver if he has some knowledge about certain parts of the car. Similarly, the CEO must have the presence of mind and ability to coordinate. He must be clear in his vision as to where he is going and how to get there. The CEO is not expected to be an expert in every specialised field of operation. He should nonetheless have a wide-ranging knowledge of subjects related to the hospital environment. Consequently, he cannot

only discuss them with his specialists intelligently, but can also guide them so that sound decisions could be made.

Qualities of the CEO

Some of the qualities the chief of the hospital must possess are:

- He/she must be well behaved and tactful in his behaviour and have a patient attitude.
- He/she must possess training in proper organisation and time-management.
- He/she must be a professional in difficult situations and carry the workload.
- He/she must inculcate positive attitude towards others and should have leadership qualities.
- He/she must possess a sense of responsibility for his/her position.
- He/she must have faith in his/her decision-making skills and look forward, toward a bright future.
- He/she must be honourable and give credit to the deserving candidates for the success of the organisation.
- He/she must be professional enough to not let emotions dominate his/her decisions.
- He/she should possess qualities for judging human nature.
- He/she should select the personnel wisely, depending on the working situation and conditions.
- Last, but not the least, he/she should be tidy enough with good posture.

Responsibilities and functions of the CEO

There are some responsibilities and duties of the CEO towards the organisation and the committee, which are:

- The CEO discharges, disciplines, and controls all the workforce.
- The CEO has the power to select and, employ the staff and fix their salaries within the approved salary scales, according to the budget of the organisation.
- The CEO has to submit an organisational plan for the hospital and get it approved from the governing board.
- The CEO formulates the rules and regulations for the proper functioning of the hospital.
- The CEO approves the charges for all the hospital services.
- The CEO also counsel the governing board on the formulation of strategies.

The CEO, in short, directs all the hospital activities and executes the established strategies. As the executive head of the hospital, he/she is answerable to the governing board for management of the hospital.

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- The CEO chooses the head of the department and assigns the part of his/her responsibility to them. The extent of the assign depends largely on the size of the hospital. Whatever the case may be, the CEO is ultimately responsible for the management of the hospital, while the department heads are directly answerable to him.
- The CEO also has to submit the periodic reports to the board on the working of the hospital. The annual report provides an analysis for the next coming year.
- The purchase department is under CEO's control. In smaller hospitals, the CEO of the hospital may perform the duties of the purchase officer. However, regardless of the size of the hospital and the officer involved with purchasing, the CEO must be actively involved in purchasing and exercise ultimate control over the department.
- The relationship between employees is also a responsibility of the CEO. In smaller hospitals, he may only perform the functions of the human resource department, but in larger hospitals, there is a fully-fledged human resource department under him.
- The CEO prepares and submits an annual budget for the governing board's approval. The budget will show all the estimated receipts and expenditure.
- The CEO serves as a channel of communication between the governing board on the one hand and various departments and other staff, on the other hand. He conveys and interprets the policies, and makes sure that they are followed.
- The duty of the CEO should not be limited to his own hospital, nor even to his own community. He has to maintain contact with the hospital associations and attend their meetings. He should write articles in professional journals, participates in the discussions and pass on the benefits of his knowledge and skills to others.
- Some progressive hospitals direct, even make it compulsory, that the CEO participates in local, regional, state, and national hospital affairs, when it is advantageous for the hospital. Additionally, the CEO must collaborate with various regulatory bodies to ensure feedback concerning the latest regulations and laws affecting health care and health care institutions.
- The CEO must exercise sound control over the business management of the hospital. Although he is not expected to take care or to have training in every field of operation, he must have sufficient knowledge of all the departments to be able to effectively supervise them and assure their management.
- The CEO of the hospital must have leadership qualities. He should identify himself with people and be actively involved in the activities of the community, particularly those, which directly or indirectly have a bearing on the work of

the hospital. He should be closely connected to the public service organisations like women's club, etc.

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Relationship of the CEO with the governing board

The relationship between the governing board and the CEO is one of employer and employee, but these should not be interpreted in the usual sense of these terms. The governing board and the CEO work as partners in some special types of organisations like hospitals. Nevertheless, the CEO is the employee, whom the board can discharge for just cause. The relationship between the CEO and the governing board has become so firm that in many hospitals the CEO is made a voting member in their governing boards. In some of the boards, he acts as a chairman, in others he has a title of president under the chairman of the board. There is a common pattern emerging in the industry, where the CEO is also a member of the board of directors and not just a hired employee.

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The transition of the CEO

The emergence and transition of the position of CEO can be described in the following manner:

- In the early times, the nurse was the top full-time officer, whose primary responsibility was the supervision of the nursing care. The other administrative responsibilities were given to other members of the governing board.
- As the demand on everyone's time increased, a business manager was appointed. He assumed more and more administrative responsibilities, while the members of the board coordinated the functions of the two positions of the nurse and a business manager.
- The third stage saw the appointment of full-time hospital superintendents and administrators. They were drawn from the ranks of physicians, trustees, and clergymen in church associated with hospitals.
- In the next stage, the governing board observed that the hospital administrator functioned both as the director of the hospital activities and the health leader in the community as well as in the wide field of health care. He was then given the title of the executive director or executive vice-president, or CEO. The board also recognised the need to treat their CEO like their counterpart in industry.
- The position of the CEO entered the final stage with the title of the president. This title of the President and Chief Executive Officer is common in many hospitals in America and now coming into use in corporate hospitals.

The CEO and his management team

The CEO of the hospital cannot be on duty continuously, even in a small hospital. Nor can he personally attend to a myriad of details, pertaining to his office and other departments. The most important thing is that he is not expected to be an

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expert and have intimate knowledge of the working of the various specialised departments. Although, he has to rely on his associates and department heads, who have specialised knowledge to manage day to day operations of the hospitals, the CEO has others to relieve him of his responsibilities during his absence at night or on Sundays and holidays. So, in many hospitals, there is a night administrator, who is in charge of the hospital at night and one of the senior administrative officers for duty on Sundays and holidays based on rotation.

Now, the question emerges, who is responsible for the operation of the hospital and who will make decisions on Sundays, holidays and after the normal working hours on weekends.

In most cases, the answer to this question is 'No one'. Anything can go wrong during these days and hours, when no responsible officer of the hospital is present. There should be an effective arrangement in every hospital, irrespective of its size, under which decisions are made and emergencies are attended to. In small hospitals, the CEO may personally take over the functions of some of the assistants. For example, he may perform the responsibilities relating to personnel management and purchase. He may directly supervise departments like food service under a manager.

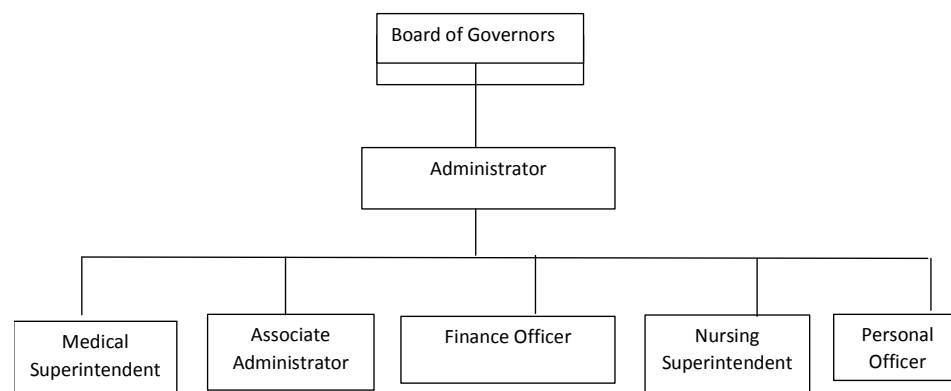
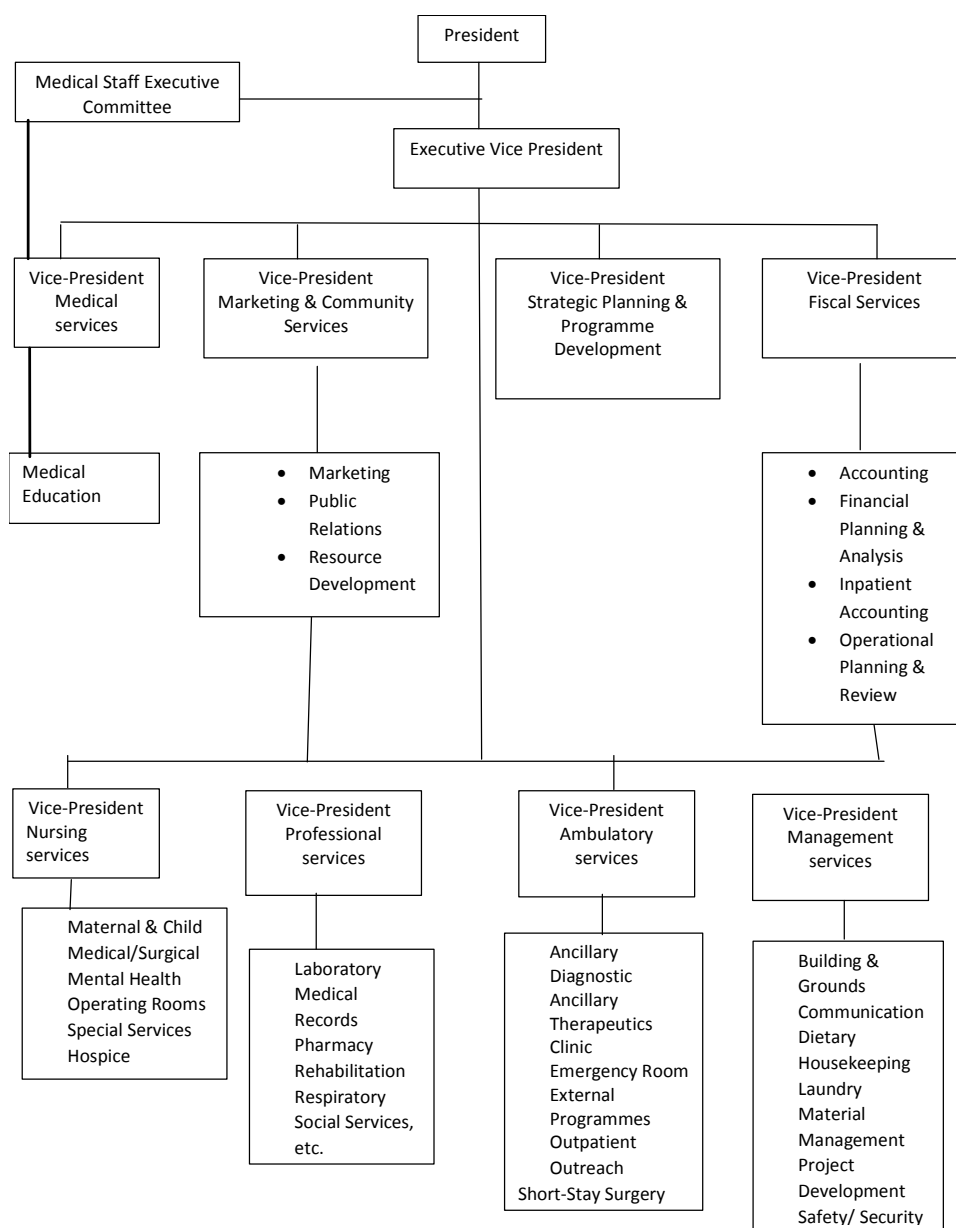


Fig. 3.1 Members of Governing Body



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Fig. 3.2 Management Structure

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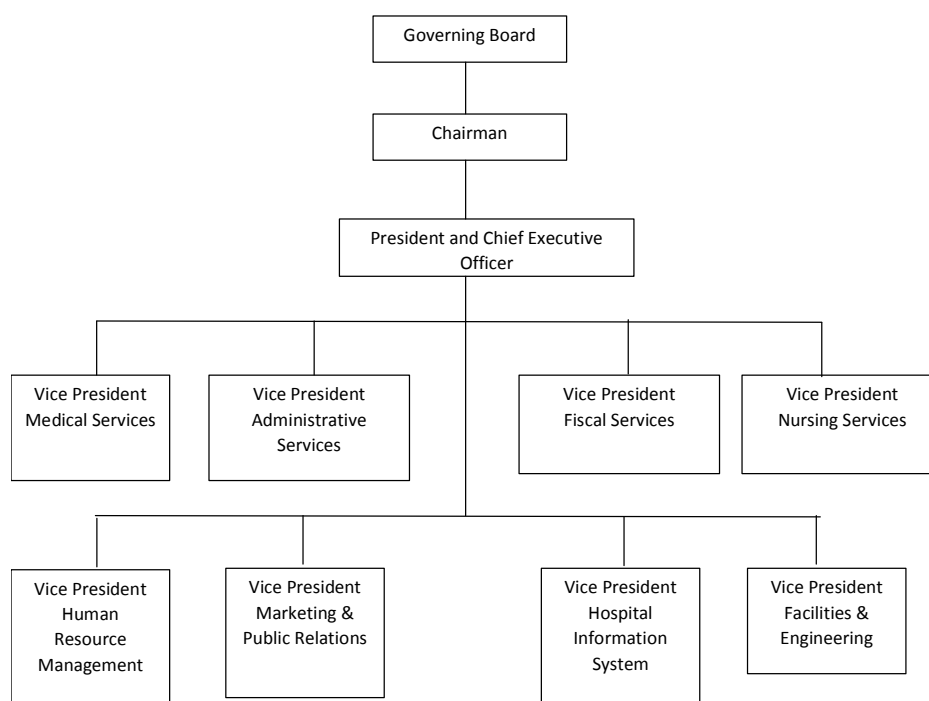


Fig. 3.3 Hierarchy in the Governing Body

Check Your Progress

4. Why is the committee needed in an organisation?
5. Who relieves the CEO of his/her duty during night and on holidays?
6. Why is the CEO expected to have a wide-ranging knowledge of hospital-related subjects?
7. State any two functions of a committee.

3.5 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The names used for the non-medical head are administrator, director, or executive director. In corporate hospitals, he/she may be called President, CEO, Managing Director, Executive Director, or General Manager.
2. The duties of a governing board are:
 - a) To raise and provide adequate funding for the proper functioning of the hospital.
 - b) To ensure control over expenditure and have detailed business records.
 - c) To increase the total assets of the hospitals in terms of finance, material, space, and personnel.

3. It is often seen that first-timers in the board are as amateurs and are easily manipulated by the senior members and made to follow their decisions.
4. The committee is needed to deal effectively with a large complex organisation, which has problems too big and complex to be dealt by one person.
5. In many hospitals, there is a night administrator, who is in charge of the hospital at night and one of the senior administrative officers for duty on Sundays and holidays.
6. The CEO is expected to have a wide-ranging knowledge of hospital-related subjects so that he not only discuss them with his specialists intelligently, but can also guide them so that sound decisions could be made.
7. The functions of a committee are:
 - a) To secure the unity of thought and action, upon which success in any firm depends.
 - b) The committee itself is a suitable body for an effective communication network.

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3.6 SUMMARY

- The administrative centre or board of directors are the point of intersection of various organisations and functioning committees of a hospital. Modern day hospitals are of a very complex nature and thus, it's not possible for a single individual to handle all the departments single-handedly.
- Every individual is specially trained in the specific areas to look upon the functioning of the various departments of a hospital and its operations, efficiently.
- The governing board is the primary authority in the organisation of a hospital. Thus, selection of the governing staff is done with utmost care.
- Governing board as the name suggests consist of certain members of the same or other departments, who work in unison on an issue and find the solution to the problems after discussions and meetings. This is done by the majority, after going through different strategies required in the process of finding a suitable answer.
- There are organisations like the committee organisation, in which work is done by the committees, which has its own advantages and disadvantages. It is basically an association of people to solve various problems.
- The hospital's CEO must be a person, who is remarkable, equipped, and ready to meet the challenges that comes in the professional and technical fields.

- The job of managing an organisation as broad and complex as the hospital, is too difficult for anyone. Thus, the CEO of the hospital is helped by associates, who are able to assist in various fields.

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3.7 KEY WORDS

- **Chief Executive Officer (CEO):** The most senior corporate, executive, or administrative officer in charge of managing an organisation
 - **Amateur:** A person who is incompetent or inept at a particular activity
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3.8 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. Write a short note on the structure of the organisation.
2. List the qualities required in a CEO.
3. How would you describe the relationship between the CEO and the governing board?

Long Answer Questions

1. Evaluate the governance of the hospitals in detail.
 2. Describe the advantages and disadvantages of the committees.
 3. Discuss the emergence and transition of the position of CEO.
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3.9 FURTHER READINGS

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UNIT 4 NURSING CARE AND WARD MANAGEMENT

*Nursing Care and
Ward Management*

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Structure

- 4.0 Introduction
- 4.1 Objectives
- 4.2 History of Nursing
- 4.3 Functions of a Nurse
- 4.4 Nursing Care: Meaning
 - 4.4.1 Role of a Nurse in the Healthcare Team
- 4.5 Ward Management: Meaning
 - 4.5.1 Components of Ward Management
- 4.6 Answers to Check Your Progress Questions
- 4.7 Summary
- 4.8 Key Words
- 4.9 Self Assessment Questions and Exercises
- 4.10 Further Readings

4.0 INTRODUCTION

Nursing is a service existing since the time immemorial. People have been involved in nursing since the primitive era, but the science of nursing is still quite young and is growing leaps and bounds. The focus of nursing is on independent and collective care of people belonging to all age groups, families, and communities. It focuses on promoting health, preventing illness/disease, and providing care to the ill, disabled, and dying people. To become a nurse, a person should have completed a basic or generalised program on nursing education, which should be authorised by an appropriate regulatory authority that recognises and permits the practice of nursing. The nurse is prepared and authorised:

- To involve in carrying out the process of general nursing, promoting health, preventing disease, and caring for physiologically and psychologically challenged, and disabled individuals of all age groups in healthcare as well as community setting
- To educate about health care
- To show full participation as a member of the healthcare team
- To undertake training and supervision of other nurses and auxiliary workers
- To get involved in research

The WHO Expert committee on nursing defines the nursing services 'as the part of the total health organisation, which aims to satisfy major objectives of the nursing services is to provide prevention of disease and promotion of health'.

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4.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the history of nursing
- Analyse the role and functions of a nurse
- Discuss the nursing care and ward management in detail

4.2 HISTORY OF NURSING

Florence Nightingale is regarded as an influential figure, who had been an epitome of strength behind the development of modern nursing. She was the one to come up with the idea of training women as nurses. The first school based on her model opened in 1870s. Earlier, nursing was only confined to hospitals, but in 1920s, it moved from hospitals to universities in terms of the education in the discipline of nursing. In India, the Indian Nursing Council (INC) is the governing and statutory body established by the act of parliament in 1947. As per the history, the first nursing school was found in 250 B.C. in India, which focused only on men being trained as nurses. According to the document, known as The Charka, the men were trained to perform all the tasks required to take care of the patient. The tasks include:

- Cooking food for the patient
- Bathing and washing the patient
- Rubbing and massaging the patient's limbs
- Assisting the patient in walking and moving
- Cleaning and making bed

Willingness to work and help with anything, when ordered 1861 was a period of major reforms in the field of nursing in India. These reforms were majorly backed by the work of Florence Nightingale and were confined to military and civilian hospitals.

A major step was initiated by St. Stephen's Hospital in Delhi, which was concerned with preparing Indian women as nurses. It was the first hospital to undertake education in nursing. This earmarked the development of nursing as a discipline in India. The very first school of nursing education for midwives was started in the Government Hospital, Madras, with only four students. During 18th and 19th century, many such schools were opened in different states. During this time, the nursing education was not governed by any organisation and there were no uniform educational standards. These standards were formulated by Unit Board of Examiners in 1907-1910. In Middle India, it was formulated in 1926, while in

Southern India in 1913. Some other important milestones of nursing education are:

- a) **1918:** First school of health visitors (Delhi)
- b) **1946:** First four-year B.Sc. Nursing in Delhi & Vellore (RAK College & CMC College, respectively)
- c) **1951:** Two-year ANM Course in St. Mary's Hospital, Punjab
- d) **1980:** First Nursing Association (Bombay Presidency Nursing Association)

The INC act was passed in 1947, which was later amended in 1950 and 1957. After that various developments took place in the field of nursing, which include the revision of syllabus of General Nursing and Midwifery (GNM) 1951, 1965, and 1986, for ANM in 1974 and for B.Sc. in 1981.

4.3 FUNCTIONS OF A NURSE

A nurse is often considered merely as an assistant of a doctor. However, a nurse is sometimes more important than a doctor. Nurses provide a diverse number of services. Their role varies from a caretaker to a leader. The advanced healthcare system has only expanded their functions, which are:

- I. Caregiver:** The nurse acts as a caregiver for the patient and assist him/her physiologically and psychologically. While in the process of delivering care to the patient, it is important that the nurse should also protect the dignity of the patient.
- II. Communicator:** Communication is the crux of nursing. The nurse communicates with the client/patient, patient's attendants, health professionals, and also with people in the community. During the communication process, the nurse listens to the patient's problems, identifies and communicate them in writing or verbally to the other members of the healthcare team. An effective communication on the nurse's part is one of the important factors in nursing care.
- III. Teacher:** As a teacher, the nurse teaches the patients about various procedures that will be performed on the patient, during the course of their treatment. They assess the learning ability of the patient and act accordingly.
- IV. Client's advocate:** They advocate the patient and express their needs and wishes to other health professionals.
- V. Counsellor:** Nurses counsel the patients and provide intellectual, psychological, and emotional support to them. This enables patient to deal with stressful situations and express their psychological or social problems.

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VI. Leader: The nurse takes the charge and acts as a leader. She influences others to work together towards a goal. The role of a nurse as a leader can be employed at various levels like individual patient, family, group of patients, colleagues, or community.

Check Your Progress

1. When was the Indian Nursing Council (INC) established in India?
2. What was the milestone in the nursing education in 1946?

4.4 NURSING CARE: MEANING

Nurses provide care to the sick, diseased, and wounded patients in the healthcare centre, where they join hands with other people of the health care team to restore and maintain health of the patients. Nurses also work and provide care to the patients in their homes, which is almost same as the care given to the patients in a hospital setting. In certain communities, where the clinics and healthcare centres are not fully equipped with many doctors, nurses also identify and care for common problems, recommend and give medicines, and can even conduct minor surgeries.

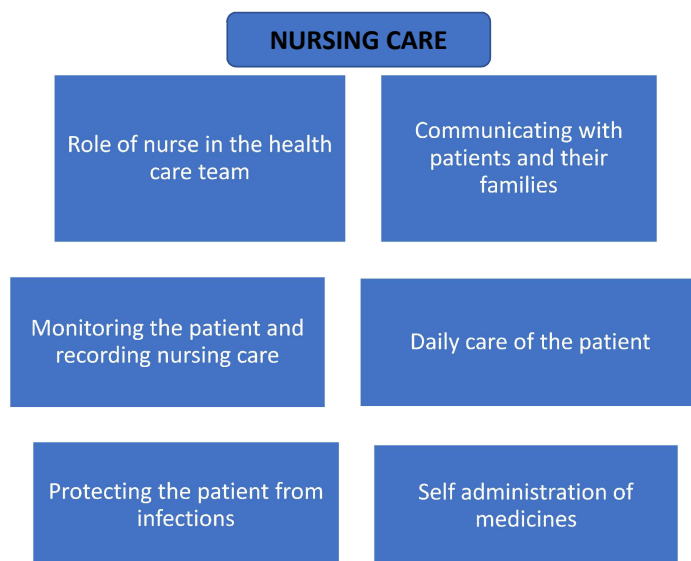


Fig. 4.1 Nursing Care

4.4.1 Role of a Nurse in the Healthcare Team

As discussed in the above section, in the healthcare team, the nurse performs many functions while providing care to the patient. Their role varies according to the situation and the requirement. They need to be well-versed in many areas of the healthcare system. Now, we will discuss these functions in detail.

Caring for the patient

A nurse takes care of the client 24 hours a day, helping them to do everything they would do for their own selves. They make sure that the patients are breathing properly, getting enough fluids as well as optimum nourishment. They ensure that the patients get proper rest and sleep, and are comfortable. Nurses help the patients to excrete waste from their body and also helps the patient in avoiding stiffness in joints and bed sores, which might happen from being immobile for long. Nurses often take autonomous decisions pertaining to patient care. For example, turning the patient every two hours to relieve pressure sores. However, the nurse might do this in compliance with the doctor so as to ensure that the patient does not suffer from anything else. Hence, the nurse utilises the know-how of the underlying condition as well as her own nursing information in caring for the patient in the hospital.

Assisting to the doctors

Doctors examines the patient for any illness or injury and diagnoses the problems, after which, they decide on the treatment. Nowadays, nurses perform a major act of ascertaining patient's problems. In fact, in certain places, nurses themselves carry out the treatment of patients and admit them in hospitals, only referring the severe cases to distant medical centres. Most of the treatment is administered by the nurse to the patient like setting up intravenous line and administering fluids, giving injections, changing dressings, monitoring healing of the wound, etc. All these things are carried out independently by the nurse after due consideration of the doctor.

Coordinating the care of the patient

The nurse coordinates and collaborates with other members to carry out the care of the patient. They put thorough efforts with doctors, physiotherapists and other people involved in a patient care. They supervise nursing assistants and keep a close eye on housekeeping staff, who clean the patient's room. Nurses also encourage the family members of patients to take care of them.

Protecting the patient

Patients' illness and sickness make him/her more vulnerable to an infection and injury, hence, it is the nurse's job to ensure a safe and healthy environment for the patient. They take all sorts of precautions to prevent the spread of an infection. They make sure that the patient has an access to a clean and tidy room, and clean water. Needles and other material used in the procedure should be sterile and the soiled materials are disposed of safely. The nurse also protects the patient's dignity and protects them from embarrassment or shame. They also ensure patient's privacy.

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Educating the patient and family members

One of the major roles of a nurse is teaching in order to restore and promote health as well as prevent illness. A nurse teaches the patient to follow certain things to speed up the rejuvenation process. For instance, teaching the patients to cough and breathe properly to prevent any lung problems, teaching them to walk on crutches, to monitor their blood sugar levels. They also teach the patient's family about cleanliness and hygiene, exercise, sleep, rest, etc.

Advocating

Nurses support the patients during their most critical times. Starting from birth and injury to the illness and their death, nurses play the role of caregiving. A patient's most intimate details lies with the nurses. Nurses stay by the patient's bed almost 24 hours a day. They are available for the patient in all conditions like feeding, bathing, and supporting them while moving. Hence, they advocate the patient, speak on their behalf and intercede whenever necessary.

Communicating with patient and their families

Along with the care, a good, effective, and clear communication is important. With good communication skills, the nurse can help the patient to pass the stressful and frightening phase of hospital easily.

- a) **Listening to the patient:** It is very important for the nurse to keep in touch with the patient; greeting them and listening to their needs empathetically. The nurse should show interest in the patient through her expressions and tone.
- b) **Maintaining confidentiality:** It is important that the nurse should convey only relevant information about the patient to the other members of the healthcare team. For example, talking about a symptom that patient has disclosed, to the doctor. Any irrelevant information about the patient should not be passed around as a gossip.
- c) **Empathising:** Showing empathy is an important aspect of nursing. A nurse should feel what the patient is feeling in order to respond emotionally to their needs and distress.

Evaluating the patient and recording nursing care

Since, nurses are in close contact with the patient, therefore, the nurse is the best person to monitor patient's progress, spot his/her problems and also ensure the specific kind of care required to resolve the patient's problems. Some basic guidelines to use to check the well-being of the patient includes:

- a. **Getting background information:** It is important to check patient's chart before going into patient's room. The chart displays the procedures done and any new information added.

- b. Observing the patient:** It is imperative to observe the patient and see any unforeseen changes which might pose a problem and prove fatal. For example, observe that the patient is breathing normally and make sure patient's airways are clear.
- c. Talking with the patient:** The nurse should talk to the patient and ask about his/her well-being, pain, the medications, and other relevant information while respecting the confidentiality.
- d. Examining the patient:** Examining the patient from head to toe and paying special attention to the problems, due to which the patient is admitted.
- e. Check any equipment in use:** The equipment being used can be 'an oxygen system, a nasogastric tube, an indwelling catheter or an IV line'. Ensure that the equipment is working properly and there are no errors in it.
- f. Assess the patient's environment:** The nurse is responsible for a clean and safe environment for the patient. The overall cleanliness of the room, clean bedding, clean toilet area, and supply of fresh water should be ensured by the nurse.
- g. Taking vital signs of the patient:** Temperature, pulse, respiration, and blood pressure are the vital signs. Any sort of a change in these signs can indicate the underlying condition. Such changes should be reported promptly.
- h. Planning nursing care:** The planning of nursing care is dependent upon the patient's status. It includes:
 - Procedures to be performed as per the doctor's orders
 - Measures to provide comfort and recovery
- i. Recording:** Recording and noting down the information is an important part of nursing care, which cannot be neglected. The records give details of the care provided to the patient and the procedures performed.
- j. Evaluating the care given:** Evaluation of the care is conducted to check the effectiveness of the care given. For example, checking on the patient after administering pain medications.

Daily care

Daily care includes caring about the basic activities of the patient's daily life. During sickness, the patient cannot perform all the activities and is dependent on the nurse for everything. The nurse should help the patient and maintain his/her well-being by doing all the chores. A high level of empathy and practice is demanded from the nurse to perform all these activities. The nurse should ensure patient's hygiene via a proper bath, oral care, hair care, etc. While caring for the patient, a nurse can give back rubs to help in recovering from pressure sores, help the patient get enough rest and sleep, ensure his/her safety, and also teach the family member of the patient about various techniques of giving care.

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Protecting against any infection

There is a higher susceptibility of infection in hospitals and the already admitted patients are more prone to infections. The spread of infections can be controlled by various methods. The germs should be destroyed, whether on hands, equipment, or furniture like patient's beds. Effective ways for destroying germs are as follows:

- Antisepsis
- Decontamination
- Cleaning
- High level disinfection
- Sterilisation

Other methods are use 'of protective clothing and safe disposal of body wastes and infected articles such as dressings'. The practice of medical and surgical asepsis should be followed by nurse and other health care professionals. Isolation of the patient in case of tuberculosis and leprosy are also essential to save the patient from catching infections.

Safe administration of medicines

The nurse is responsible for preparing and giving drugs safely as prescribed by the doctor to the patient. If not given properly, these medicines may harm the patient and prove to be fatal. The nurse should keep in mind the following points before administering the medications to the patients:

- Safe dose of the drug to be administered
- Dose prescribed by the doctor
- Method of administration
- Action of the drug and expected effects
- Possible side effects
- Any allergies that the patient might have

Check Your Progress

3. Mention the points that should be kept in mind by the nurse before administering the medications to the patients.
4. How do the nurses coordinate the care of the patients?

4.5 WARD MANAGEMENT: MEANING

A hospital ward can be defined as a suite of rooms or a block, forming a division of hospitals, which is shared by patients who need similar kind of care.

Objectives of ward management

- To provide best quality nursing care to the patients.
- To ensure provision of sanitised, clean, and properly ventilated surroundings for the patient and their protection from infections, accidents, and potential hazards.
- To Help staff attain job satisfaction
- To provide facilities to patients and their attendants.

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4.5.1 Components of Ward Management



Fig. 4.2 Components of Ward Management

Patient care

This includes all the activities, which are necessary for providing nursing care. The main principles of patient care revolve around providing comfort to the patient and maintain patient's health. The patient care revolves around the following aspects:

- Admission and orientation of the patient
- Assessment of patient's needs and care
- Progressive patient care
- Assigning personnel to care for the patient
- Plan the time and work schedules
- Ward Rounds
- Records and reports
- Management of emergencies

- Client/patient education
- Appraisal of nursing staff

Personnel management

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Everything that concerns staff is included in personnel management. It involves:

- Orientation of the new personnel or staff
- Supervising the personnel and delegating authority to them
- Establishing inter-personal relationship
- Evaluating the staff/personnel
- Staff conferences
- Conducting staff development programme
- Ensuring health, safety, and welfare of the personnel

Personnel Management in a ward includes assignment of personnel for patient care. It also includes ward rounds. The ward sister is responsible for taking ward rounds. It is a tool of supervision, evaluation, and teaching. The different types of ward rounds include:

a. Doctor's Round: For doctor's round, the following preparation is required:

- Clean and tidy ward
- Ensure the ward is not overcrowded with patient's attendants
- Update patient's chart and include relevant information about the patient
- Prepare diagnostic tray for use
- Rounds should not be conducted during lunch timings or visiting hours

b. Matron's Round: A matron is considered as an adviser. Her work is to guide the nurses and ensure that they visit the patients regularly.

c. Ward Sister: A ward sister is required to take ward rounds with staff nurses. Her work is to observe critically ill patients and the ones who require special attention. She is also required to note the available stocks and equipment.

Environment cleanliness

This includes sanitation and provision of therapeutic environment. It is the patient's right to have access to a safe and sound environment, which should be provided by the hospital administration and staff. A safe environment includes:

- Regulated temperature
- Light

- Removal of unpleasant odour
- Controlling dust
- Safely disposing off excreta
- Ensuring safe water supply

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Supply and equipment

Management of supplies include checking the condition of equipment, managing shortages, checking the supplies for effectiveness, and ensuring the safety of expiry date as well as safe handling of the supplies. Basically, all the decisions relating to the supplies and services comes under the domain of supply management. The various actions to be taken for ward management should be at par with the supply. Supply implies services of the personnel or staff as well as the services in terms of equipment, medicines, and other fringe commodities which might be useful for patient care, either directly or indirectly. This demands various actions to be taken, which are demonstrated in the Figure 4.3.

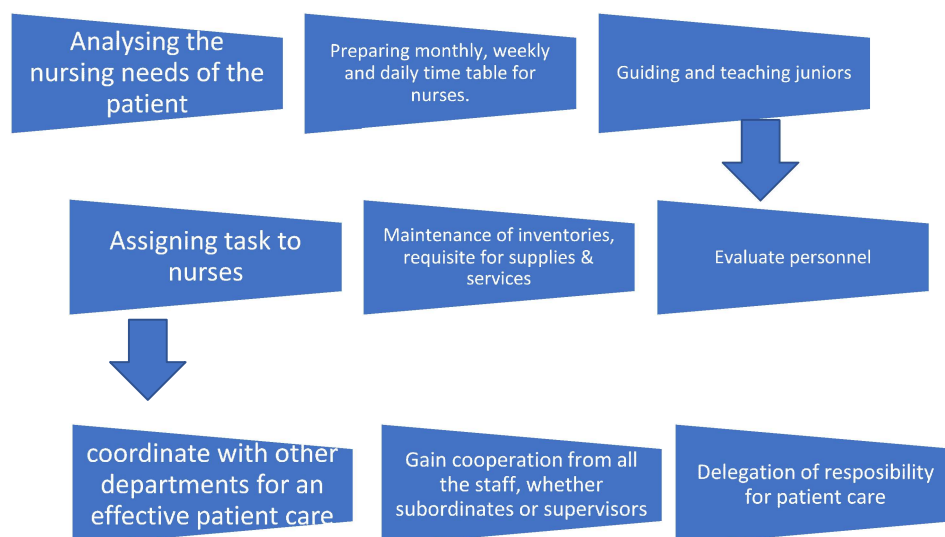


Fig. 4.3 Supply Management

Check Your Progress

5. List any two objectives of the ward management.
6. List any three aspects of patient care.

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4.6 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The Indian Nursing Council (INC) was established by the act of parliament in 1947, in India.
2. The milestone in the nursing education in 1946 was the first four-year B.Sc. Nursing in Delhi & Vellore (RAK College & CMC College, respectively).
3. The points that should be kept in mind by the nurse before administering the medications to the patients are:
 - a) Safe dose of the drug to be administered
 - b) Dose prescribed by the doctor
 - c) Method of administration
 - d) Action of the drug and expected effects
 - e) Possible side effects
 - f) Any allergies that the patient might have
4. The nurse coordinates and collaborates with other members to carry out the care of the patient. They put thorough efforts with doctors, physiotherapists and other people involved in a patient care. They supervise nursing assistants and keep a close eye on housekeeping staff, who clean the patient's room. Nurses also encourage the family members of patients to take care of them.
5. The objectives of the ward management are:
 - a) Helping staff attain job satisfaction
 - b) Providing facilities to patients and their attendants.
6. The aspects of patient care are:
 - a) Assessment of patient's needs and care
 - b) Progressive patient care
 - c) Assigning personnel to care for the patient

4.7 SUMMARY

- Florence Nightingale is regarded as an influential figure, who had been an epitome of strength behind the development of modern nursing. She was the one to come up with the idea of training women as nurses.

- A major step was initiated by St. Stephen's Hospital in Delhi, which was concerned with preparing Indian women as nurses. It was the first hospital to undertake education in nursing. This earmarked the development of nursing as a discipline in India.
- Nurses provide a diverse number of services. Their role varies from a caretaker to a leader. The advanced healthcare system has only expanded their functions.
- Nurses provide care to the sick, diseased, and wounded patients in the healthcare centre, where they join hands with other people of the health care team to restore and maintain health of the patients.
- A hospital ward can be defined as a suite of rooms or a block, forming a division of hospitals, which is shared by patients who need similar kind of care.

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4.8 KEY WORDS

- **Midwifery:** The health science and profession that deals with pregnancy, childbirth, and the postpartum period, along with the sexual and reproductive health of women throughout their lives
- **Counsellor:** A person who works with individuals and groups to promote optimum mental and emotional health
- **Antisepsis:** Antimicrobial substances that are applied to living tissue/skin to reduce the possibility of infection, sepsis, or putrefaction
- **Sterilisation:** Any process that eliminates, removes, kills, or deactivates all forms of life and other biological agents

4.9 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. What were the tasks performed by male nurses according to 'The Charka'?
2. How can a patient be protected against infection?
3. Write a short note on supply management.

Long Answer Questions

1. Describe the role and functions of a nurse.
2. Discuss the basic guidelines for nurses to check the well-being of the patient.
3. Evaluate the component of personnel management in ward management.

4.10 FURTHER READINGS

NOTES

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BLOCK - II

LABORATORIES AND RULES

*Importance of
Laboratories*

UNIT 5 IMPORTANCE OF LABORATORIES

NOTES

Structure

- 5.0 Introduction
- 5.1 Objectives
- 5.2 Clinical Laboratories
- 5.3 Duties and Responsibilities of Laboratory
- 5.4 Organisational Hierarchy of a Laboratory
 - 5.4.1 Measures
- 5.5 Laboratory Procedures
 - 5.5.1 Workflow
 - 5.5.2 Information Flow
- 5.6 Obstacles
- 5.7 Quality Control and Assessment in a Laboratory
- 5.8 Ethics and Laws of Laboratory Operations
 - 5.8.1 Bio-Safety Programme
- 5.9 Documentation and Records in a Clinical Laboratory
 - 5.9.1 Requisition Record
 - 5.9.2 Accession Record
 - 5.9.3 Laboratory Record Book
 - 5.9.4 Retention of Records, Reports and Specimens
- 5.10 Answers to Check Your Progress Questions
- 5.11 Summary
- 5.12 Key Words
- 5.13 Self Assessment Questions and Exercises
- 5.14 Further Readings

5.0 INTRODUCTION

A laboratory is an installation, which provides a controlled atmosphere in which a scientific or technological researches can be carried out, by performing certain experiments or measurements. There are many different kinds of laboratories, each to necessitate different approaches. The laboratory is a major auxiliary unit of the hospital. It requires highly trained employees and can be a major turnover area. The prime function of the laboratory is to execute laboratory test for each of the patient. Besides, the laboratory can also coordinate the procurement of the test from the laboratory located outside the hospital.

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5.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the importance and responsibilities of the laboratories
- Analyse the procedures in the laboratories
- Discuss the documentation and record system in the laboratories

5.2 CLINICAL LABORATORIES

Out of many laboratories, one of the most important is clinical laboratory. Clinical or Medical laboratory is of great importance in a healthcare delivery system. Clinical laboratory tests are an essential part of most of the diagnostic procedures. Laboratory testing allows the healthcare provider to obtain information, which is unattainable through the patients' history or physical examination. Physicians or specialists may request for laboratory tests for one or the more reasons, which are as follow:

- I. To designate a diagnosis and prognosis:** Laboratory testing can be used to diagnose the patients, according to their signs and symptoms. It can also be used to assure a clinical diagnosis like diabetes or myocardial infarction. These details can later be used to forecast the course of the disease or disorder. While diagnosis refers to the spotting and determination of the disease affecting the patient, the prognosis refers to supposition regarding the personal outcomes for the patient, including but not only limited to existence, life expectancy, quality, functional capacity, and many more.
- II. To develop a treatment strategy:** A resolute diagnosis aids the physicians or specialists to ordain an operative treatment strategy for the patient. Conditioned on the results of the laboratory testing, the specialists or physicians may insist upon behavioural modifications, prescribe a suitable medication, and so on.
- III. For preventing or early detection of the disease:** Screening tests can aid in early detection of a disease or disorder, which is usually symptomless in the initial stages. Regular screening may detect or probe a disease in its early stage, allowing or granting timely initiation of the treatment to avoid the complications and maintain the patient's health.
- IV. For on-going assessment and follow-up of a treatment:** The efficacy of the treatment must be closely watched. In many cases, signs and symptoms may not completely show the proper response to the therapy. In foresaid cases, laboratory testing may furnish the further assessment of the status of the prevalent disease or disorder, or it may aid in assessing the complexity of the therapy.

5.3 DUTIES AND RESPONSIBILITIES OF LABORATORY

The major duties and responsibilities of a medical laboratory is to perform laboratory examinations in six major areas:

- Bacteriology
- Histology
- Biochemistry
- Haematology
- Serology
- Cytology

Laboratory tests aid medical personnel in the formation of a firm conclusion and then the treatment and cure of the disease or disorder. The modern healthcare is getting more and more relied on a laboratory test. Certain laboratory examination are used to carry out the specialised tests. Besides the general examinations, the obstetric women may need the test for Rh factor and blood as well. For the patients, who have undergone a surgery, the test of the removed tissue (during the time of the surgery) is also needed. The laboratory also plays a very crucial role in hospital's surveillance program and infection control program. With increasing pace of technology, the modern medical care system can be said to have taken a step in the era of laboratory medicine.

A couple of years back, laboratory examinations were done manually, which could only cover the basic tests required for patients' evaluation. However, now many of the laboratories are specialised, with refined automated machines, for example, certain electronic devices and automated analyser, which have increased the rate of the productivity. Tests are conducted in a matter of minutes with a guarantee of highest degree of reproducibility and accuracy. The distribution of the tasks of the laboratory and the amount of the tests to be performed have doubled with the time.

Nowadays, in large hospitals, a major department or unit of the laboratory is specialised to such an extent that they only select the skilled staff to carry out each and every test and examination regarding their respective discipline or field within that unit. The total number of staff has also increased. Nevertheless, in smaller hospitals, a few sections are still combined and the employees are cross-trained to carry out the procedure of more than one section. Some of them can be trained to work in haematology, biochemistry, serology, blood bank, urinalysis, microbiology, and histology. It is futile to look for the smaller hospitals, which prefer to form and employee some units such as cytology and histology, but they pay get the results from the larger establishments.

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Large laboratories provide non-stop 24x7 service in smaller hospitals and is usually on the call service during night shifts. With the separate departments, the laboratory forms a composite section. Its intricate style of work can be better comprehended by the functions which are performed by its major units:

- The haematology unit analyses and examines the blood samples and provides with results, which help in treating or curing the disease.
- The blood bank is the inseparable part of the haematology department of a laboratory. It preserves and stores the blood after obtaining it either from other blood banks or directly from the donor. The blood bank is also a major participant in the blood donor programmes, cross-matches typing, and processing and so on.
- The biochemistry unit carries out all the qualitative and quantitative analyses of the different body fluids and substances present in the tissues and certain secretions.
- An additional clinical pathology unit carries out a large number of researches, which include urine analysis, semen analysis, and so on.
- The histopathology unit of the laboratory prepares and performs tests for different tissues of the body, in order to give a correct data on the course and progress of a certain disease or disorder. It deals with all the microscopic determinants of tissue pathology.
- The bacteriology unit identifies the microorganisms present in the body fluid, surgical samples, and scraping of the skin. It deals with the correct identification of the microbe and then performing sensitivity tests on the microbes for different antibiotics.
- The cytology unit deals with the examination of the body cell for identification of malignancy and certain other diseased conditions, hormonal effects, etc.
- The clinical laboratory also performs certain other functions, for instance, research programmes and teaching. Many hospitals are also running schools for medical technicians.
- Autopsy is also one of the prime responsibilities of the pathology unit of the laboratory.

Check Your Progress

1. In which areas does a laboratory perform examinations?
2. Why is the Clinical or Medical laboratory important?

5.4 ORGANISATIONAL HIERARCHY OF A LABORATORY

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A clinical laboratory must have a clear formulation of organisational structure in order to execute its job efficiently. The laboratory can either be attached to a medical institution or can operate independently. Each hospital must have a clinical laboratory to carry out emergencies as well as assist surgical decisions. The hospital categorise the clinical laboratory as a part of its unique diagnostic service. The pathologists, as the conductor of the laboratory, have a medical degree with a certain specialised training. The bench workers are college graduates, a technologist has a higher degree, it can be a master's degree or a Ph.D. and he/she should have a higher level of laboratory skills so that he can also perform a supervisory role.

In all the situations, a pathologist inspects the laboratory and hence is answerable for all the operations of its different departments. The departments, however, may function freely under the supervisor, who monitors everything. Still, the pathologists remain responsible for taking major decisions. The informants, through experiences with the hospitals, have found that the pathologists primarily use their time in the microscopic tests of different body tissues and hence, the observations of the pathologist are crucial. Other laboratory employees follow the conformance methods and uphold the precise records.

Technologists at least have a degree of baccalaureate if he/she did not have a degree of master's or a Ph.D. In addition to this, he/she must have completed a minimum of three years of technical schooling in some university. Every technologist is responsible for entire activities that take place within his unit of the laboratory, which include the distribution of every task, decision making on a daily basis, and problem solving. They handle all the external operations and aid the pathologist, when required. They also do all the paperwork, look after the order supplies, purchase, and maintenance of all the types of the equipment, compliances, keeping in mind all the safety rules. They get higher salary than a technician. As a supervisor, they can also give official approvals on the test reports gathered by the technicians. The pathologists can decide to communicate the results with the technologists prior to signing of the reports and forwarding it to the physicians. This interlocution has been aided by the use of computers in the contemporary laboratories. The final reports are then, signed by the pathologists. Only the pathologists are allowed to further pass the reports to the clinical physicians. Only they can disclose the confidential laboratory researches, regarding the patients to the physicians.

The core or the essence of the laboratory is the laboratory technician. Although the pathologist, a medical scholar is accountable for all the operations of the laboratory, he/she rarely engages with the bench work. The technicians who work in background, prepare the crucial data, which is then forwarded by the laboratory for further identification. The experts claim that the potential of the laboratory medicine in patients' care is mainly conditioned on the quality of the education and

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ethics of the technicians. The technicians, as the bench workers, are workhorses or draft-horses of the laboratory. They have a degree of baccalaureate science or a higher secondary degree and have typically received one or two years of laboratory training from an authorised technical school. A few technicians are ‘on the job trained’ (OJT). While, the rest are the high school or plus-two pass outs in the technical training with respect to the laboratory. The technicians perform examinations, record reports in a record book, and forward these outcomes, along with his dated signature to his supervisor. Laboratories tries to maintain low expenses and increase the efficiency of the worker. Therefore, a technician must be aware of more than one field to execute the laboratory activities.

It should be kept in mind that the salary of employees takes up the highest cut from the expenses of any laboratory. Other than technicians, the workforce of the laboratory involves certain non-technical personnel, who gets involved in less skilled technical operation, for instance, receiving the specimen, processing the sample, autoclaving, disinfecting, sterilising, distillation, preparation of swabs, and many more. Laboratories of the countries which are on the path of expanding their economy, seldom enjoy the ideal conditions or situations because, most of their workers are OJT (on the job trained). In laboratory located in remote area or the outskirts, the pathologists may not stay present at all times, hence the laboratory employs two employees; one with a high school diploma, skilled in general laboratory operations, and the second one with minimal education, who is able to perform all the non- technical errands.

In order to be efficient and accurate, a laboratory is categorised into certain units. These unusual divisions are made for the ease and therefore, can be differentiated based on the laboratory work and the number of the workers. The laboratories usually make strenuous efforts to maintain the lower costs and the higher worker efficiency. Hence, employee’s salaries are crucial for any laboratory. In short, the administrator of the medical laboratories must keep in mind that outcome of any operation and effectiveness of a laboratory are dependent on modern equipment, skilled staff, a well–designated physical atmosphere, and a good administration team.

5.4.1 Measures

There are a number of counterfeit training centres all over the country. Superintendents must look at all the documents of the applicants for the job cautiously. Applicants, who are trained under the CMAI programme or in any recognised medical institution should only be employed. It is observed that, in most of the places the laboratory activities are usually carried out by unskilled workers, who are improperly trained or have a little knowledge about the laboratory activities. This negligent behaviour not only affects the accuracy of the staff members in preparing the final results but also lead to risky outcomes in patients’ proper care.

Requests for the examination of the sample arrives at the laboratory quite arbitrarily with deadlines for the results. There are multiple requests requiring different examinations, needing processing at various workstations. Thus, a laboratory should be managed properly to work efficiently.

In order to maintain the system of smooth workflow, cardinal aspects must be kept in mind. One of which is concerns collecting and forwarding of the samples. The outpatients usually go to the laboratory and submit the sample themselves. But in case of the inpatients, technicians must go themselves, collect the specimen, and take them to the laboratory. The second concern are the tests, which should be carried out according to the pre-decided procedures, which are based on various factors. There are two prime factors, which determine the laboratory's level of merit, fidelity, work pressure, and efficiency, the extent of automation in laboratory and the extent of the computerisation of the laboratory and its several activities.

The complexity of operating automated machines differs from hospital to hospital. Haematology and biochemistry units of the laboratory are the two greatly automated sections. A number of hospitals now keep the auto-analysers in chemistry labs, the equipment, which perform low-profile tests, have a pre-established set of dozen or more evaluations from a patient's single sample. Some complex apparatus can even perform multiple tests in several combinations. The laboratory procedures can be manually, partially or fully computerised. In the primary system, everything is carried out manually-orders are received in writing and are forwarded to the laboratory via messengers from various spots of the hospital. These are then noted in the maintained records and passed further to the sections dealing with it or sent to be filed in the records of the patients.

A semi-computerised system requires a minicomputer system, with information terminals placed in the laboratory and other departments of the hospital. They can be required for a large number of applications for instance, order entry, reporting, charging, and so on. The system replaces the paperwork and keeps a check on all the missing charges. Through this system, a lot of time is saved and used in more urgent matter. It goes without saying that the computer systems have largely improved the quality of the processing in the laboratory.

The fully computerised system leads to smooth workflow, reports information, gives work assignment, and permits quick on-line access to information. The advantages include higher yield at lower scale unit cost, except the cost of computerization and availability of comprehensive reports, statistic, and data. The disadvantages of a fully computerised system is that it may have high expenses than the total production. In other words, its costs may exceed its benefits. Hospitals should take into consideration the computerised laboratory for higher accuracy, less precision, high efficiency, excellent pace, and convenience.

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Check Your Progress

3. What are the activities which only a pathologist can do?
4. What are the educational qualifications required of pathologists, technicians, and technologists?
5. What negligent behaviour affects the accuracy of the staff members as well as the final results?

5.5 LABORATORY PROCEDURES

There is a common procedure followed in every laboratory despite some differences:

- Request of the test must be present in writing.
- The specialist or the physicians, nurse, or the laboratory technician are responsible for receiving the samples.
- The time schedule of the institution for receiving several types of the specimens will also affect the operations carried out in the laboratory.
- Emergency requests must be entertained during certain time period, and must be given priority over all other requests. The hospital must, nevertheless check that this facility must not be misused in any situation. Misuse of the laboratory's utilities occur only when specialist or the physicians label tests as urgent even when they are not. More often than not, the tests are ordered on urgent basis when they could have been placed previous day. There is a chance of laboratory staff losing urgency in actual emergency cases.
- Samples must be sent to the laboratory within a specific time period in a fully organized manner with labelled container.
- It is necessary for the nurses to be familiar with the proper time for collecting sample (for instance, the conditions of fasting or non-fasting) and the proper amount of the necessary sample should be collected in the proper container and labelled accordingly. A list of all prescribed tests with all the required details should be sent off and be readily available at a nurse's station.
- All the order information should be maintained in well-organised request forms, with uniformity maintained throughout.
- Various colours can be used for distinctive types of the forms for easy identification. It should also be easy for the doctors and nurses to locate it.
- In case of the inpatients area, the caretaker or nurse is accountable for preparing all the formal papers according to the requirement of the request. If the laboratory technicians collect the samples, the nurse will pass on the form to him.

- No samples or orders should be left in the laboratory casually. It should be kept under the custody of the amenable laboratory staff, who could maintain it and enter it in the record register. A numbering system must be followed to determine the different samples.
- Final reports of the investigation are entered in both copies of the request form. The true copy is then sent back to the nursing section which is to be attached with the patient's records. The final results are also maintained in the daily record register.
- The laboratory results must be passed on to the authorised or authenticated medical specialist or can be sent to be attached to the patients' reports. The reports should never be passed directly to the patients, except when requested by the doctor.

Nowadays many of the foresaid steps are completed on-line in the fully computerised hospitals. Certain basics principles for an authenticated laboratory are:

- Licensing
- List of the apparatus required and the supplies needed for the blood bank
- Refreshment services
- Reagents
- Expiry date
- Trained personnel
- Laboratory equipment
- Records, labels and labelling
- Basic supplies
- Testing of the whole human body

5.5.1 Workflow

The management of the workflow is demonstrated by two major functions. One, the management of the sample and its transportation to the laboratory for analysis following the established procedure. Second, the laboratory investigations are carried out according to the procedures which are predetermined based on the type and number of the requested tests, the time they are requested, and the accessibility of the automated apparatus. These factors define the method by which the investigation will be done and the skills, with which it will be performed.

The prime factor, which determines the work load rendition, is the extent of the mechanisation in the cycle of yield. The type and complexity of mechanised equipment in the laboratory differs from hospital to hospital. The most mechanised departments of testing are haematology and chemistry. At the very least, small hospitals have chemistry analysers which fit on the bench top and deals with complex

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procedures. Many laboratories have the apparatus which is able to perform low-cost profile testing, which implies that it performs a pre-organised set of tests, only from a single sample collected from an individual (patient). Many complex equipment can occupy the entire allotted room and perform a number of tests in various combinations. The laboratory can have its own minicomputer, which connects to the hospital's master computer system for dealing with the billing and the demographics data and the report of the final result. Laboratory staff usually favour computerised work over manual one as it provides with more detailed data than the manual one, especially during heavy workload.

5.5.2 Information Flow

Information flow have two prime levels. The first level includes a process in which the investigation is requested. The request must imply if the final result is supposed to be reported to the nursing work stations for filling out the patient's chart or to the requesting doctor. The second level of information flow is the entry of the result and final report of the investigation. The medical laboratory procedures can be performed manually or mechanised partially or fully. Whether manual or automated, the statistics of the reporting process are used in most of the laboratories to supervise the work load and efficiency, which is defined by the regulatory body.

Requests for the procedures are collected in writing or received over telephone from the nursing work station, doctor's office, and the emergency unit. Outpatients can also carry orders and samples to the laboratory, when conditioned to do so by their heads. The orders for the investigation are noted in the logs and the tasks requested are further forwarded to the technologists. Once the tests have been performed successfully, the final reports transferred to the doctor or to the nursing work station. The daily, weekly, and monthly statistical organized reports are also made from this data and the worksheets.

A semi-computerised system usually requires minicomputers, with a number of information terminals present in the laboratory and also throughout the hospital. The system is needed for feeding the request, reporting, and replacing most of the paperwork. It also contains all the charges at the point of time when the request is noted at the terminal, which could aid to eradicate lost charges. The advantages of a semi-automated laboratory system is elucidated below:

- Saves the clerical time
- Timely response to the emergencies requests
- Request requisition can be received directly in the laboratory
- Labelling of the sample containers can also be generated by the computer, thereby, making it easier to identify the samples.
- Routine statistical reports can also be produced automatically.

The prime limitation of such a semi-computerised system is basically that although it is able to carry out all the clerical activities with greater efficiency, it still

cannot be used for performing the most basic of laboratory errands such as on-line collection of final reports.

*Importance of
Laboratories*

Fully computerised medical laboratories are devoted systems, which are accessible as stand-alone minicomputers or as the computer systems, which are mutually connected with other laboratories. The computer carries out most of the possible clerical tasks that would otherwise be carried out by the technicians and other supporting staff members. This system is responsible for managing the laboratory's work flow, providing work assignments, reporting information, and allowing immediate access to the information. It offers the following advantages:

- Higher production at a lower unit expense (except computer cost).
- Quick response time.
- Accessibility to a comprehensive information and the reports.
- Enhanced level of quality control in the laboratory.

Other advantages of the system include its adaptability in generating the computer printouts in place of manually organised log books, illustrated information, on-line acquisition of the information from automated equipment, and label printing. However, the system's adaptability to use customised software is one of the major drawbacks in that the expenses of any comprehensive system can far exceed its advantages. This is one factor, which leads the hospitals to develop a shared laboratory.

Check Your Progress

6. Mention any five basics principles of a recognised laboratory.
7. What is the prime factor, which determines the work load rendition?
8. What are the levels of information flow in the laboratory?

5.6 OBSTACLES

When pathologists practise with a team and do have contacts with hospitals, they can be given a fixed pay for the services which they provide, a percentage of the laboratories investment, or a combined form of both of them. When pathologists are serving the hospital directly, they could be given the salary in a conventional sense or a percentage of the departmental revenue. Therefore, pathologists believe that their restitution greatly depends upon the the quantity of the laboratory work, on which they have no control.

On account of the hospital's expenses, the requirement for meeting 24*7 hours service, providing the much needed care of severely ill patients, and carrying out the laboratory procedures, generally costs much more than in an independent commercial laboratory. A few hospitals have attempted to become financially capable

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by establishing a cost centre, which operates as a commercial unit laboratory or by setting up a disunited corporation to meet all the requirements of a functional laboratory. Due to this, when the physicians recommend their outpatients to these commercial laboratories for examinations, which are not connected with the hospitals, both, the pathologists and the hospitals, find their work load getting decreased.

Suitable test utility is still rises up as the prevalent problem. Batteries of the examinations can be economical but they cannot become popular among the physicians. Some of the doctors or specialists believe that these test batteries can result in futile investigations and they can encroach on the physician's medical opinion with regards to the suitability of the tests for their patients.

5.7 QUALITY CONTROL AND ASSESSMENT IN A LABORATORY

A laboratory record must be authentic or else it will mislead the physician and threaten the life of the patient. This requires not only the proper Quality Control (QC) of the laboratory results, but should also involve sample handling and delineation. Therefore, well known laboratories run Quality Assurance (QA) programs to ensure the accuracy of the outcomes to the physicians. A while ago, the term Quality Assurance (QA) got replaced by a new term, called Quality System (QS), which is chosen to describe all the policies, processes, procedures, which are required to attain the high quality testing. The greatest purpose of the quality program is to stipulate excellence in the healthcare that is safe, quick, patient-centred, impartial, and operative. Quality programmes aid in attaining these objectives by providing guarantee performances, excellence, and securing laboratory test reports.

The final conclusions of the laboratory in many developing countries depend greatly on the productivity of the laboratory technicians than in developed countries, where mechanisation has completely removed the chances of human error. According to P.G. Hill and A.S. Kanagasabapathy, 'one half of all the biochemical results from 60 laboratories (of India) were incorrect by 15% or more and many results were clinically misleading'. Therefore, it became mandatory to adopt a quality control program in the laboratory of the countries which are on the verge of development. There were certain recommendations, which were prepared by the founders to ameliorate the conditions:

- Better training of technicians to provide ample opportunities for the possible improvement of the current data.
- Ensure the established processing, which can also be carried out in local situations, noting the record in such a language, which a technician can comprehend and duplicate it, without any error.

- Rules by the government and professional organisations, ordaining a laboratory in order to make it authentic so that the results are reliable for the proper diagnosis.

*Importance of
Laboratories*

5.8 ETHICS AND LAWS OF LABORATORY OPERATIONS

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Although, legal and moral principles for a laboratory are somewhat mellow in developing countries, they cannot be overlooked when it comes to patient's right to information, agreement forms, briefing, and imparting proper information related to the chain of tutelage requirements and the necessity for describing threats as well as the need of the tests, before it is conducted. Respect towards the reality of an individual shows basic ethical cogitation. The patient and his family have a right to accord, inquire, appeal for other perspective, and turn down the diagnostic investigations. Similarly, caretakers have equal rights to be informed about the diagnosis of the patient in order to be informed about the risks to their selves and act accordingly.

One has to keep in mind that improved and impregnable laws do not make a good laboratory technologist. Just like other jobs, this one also has its level of professionalism built on certain moral duties towards fellow human beings. Such rules and regulation for conducting, personal and professional duties are termed as ethics. The most crucial consideration of the professional morals is termed as honesty. The information given by a technician has to be reliable. The tragic flaw lies in the fact that a large number of the technicians are not aware of the fact that 'they do not know all the things'. Therefore, properly organised quality control measures are very important for the improved procedures. The details of the patients must not be revealed to anyone, the working stations should be kept clean, equipment or apparatus must be checked properly from time to time, and any mechanical failure must be immediately reported to the supervisor.

5.8.1 Bio-Safety Programme

Some of the bio-safety programmes necessary for a laboratory are:

- I. Responsibility for the safety:** Both the employee and the supervisor are accountable for the safety of the clinical laboratory. The supervisor also has ethical and legal duties for looking after the safety of the working environment. The staff must be aware of their role in their own safety and also the safety of their fellow colleagues. There must be a mandatory laboratory safety programme, with clearly stated obligations of the management and the staff.
- II. Responsibilities of the organisation:**
 - Developing a safety programme in order to inform about the laboratory work methods and safety policies.

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- Establishing a safe zone for working staff, including the supplies of proper safety apparatus and the personal protective equipment, wherever required.
- Establishing on the job training (OJT) for the workers so that they can learn the proper safety practices and the provision of guidance and supervision to the laboratory staff, whenever required.
- Developing and executing an effective infection control programme.

III. Responsibilities of the Employee:

- Proper knowledge and awareness, compliance with the laboratory work practices.
- Conducting safe work practices and utilising personal equipment provided for safety.
- Prompt notification of hazardous condition to the immediate supervisor available on duty.
- Positive behaviour towards learning, training, provided facilities, co-worker, and the heads.

Check Your Progress

9. How can hospitals become financially capable?
10. Why do laboratories run Quality Assurance (QA) programs?
11. Mention any two responsibilities of an organisation for the bio-safety of a laboratory.

5.9 DOCUMENTATION AND RECORDS IN A CLINICAL LABORATORY

The significance of maintaining laboratory records can never be overstated. The properly or correctly maintained records assist in finding the patient's history, upholding correct or right stats, tracking previous errors in order to improve them, and preventing each and every kind of confusion; thus, providing total clarity to the laboratory staff, the technician, and the physician.

5.9.1 Requisition Record

A physician conducting a physical test of the patient prepares an investigation request. This request form carries all the necessary information of the patient like patient's ID and the name of the physician and the doctor who referred the patient for the tests. A table of content of all the diagnostic examinations rendered by the laboratory is already enlisted in the requisition form for the physician to see. A correctly filled requisition slip is sent to the ward for compiling the apt samples or

the sample is collected in the laboratory itself. The second method is usually performed, for outdoor patients.

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Laboratories*

The aforesaid requisition slip can also be used for recounting the final results. The doctors are connected with the laboratory staff, with the help of the requisition slip only. Doctors use the slip to give the interim or tentative diagnosis grounded on the basis of physical tests and medical history of the patient. This will aid the technician and the laboratory staff to be informed of the doctor's suppositions, which in turn, will help them decide, a definite course of examination.

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5.9.2 Accession Record

Led by the doctor, the patient reaches the laboratory with his requisition form for providing his sample, which will be required by the laboratory to conduct the test. A number of examinations require patients to be prepared, whose information is written in the requisition form. A patient must know his/her part in the preparation and should act accordingly, so that the test can be carried out successfully. This knowledge of the patient is one of the important parts in getting accurate results. The patient must know that an incorrect or improper sample will always give an incorrect outcome. The laboratory also receives samples, collected by external agencies or units. All of them should have their unique identity and a requisition form.

The registration desk notes the date, name of the patient and the referring doctor, and the tests which are to be performed. Only samples of urine and blood are collected at the site of the collection, in the laboratory whereas, other samples are collected at home or the wards itself and transferred to the laboratory for carrying out the requested test. Every sample is given a unique acquisition number allocated by the registration desk, if it is not provided by the hospital. All the laboratory records should have the specific number provided by the hospital, even in the cases when laboratory uses its own code system. The phlebotomist scrutinises the test to be performed and collect the samples in suitable receptacle. In most of the cases the registration desk provides the printed labels. In case the laboratory receives a sample from any external agency, then the sample should have the identity number from that source laboratory, with the information of the patient such as the name of the patient and the requisition form from the doctor. The laboratory staff is now responsible for preparing a note of the sample identity, given by that external source and then giving their own unique code for internal affairs. The requisition form contains the laboratory acquisition number for further references.

The log book of the registration desk contains the acquisition numbers that are given to each sample before collecting it. Patient's ID (including: name, date of birth, and sex) accompanies the acquisition number. Other records of the laboratory log book involves:

- Type of the patient
- Name of the referring doctor
- Date and time of receiving the sample

*Self-Instructional
Material*

- Type of sample received
- Type of the test demanded

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The laboratory clerk, who receives the information can print out a few stickers that has to be attached with each sample, which is then transferred to the phlebotomist. All the samples, that require immediate attention, termed as STAT, are noted and tested on priority basis. The samples, along with the requisition form, is divided among the different units of the laboratory for proper completion of the investigation. It can take a day or a few days depending on the type of the test to be completed. After successful completion of the test, the conclusion is drawn and noted in the technician's record book with proper date and time of the results and then transferred to the supervisor for further proceedings. In case of any additional data about the sample or the patient, the receiving clerk needs to record that as well in the laboratory log book under the name of 'comments'. This will help to point out any inconsistency in the data.

5.9.3 Laboratory Record Book

The main laboratory register gives an acquisition number to the request slip, which is consistently recorded in a log book. The log book upholds the record of urgent sample separately. The log book should have patient's ID, patient type, referring doctor, date, time and type of the sample collected, and the type of the test demanded. Samples with additional requisition slip are divided into different departments. Diagnostic investigations are performed in different units, which maintain their own records separately. But all of these units use the same acquisition number for the given sample. A large number of variations can be seen with regards to maintaining the records.

Technicians' record book

This is a record register maintained solely by the technician. It maintains all the basic information, calculation, and final reports, accompanied by acquisition number given by the laboratory. The technician's diary must contain the results of the quality control (QC) and any other additional maintenance reports. When the technician submits the final results to the central desk via requisition form, then those final reports are entered in the main central laboratory register. The receiving clerk further transfers the request bill with the attached report to the pathologist for his approval and signature. This report is then sent to the doctor who initially, demanded the tests. In case the patient requires a copy, he/she can easily ask for it, from the physician. The laboratory never sends the results to the patients directly, unless the case is crucial.

Records of laboratory procedures

The laboratory maintains a hard copy of all the procedures carried out in the laboratory. Whenever, any new procedures are introduced in the laboratory, it

has to get approved by both, the supervisor and the pathologist. The laboratory records also provide with the working plan and the responsibilities of each and every worker. This assists the new workers in understanding the proceedings of the laboratory, thoroughly. The records are also responsible for providing the correct rules for dealing with a sample. For instance, if a certain sample is not analysed within two to three hours of receiving, then it may become unsuitable for conducting the tests. Therefore, it is necessary to preserve the sample correctly within the specified time limit. As a set policy, the laboratory should perform investigation, only after receiving the written or electronic order from an authorised body. Moreover, the test request or clinical chart-sheet having the request must be valid for at least two years.

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Safety records

The medical laboratory is one of those places, where the risk of catching infections and the chances of laboratory accidents are quite high. Therefore, the laboratory must maintain rigid rules and regulations for safety of the staff. These principles must be recorded for references and the staff should be reminded about them from time to time. Safety workshop, fire drills, and certain other methods should keep the laboratory workers, informed about the threat involved. One must remember that hospital acquired infections is way much dangerous than the naturally acquired one.

Quality assurance record

Quality assurance is an on-going process in any laboratory; hence, it should be recorded properly. Well reputed laboratories also take part in the commissioned competence programmes, in which they are tested on all possible grounds, following which they get certified.

Record of laboratory Personnel

Educational qualification of the laboratory staff should be recorded and availed to the government agencies, whenever required. This is because, the laboratory deals with the public, which makes the government accountable, if any controversy occurs. Unqualified and unskilled staff of the laboratory could be subjected to the legal issues. In such cases, the pathologist is held accountable. Continuity in the education of the laboratory staff, including the pathologist and the supervisor, is a crucial step for carrying out the laboratory procedures efficiently.

Each and every staff member should be encouraged to attend the professional workshops, whenever it is possible, keeping in mind its necessity, availability, and affordability. This type of involvement keeps the staff updated with the recent techniques and procedures, which is very much needed in the medical field. The participation of the staff members in such events must be recorded in their personal files. The staff should focus on the 'methodology based training programme' instead of academic achievements.

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5.9.4 Retention of Records, Reports and Specimens

The records can be used for reference at the time of inquiries and follow-ups. A few valuable reports can be preserved for twenty years whereas, the others can be kept temporarily. Nowadays, with increased computerisation, each and every report is kept permanently unless the system gets crashed. Therefore, it is advised to maintain a hard copy of each report, whose retention time can be set. For various safety reasons, all the records are kept separately in a hard disk and not in the memory of the system.

Check Your Progress

12. How can a laboratory get certified?
13. Who approves new procedures introduced in the laboratory?
14. What is STAT?

5.10 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. A laboratory performs laboratory examinations in six major areas:
 - a) Bacteriology
 - b) Histology
 - c) Biochemistry
 - d) Haematology
 - e) Serology
 - f) Cytology
2. Clinical or Medical laboratory is one of the most prominent laboratories. Clinical laboratory tests are an essential part of most of the diagnostic procedures. Laboratory testing allows the healthcare provider to obtain information, which is unattainable through the patients' history or physical examination.
3. Only the pathologists are allowed to further pass the reports to the clinical physicians and disclose the confidential laboratory researches regarding the patients to the physicians.
4. Pathologists as the conductor of the laboratory, should have a medical degree with a certain specialised training. The bench workers should be college graduates, a technologist should have a higher degree, which can be a master's degree or a Ph.D. and he/she should have a higher level of laboratory skills so that he can also perform a supervisory role.

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5. It is often observed that, in most of the places the laboratory activities are usually carried out by unskilled workers, who are improperly trained or have a little knowledge about the laboratory activities. This negligent behaviour not only affects the accuracy of the staff members in preparing the final results but also lead to risky outcomes in patients' proper care.
6. The basic principles of a recognised laboratory are:
 - a) Licensing
 - b) List of the apparatus required and the supplies needed for the blood bank
 - c) Refreshment services.
 - d) Reagents
 - e) Expiry date
7. The prime factor, which determines the work load rendition is the extent of the mechanisation in the cycle of yield.
8. Information flow have two prime levels. The first level includes a process in which the investigation is requested. While the second level is the entry of the result and the final report of the investigation.
9. A few hospitals have attempted to become financially capable by establishing a cost centre, which operates as a commercial unit laboratory or by setting up a disunited corporation to meet all the requirements of a functional laboratory.
10. Laboratories run Quality Assurance (QA) programs to ensure the accuracy of the outcomes to the physicians.
11. The responsibilities of an hospital organisation for bio-safety of the laboratory are:
 - a) Developing a safety programme in order to inform about the laboratory work methods and safety policies.
 - b) Establishing a safe zone for working staff, including the supplies of proper safety apparatus and the personal protective equipment, wherever required.
12. A laboratory can get certified by taking part in the commissioned competence programmes, in which they are tested on all possible grounds.
13. New procedures introduced in the laboratory have to get approved by both, the supervisor and the pathologist.
14. All the samples, that require immediate attention, termed as STAT, are noted and tested on priority basis.

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5.11 SUMMARY

- A laboratory is an installation, which provides a controlled atmosphere in which a scientific or technological researches can be carried out, by performing certain experiments or measurements.
- Laboratory testing allows the healthcare provider to obtain information, which is unattainable through the patients' history or physical examination. Laboratory tests aid medical personnel in the formation of a firm conclusion and then the treatment and cure of the disease or disorder.
- A clinical laboratory must have a clear formulation of organisational structure in order to execute its job efficiently. The laboratory can either be attached to a medical institution or can operate independently. Each hospital must have a clinical laboratory to carry out emergencies as well as assist surgical decisions.
- The management of the workflow is demonstrated by two major functions. One, the management of the sample and its transportation to the laboratory for analysis, following the established procedure. Second, the laboratory investigations are carried out according to the procedures which are predetermined based on the type and number of the requested tests, the time they are requested, and the accessibility of the automated apparatus.
- A laboratory record must be authentic or else it will mislead the physician and threaten the life of the patient. This requires not only the proper Quality Control (QC) of the laboratory results, but should also involve sample handling and delineation.
- The significance of maintaining laboratory records can never be overstated. The properly maintained records assist in finding the patient's history, upholding correct or right stats, tracking previous errors in order to improve them, and preventing each and every kind of confusion.

5.12 KEY WORDS

- **Myocardial Infarction:** Commonly known as a heart attack, occurs when blood flow decreases or stops to a part of the heart, causing damage to the heart muscle
- **Obstetric:** The field of study concentrated on pregnancy, childbirth, and the postpartum period
- **Rhesus (Rh) factor:** An inherited protein found on the surface of red blood cells

- **Serology:** The scientific study of serum and other bodily fluids
- **Autopsy:** A surgical procedure that consists of a thorough examination of a corpse by dissection to determine the cause, mode and manner of death or to evaluate any disease or injury that may be present for research or educational purposes
- **Baccalaureate:** A university bachelor's degree

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5.13 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. State the reasons for which, physicians may request laboratory tests.
2. Write a short note on the technologists.
3. What is the importance of the laboratory technicians?
4. Why is quality control important in laboratories?

Long Answer Questions

1. Evaluate the functions performed by the major units of laboratory.
2. Discuss the obstacles faced by a laboratory.
3. Describe the procedure followed in a laboratory.
4. Analyse the importance of the different kinds of records in the laboratory.

5.14 FURTHER READINGS

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UNIT 6 EMERGENCY SERVICES

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Structure

- 6.0 Introduction
- 6.1 Objectives
- 6.2 Components of Emergency Services
- 6.3 Aims and Functions of the Emergency Department
- 6.4 Organisation
 - 6.4.1 Models and System
- 6.5 Duties of Staff in Emergency Services
- 6.6 Types of Emergency Services
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 - 6.9.2 General Rules
 - 6.9.3 Examination and Diagnosis
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 - 6.9.5 Aids to Diagnosis
- 6.10 Answers to Check Your Progress Questions
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- 6.13 Self Assessment Questions and Exercises
- 6.14 Further Readings

6.0 INTRODUCTION

Medical emergency is a state, in which the patient requires critical and high quality medical care to thwart loss of life or limb and/or to initiate action for the restoration of life. Emergency medical situations usually occur through an unexpected affront to our body or mind, through injuries, infections, obstetric complications, or chemical imbalance. It may commonly occur as the result of tenacious neglect of chronic conditions such as diabetes. Emergency medical services (EMS) provide quick assessment, timely provision of suitable interventions, and early transportation to the nearest appropriate health facility by the best possible means to enhance survival, control mortality, and avoid disability.

The main goal of effective EMS is to provide emergency medical care to those who are in need of it. With the rapid development in medical care and

technology in recent decades, the emergency services have received a boon in helping people and saving their lives. These services have spread to transportation and pre-hospital treatment and care. Even with the best services and efforts made by the emergency services, there are still some emergency cases, which are unsuccessful. Emergency medical care is very crucial for the people ranging from rural to densely populated slums areas and cities.

Emergency services involve trained emergency professionals, laymen, and taxi drivers. But in most cases, health professionals occupy the role unless out of staff. In order to meet the demand of diverse circumstances in different health sectors, both private and government, innovative ideas and orientation of public health planning is need of the hour. It's a misconception that the emergency services are only for the low income group and it's still lacking in the underdeveloped countries, especially in populated countries. These ideas include equating emergency care with ambulance transportation, neglecting the role of the community and facility, and assuming that emergency departments and physicians are only acute care resources. Such a narrow viewpoint, ignores the important contributions of other disciplines, skills, and personnel. Perhaps the most common myth is that emergency medical care is inherently expensive and requires high-technology interventions as opposed to simple and effective strategies.

Emergency care when delivered in the disaster situation, with weak planning and ineffective use of the resources may have poor outcome. For dealing with the emergency cases, few resources are set aside in different hospitals and are directly deployed, when required. Efforts to improve emergency care, however, do not necessarily increase costs. Emergency services are a vital part of a hospital because of the following reasons:

- Acts as a miniature version of the hospital
- Front door of the hospital
- Portal of entry that interacts with the highest volume of patients, requiring critical care.

6.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the structure of the emergency services
- Analyse the importance of ambulatory services
- Discuss the components of first aid kit and other equipments

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6.2 COMPONENTS OF EMERGENCY SERVICES

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No emergency service can work efficiently without its essential components. These components, including staff, care, triage, etc. make the emergency services what it is. These components are discussed below:

- I. Emergency care:** Emergency medical care is that care, which is accessible to the patients in the first few hours of serious medical complications, injuries, and onset of an acute medical or obstetric problem.
- II. Paramedical personnel:** Those health workers, who are involved in transportation and care of patients in the emergency situation are paramedical personnel. The time period of training length and quality of training vary, from highly specialised personnel with capabilities to handle advanced life support to those with simple first-aid training and limited field experience.
- III. Triage:** Triage is the assessment and screening of patients on the basis of priority for treatment. It is typically necessary, when there are multiple casualties requiring care. It categorises patients into three groups: those very unlikely to survive even with the treatment; those who will recover without the emergency support services; and those who will survive with the timely emergency care. Patients in the last category form the highest priority for emergency care.
- IV. Emergency medical systems (EMS):** They deal with various kinds of diseases like communicable infections, non-communicable conditions, obstetrics, and injuries of various levels. All of these conditions may present itself to the EMS in its acute stages (for example, Diabetic Hypoglycemia, Septicaemia, Premature Labour, or Asthma), or they are acute in their natural presentation (for example, myocardial infarction, acute haemorrhage, or injuries). EMS involves acute event on-site management transportation health facility care, whose functions include recognition triage, stabilisation, safe and efficient transportation prompt, appropriate and quality care, trained personnel, equipment, identification referral system, etc.

Emergency care unit as a whole is interdependent and includes pre-hospital care, hospital care, and the transportation. These three components should work together to make the working situation of the hospital effective. Generally these three components are different in different countries. Due to the lack of good pre-hospital transportation or absence of transportation altogether, death of the patients can occur like maternal death. Pre-hospital care consists of providing the care to the communities (injuries or illness at home, school, or other location) until patients receive proper treatment from the hospital. Pre-hospital care should be basic, cheap, and efficient because some resources are unavailable within countries. In places with no formal set up of the pre-hospital system, any layman can be trained and any drivers having the basic knowledge, can get involve in giving care to the patients.

Transportation of the patient from a specific place to the desired location is very crucial. Due to the untimely transportation of the patients, so many deaths and disabilities often occur. Moreover, the failure to provide the emergency care is due to the lack of proper transportation. In some local areas even bicycle ambulance has been initiated for the obstetric patient, which is used even by patients with injuries and medical emergency. The quick response from the transportation system is very important and an ambulance well-equipped with proper facilities and the trained staff gives a better outcome.

People who come for the emergency services are those, who need immediate medical attention and therefore, emergency department is responsible for the quick assistance to the patients and determining the appropriate place for the treatment. People directly look for the emergency department to provide them with all the medical services. Following are the factors which are linked to the increased demand:

- Awareness and confidence of getting service on priority basis in case of unexpected illness and injuries
- People see hospital as the place to treat the life threatening diseases and chronic diseases
- Some people cannot afford the private hospital
- The increasing rate of the accidents
- Shortage of physicians and health staff at night

6.3 AIMS AND FUNCTIONS OF THE EMERGENCY DEPARTMENT

The basic aims of an emergency department are common to almost all areas:

- Right treatment
- Right time
- Right place
- Right resources
- Speed
- Sympathy
- Science

While the functions of an emergency department include:

- Delivery of instantaneous and proper lifesaving services to the patients at any time
- Rapid institution of BLS to critically ill at site, en route, and in hospital
- Assistance to the ones, who are in need of the medical treatment
- Provision of an effective care and management during disaster situations.

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NOTES**Check Your Progress**

1. List any two functions of an emergency department.
2. What is triage?

6.4 ORGANISATION

In this section, you will study about the organisation of emergency services.

Health workers in the emergency department include:

- Physicians
- Physician's assistant
- Nurses
- Nursing aides
- Department managers
- Orderlies
- Technicians
- Receptionist
- Clerks

The initial assessment of the patient is done by the nurse, who arranges appropriate things for the treatment, so that physicians can directly perform treatment without wasting time. The physician's assistant are seen in the big hospitals, where there are numerous patients. All of the reports made by nurses and physicians are reported to the head nurse of the department and hospital administration respectively. Different hospitals follow different organisational models:

- Emergency service staff (physicians, nurses, clerical) report to the administration directly.
- The very first reports are submitted to the physicians by the staff and finally to the administration.
- In big hospitals, nursing staff reports to the head of the nursing department, while physician to the administration department.
- Staff working in the laboratory, radiology and electro cardiology submit report to their respective departments.

In some big hospitals, part timers are recruited for peak hours. In the emergency department, there should always be at least one physician and one nurse on duty round the clock. Emergency department is the base platform in the hospital, which also includes radio communication centre for providing clinical advice to the paramedical units or remote treatment centres.

6.4.1 Models and System

The models and systems found in the hospital are set up in such a way that priorities are set for the treatment of patients and for ensuring the efficient charging, registration, and record keeping.

Priorities

The emergency department receives patients, who require treatment in several areas. Since emergency area is the primary area for the treatment therefore, it needs additional diagnostic services such as radiology, laboratory, and electrocardiography. In order to meet such requirements the policy of the hospital is the main issue, which should be put into an account.

The staff, working in the emergency department, are all trained to assess the severity of the admitted cases and are also taught about the legal requirements related to the patients. The patients are assessed or recognised by the degree of illness and threat to one's life. Patients are classified on the basis of priority in the following ways:

- a) **Emergency:** Cases requiring immediate examination and treatment like serious injuries and illness of a traumatic nature.
- b) **Urgent:** Those cases like stomach pain, headache, suture removal, dressing changes, though not urgent but still require immediate attention.
- c) Employees and staff of one's own hospitals.
- d) **Non-emergency:** Minor cases, which are not acute. They visit hospital based on the physician's recommendation.

Patients are also described according to the type of medical problems, that is, if they have surgical, medical, obstetrical, or orthopaedic problem. Physicians are then allocated to them for the treatment accordingly. The policies and the procedures of the hospital, the availability of the emergency equipment for the examination and treatment, staff number, and time of patient's arrival will affect the time to treat the emergency patient.

Charging, registration and record keeping

Charging system differs from hospital to hospital. Following models are usually utilised for the charging system:

- a) **Flat rate:** Visiting patients are charged the usual consulting fees and for the treatment
- b) **Sliding rate:** Patients are charged based on the level of treatment they receive.

In case of hospital physicians, the charges are made by the hospital while the private physician charges are billed separately by the physician. All the manual registrations and charge system are now computerised. People working in the

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registration department do not belong to emergency staff, but are part of the whole organisation. There is a specific department for the record keeping, which is responsible for retaining the data and records of each patients. Apart from it, those who work in the record keeping department should also maintain separate log and update all the cases about the patients.

Check Your Progress

3. Define sliding rate.
4. List all the health workers in the emergency department.

6.5 DUTIES OF STAFF IN EMERGENCY SERVICES

The duties of staff in emergency services depend on the position assigned to the person:

Department Chief Medical Officer In-charge

- Checks the punctuality of the other Medical officers and Resident Doctors
- Prepares the duty roster for the doctors
- Takes supervisory rounds of equipment and drugs
- Checks registers, documents, etc.

Medico legal register and registration register

- Supervises the maintenance of sanitation and Bio-medical waste management
- Responsible for training re-organisation classes of all categories.
- Conduct drills for emergency management/disaster action plan

Senior Resident

- Examines and treats promptly
- Responsible for certifying death certificates and getting it countersigned by CMO IC of casualty

Junior Resident

- Carries out treatment as advised by the senior doctor
- Takes rounds along with senior resident
- Performs minor operations and write patient's case sheet in legible handwriting.

Specialist

- On regular duty or call
- Puts his notes in the case sheet, when he is consulted
- Gives expert guidance
- Guides his subordinate staff
- Trains junior doctors in handling emergency cases

D.N.S/A.N.S/SISTER-IN-CHARGE

- Responsible for the efficient working of the Emergency Department
- Prepares duty roster
- Stores essential drugs, i.e. fluids, ensure all the equipment's are in working order.
- Ensures that all equipment's are in working order. E.g., suction apparatus, Central oxygen supply, Boyle's apparatus.
- Nursing Personnel will seek guidance of Nursing Superintendent and appraise her day to day problems.

Nursing staff (Nursing Attendants/ward boy/*Ayas*)

- Carries out doctor's orders
- Maintains the vitals of a patient
- Packs dead bodies
- Assists nursing staff in patient care
- Gets the indent from stores and sterilized items from C.S.S.D
- Transfers patient to other ward/units
- Assists nursing personnel in packing dead body
- Carries out any work assigned by superiors

House keeper

- Keeps the area neat and clean
- Performs all the duties as required by the supervisor.
- Changes soiled linen
- Transports dead bodies to the mortuary
- Stretcher Bearer
- On duty near the entrance of the casualty
- Assists in transferring the patients from Ambulance to Casualty
- Trained in first aid treatment

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Security guards

- Duty as per roster prepared by the Security officer/CMO
- Regulates the flow of patients or their attendants
- Answerable to CMO in case of any casualty or untoward incidence
- Performs any other duty as required by his supervisor/security officer

6.6 TYPES OF EMERGENCY SERVICES

Depending on the size of a hospital, nature of injuries, population, and catchments area the emergency services can be categorised into:

- Emergency major emergency and disaster management
- Referral emergency
- Basic emergency
- Stand by emergency

6.6.1 Departmental Analysis

Emergency department should mainly focus on following factors:

- Staff:** It is necessary to gauge the productivity and the utilisation of the staff. Identification of the type of skills required, whether it is clinical or nonclinical and the number of full timers and the part timers, along with the shifts of their duty is a must.
- Treatment room:** Room for the patients to undergo treatment should be determined properly.
- Patients:** Keep in touch with the number of the patients admitted and their daily treatment routine.
- Minor surgery:** Study the time required for the surgery and appointment of doctors for the surgery.
- Materials management:** Review all kinds of materials for maintenance, distribution, and transportation.
- Ambulance service:** Check for the availability of the ambulance service.
- Staff education:** Conduct trainings, interviews, and surveys to measure the staff skills and knowledge.
- Charging procedures:** Always review the charging system and cost.

The above-mentioned areas are the main factors, which should be analysed thoroughly because any problem in one of them may affect the emergency services.

6.7 CRISES IN THE EMERGENCY DEPARTMENT

The emergency department with its technology and skills of staff, is also not free from crises. Being an essential part of a hospital, an emergency department often remains over-crowded with patients. Thus, there is good chance of crises happening in this department. Some of the crises are:

I. Overcrowding: Too many patients leads to

- Long waits for inpatient beds
- Ambulance diversions
- Errors
- Adverse outcomes

II. Nurse shortages

III. Physician shortages

IV. Uncompensated care: More and more patients show up for these services, without the means to pay for it. This creates burden on the tax payers, whose money government uses to maintain the emergency services.

V. Overworked staff: Due to constant influx of patients into the emergency wards, the staff is overworked, prone to error and indifference towards patients' suffering.

VI. Lack of specialised training: Despite being the most essential requirement, the staff is not properly trained with respect to their location, to handle the severe cases. As can be seen in India, which is prone to tropical diseases, cardiac complaints and trauma, the staff lacks the proper skills and equipment to handle such cases.

Check Your Progress

5. How can we categorise the emergency services?
6. List any two duties of Department Chief Medical Officer In-charge.

6.8 AMBULANCE SERVICES

An ambulance is a transporter for carrying the sick or injured people to the hospital from patients' homes or any spot, for treating an illness or injury. The term ambulance does not only mean the vehicle with the flashing lights and sirens. The term also includes a large number of non-urgent ambulances, which are for the transport of patients without an urgent acute condition. The word, Ambulatory, literally means an injured person who is able to walk. Basically, it highlights the

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mobility of ambulatory services, which is accessible to all. These services are beneficial as:

- Inpatient care is so expensive that patients cannot afford to pay for an overnight stay, if it isn't medically necessary
- Hospital-based outpatient care is more expensive than non-hospital based outpatient care

Hospitals must maintain all the expensive services and technology in order to:

- Be properly accredited and licensed
- Attract various insurers, patients, and good physicians
- This high overhead puts them at a competitive cost disadvantage relative to free-standing, single focus ambulatory surgical-centres or imaging centres

In the past, hospital ambulatory amenities were meant to offer necessary health care to the impoverished population. Since, such people were highly concentrated in the urban regions, outpatient clinics were mainly situated in or associated with large public hospitals. Clinics were able to continuously run at minimum cost using medical interns and residents to give the professional care, while at the same time building a practical training ground for physicians. Operating revenues were mostly funded from tax revenues of the local, city, or government authorities and occasionally from contributions by private or governmental organisations.

The progressively extensive movement in the provision of health care in ambulatory settings has caused the growth of outpatient clinics in recent years. The effect of prospective payment systems has further boosted this system. Though created with a constrained scope, outpatient clinics are expanding their facilities, setting up in unreachable places, and augmenting their population base. Such clinics need to be considered distinct from the emergency departments as they are not necessarily located within the hospital complex, which will substitute for the outpatient clinics. Various ambulatory services offered in the hospital are different in different set ups. In terms of physical structure it should:

- Have proper parking space
- Have separate entrance
- Have a spacious parking area
- Be adjacent to OPD
- Have steps/ramp

Following are the areas, in which, services are provided by the ambulatory system:

- Allergy
- Hand
- Hypertension
- Dental
- Nutrition
- Psychiatry
- Medical walk-in
- Eye and ear
- Dermatology
- Infectious disease
- Obstetrics
- Diabetes
- Gynaecology and women's health
- Internal medicine
- Paediatrics
- Neurosurgery
- Sports medicine
- Language disorders
- Endocrinology
- Pulmonary
- Learning disorders
- Gastroenterology
- Substance abuse
- Orthopedics
- Social service
- Neurology
- Renal

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Advantages of ambulance services

- Ambulance services provide quick arrival of patients to the hospital, saving their lives.
- Ambulance services are cheap in general and are generally preferred on road.

- Services are provided for all the people irrespective of their religion, caste and income.

Defining characteristics

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- Ambulatory physicians and nursing staff require critical reasoning and judicious clinical judgment to further provide appropriate care and treatment, because some patients may develop complications.
- Provide care to all the people in all the locations.
- Ambulatory care staff are mainly present in the clinic and the hospital centre, where there is need for an immediate assessment and the urgency level is high.
- Telecommunication services are very important in the hospital department, especially for the nurses, if they want to form a long term relationship with the patients.
- Telehealth nursing is the primary component of professional ambulatory care that utilises a variety of communication technologies to manage the health of the patients, for consultation and for the regular check-ups.
- The ambulatory care registered physicians and nurses focus on patients' safety and the quality of nursing care offered to the patients with different diseases.
- The physicians and other staff act as partners and advisors to assist and support patients in order to motivate patients to work on their medical problems.
- Knowledgeable staff with leadership qualities are usually assigned to the ambulatory department to help the patients effectively.
- Ambulatory care registered nurses plan, manage, and assess nursing services within the organisation in accordance with relevant federal requirements, state laws, and acts.
- Ambulatory care registered nurses are responsible for the coordination of nursing services, including the assignment of appropriate job for a particular skill and delegation of roles and responsibilities for licensed and unlicensed nursing personnel.
- Ambulatory care registered physicians and nurses follow lifelong learning process that appraises and multiplies their clinical and professional roles and responsibilities.

Ambulatory care is evolving as the real and focal policies and procedures serve as operational guidelines for providing proficient care to patients. These standards include maintaining a safe, controlled, and constantly monitored environment and keeping precise records, along with active quality assurance programmes by organising a defined committee to promote excellence in ambulatory

care. Many different surgeries can be performed in an ambulatory department, which leads to less hospitalisations and susceptibility to nosocomial infection. Six surgical procedures performed in ambulatory setting are:

- Biopsy
- Dilatation and curettage
- Excision of skin lesion
- Tubal ligation
- Cataract extraction
- Inguinal herniorrhaphy

Other non-surgical procedures, which can be performed on day care basis are:

- Aspiration of breast mass
- Bladder irrigation
- Bone marrow aspiration
- Bronchoscopy
- Cardiac arteriography
- Antineoplastic and antibacterial chemotherapy
- Cystoscopy
- Colonoscopy
- Esophageal variceal sclerotherapy
- Intercostal tube drainage
- Liver biopsy (percutaneous)
- Lumbar puncture (Diagnostic)
- Renal biopsy
- Sigmoidoscopy
- Thoracentesis

Ambulance team

The ambulance team consists of following members:

- Physician
- Nurse
- Paramedic/senior paramedic
- Ambulance care assistants
- Ambulance driver
- Call handler/emergency medical dispatcher

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6.8.1 Types of Ambulance

Ambulance services are classified based on whether they transport patients or not, and under what conditions.

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- I. Emergency ambulance:** It is the most common type of ambulance, which provide care to patients with an acute illness or injury. These can be vans, boats, helicopters, or fixed-wing aircrafts (known as air ambulances).
- II. Patient transport ambulance:** A vehicle, which has the job of transporting patients to, from, or between places of medical treatment, such as hospital or dialysis center, for non-urgent care, are known as patient transport ambulance. These can be vans, buses, or other vehicles.
- III. Response unit:** Also, known as a fly-car or a [Quick Response Vehicle], it is used to reach an acutely ill patient quickly and provide on-spot care.
- IV. Bariatric ambulance:** A special type of patient transport ambulance designed for extremely obese patients equipped with the appropriate tools to move and manage these patients.

Cardiac care ambulance

Cardiac care ambulance provides a network of fully equipped life support ambulances in the metro city. In case of a medical emergency, any nearby resident may access a fully equipped ambulance, within the shortest possible time. These ambulances are located in strategic locations. After receiving the call, the nearest free ambulance reaches the place to rescue the patient/victim in the shortest possible time. Cardiac care ambulances are equipped with international quality instruments for international standard of service and care. The emergency physicians on board are usually trained in the city's ICU and emergency department and are adequately experienced to handle all kinds of emergencies. Cardiac care ambulance consists of:

- Doctor and Trained Nurse
- E. C. G.
- Cardiac Monitor
- Defibrillator
- Portable Ventilator
- Suction machine
- Pulse Oximeter
- Oxygen
- Emergency Medicines
- Resuscitation kit
- Nebulizer

- Syringe Pump
- BiPAP
- Pace Maker
- Spine Board

Types of ambulatory care settings

Ambulatory care can be categorised on the basis of setting:

- Hospital emergency departments (EDs)
- Hospital-based outpatient departments (OPDs)
- Outpatient offices Physician private practices
- Community health centres
- Free-standing
 - a) Ambulatory surgery centres (ASCs)
 - b) Imaging centres
- Urgent care centres
- Retail clinics
- Concierge medicine
- Complementary and alternative medicine

Professional ambulatory care is a multifaceted, polygonal specialty that covers autonomous and joint practice. The broad practice of ambulatory care is based on an extensive knowledge of nursing and health sciences, and applies clinical expertise rooted in the nursing process. Physicians and the other staff in the health department use the information of all the patients to provide safety and achieve the organisation's goals. Ambulatory care in the medical department includes those clinical, organisational, and professional activities, which involve both the health workers and the patients, who seek the assistance for their immediate health problems. Physicians, nurses, and supporting staff are involved in the management of the acute illness, mitigating the chronic pains, creating awareness about the health issues, and to provide support in end of life care. Those physicians, nurses and supporting staff are legally registered and are responsible for providing the care in accordance with the requirements, state laws, and other professional standards.

Ambulatory care is ultimate arm of health care as it is provided at all levels of health care system i.e., from sub-centre to tertiary care hospital. Ambulatory service is the area, where most of the health revolution is taking place, today. Ambulatory care is the care offered to the patients, who are not bed ridden. These services are usually offered, when the treatment can be performed in the clinic or the hospital centre. With the increasing number of the population and accidents, ambulatory services should be well equipped and should be of good quality. In the

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populated countries, having the good ambulatory services can be of great economic value.

6.8.2 Problems Faced by the Ambulatory Services

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Ambulatory department is one of the most important services in the medical department. It requires commitment and the involvement of physicians, and the trained staff to deliver proper care to the patients. Due to the difference in the goals and the objectives of the physicians and the other hospital staff, numerous problems arise in the hospital. In outpatient clinic, the close communication and mixing of the physicians with staff is must.

Sharing the budget between outpatient and inpatient should be considered by the hospital practically and objectively, for its convenience. The specialised equipment present in the outpatient department should be present in the inpatient department as well, which will result in smooth and flexible hospital functioning. The number of patients visiting the hospital is directly affected by the variable known as centralisation and decentralisation in making appointment and registration. These two systems will affect the organisational structure and the physical location of the ambulance. These two systems have both the advantages and the disadvantages. The advantages of the centralisation are:

- Bulk examinations and the use of the ancillary services are effective
- Simplifies the location and process of getting the medical records
- An efficient utilisation of the registration personnel.

The advantages of the decentralisation:

- Direct access to the staff at the clinic for the services
- Reduction in the confusion at the registration desk
- Provision of the quick schedule changes for the patients.

With the use of computerised system, the registration and the appointment is made a lot easier. It also helps the ambulatory service department to run effectively by using both the centralisation and decentralisation system. Patients are directly registered in the central registration system, after which, they visit the ambulatory service department. When the same patient keeps on visiting, then he/she is registered for the repeated visits. For some low level ambulatory services, it is always wise to register oneself in the central registration, whereas in the bigger set up registration is done in the ambulatory department itself. The biggest problem faced by the managers are the absent patients and the walk-in patients. The walk-in patient is the one, who does not have appointment and generally seen by the physician. The absent patients do not turn up for the appointment.

Check Your Progress

7. State any two advantages of the decentralisation.
8. What is Cardiac Care Ambulance?
9. What does the word, Ambulatory, mean?

NOTES**6.9 FIRST AID**

First Aid is the most important branch of the medical science. It is both an art and a science. First aid is the immediate and temporary care given to the victim of an accident or in case of a sudden illness to a person. Its main purpose is to save life, assist recovery, and prevent any aggravation of the condition, until the service of a doctor is obtained. The St. John Ambulance Association adopted the term, first aid, officially in England in the year 1879. In 1920, Red Cross Society of India was established. This was the time, when technology has produced complicated machinery and swift means of transport. Therefore, from this period accidents started happening frequently, producing devastating results with loss of life or injuries to the body and mind. Under these circumstances, the first aid gained much significance for the injured person.

When the first aider reaches the spot of the accident, the causes of the accident may still be present and continues to exert harmful effect. The term 'Medical First Aid' is used to indicate the treatment by a doctor either on the spot, in the home of a person or in the hospital. The first aider's responsibilities end as soon as medical aid is made available to the person; but he/she still need to stand by, after making his report to the doctor, in case he/she can be of further assistance during treatment of the patient. First aid is definitely limited to the assistance rendered at the time of the emergency with as much material as may be available. The first aider is not supposed to replace the doctor. It must be clearly understood that the redressing of injuries and other such treatment are outside the scope of a first aider.

Diagnosis

In deciding the nature of the case or accident, the first aider must consider its 'History, Symptoms and Sign'.

- I. History:** This is a full story of how the incident occurred or the illness began and should be taken directly from the casualty or a responsible bystander whenever possible. Never hurry the casualty and remember to pass on all the information that you have obtained to the skilled help.
- II. Symptoms:** These are the sensations that the casualty feels and describes to you, the most useful among which is pain. If the casualty is unconscious

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or unreliable, their diagnosis cannot be based on symptoms but has to be based on information obtained from bystanders and sign.

III. Signs: These are details ascertained by you using your senses-sight, touch, hearing, and smell. These may be sign of injury such as bleeding, swelling, deformity, or signs of injury such as raised temperature and a rapid or irregular pulse.

Treatment

If the cause of the condition still exists in the body, then it should be removed as soon as possible, e.g., log of wood on leg or the victim themselves should be removed from the site of the accident, e.g. gas filled room. The proper treatment should be provided to preserve life, promote recovery, and prevent the condition from becoming worse. Special attention needs to be paid towards the treatment of failure of breathing, severe bleeding, and shock. When there is a slightest doubt as to whether a casualty is dead or alive, continue treatment until medical aid is obtained.

Disposal

The speed with which a casualty is brought under medical case is of tremendous importance in the recovery. Unless the casualty is examined by a doctor on the spot, it is the responsibility of a first aider to ensure that the victim is conveyed to his or her home, other suitable temporary shelter, or hospital as soon as possible in the manner most suitable in the circumstances. A tactful message should be sent to the casualty's home or relatives, describing the circumstances in a general way and the casualty's destination.

Golden rules of first aid

- Do 'first things first' and without any fuss or panic in the situation.
- 'Give artificial respiration from the mouth' if the breathing has stopped as every second counts.
- Stop any bleeding by the means of medicine or bandage.
- 'Guard against or treat for shock' by moving the body of a casualty as little as possible and handling him/her carefully.
- 'Do not attempt too much', do the minimum thing that is essential for saving the life and prevent the condition from becoming worse.
- 'Reassure the casualty' and those around the casualty to help lessen anxiety.
- Do not allow people to crowd around the casualty, as fresh air is essential for him or her.
- Do not remove the clothes unnecessarily.
- Arrange the means for taking the casualty to the care of a doctor or hospital as soon as possible.

Action at an emergency

The basic principles of first aid apply to all injuries or illnesses regardless of severity of the casualty. Irrespective of the kind of the incident, it is the first aider's responsibility to act quickly, calmly, and correctly in order to:

- Preserve life
- Prevent deterioration in the casualty's condition
- Promote recovery

These objectives are best achieved by:

- A rapid but calm approach
- A quick assessment of the condition based on the history of the incident, casualty's history, symptoms, and signs shown by the body
- Immediate and appropriate treatment of any conditions diagnosed.
- Proper disposal of the casualty according to the injury or condition.

6.9.1 First Aid Kit and Emergency Equipment

In order to make one's job easy, they should be well equipped. Whether at home, school, buses, factories, or cars, there should be a first aid bag or kit kept ready for emergency conditions. The contents of a first aid bag varies according to the duties to be performed and the special needs of the occasion.

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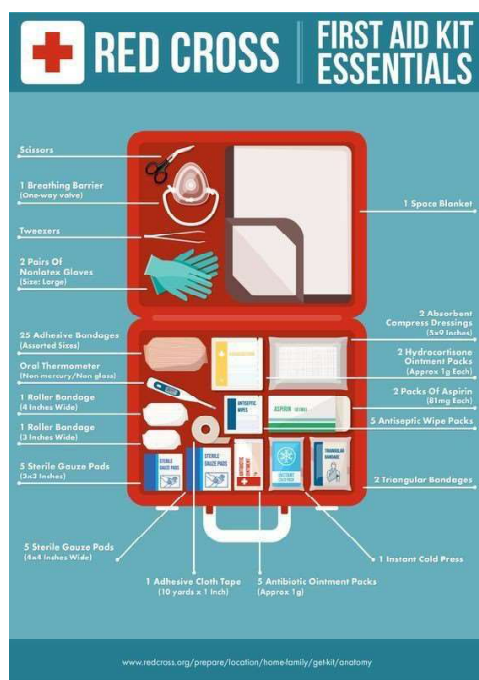


Fig. 6.1 First Aid Kit

The following list provides a basic guide for selecting the equipment for first aid kit:

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a) First Aid Kit:

- Dressing
- Roller bandages
- Cotton wool
- Triangular bandages
- A pair of scissors
- A set of large safety pins
- Dissecting forceps
- Graduated medication glass
- Torch
- Glucose, sweets, or lump of sugar
- A cake of soap
- A small note book and pencil
- Smelling salts

b) Equipment:

- Stethoscope
- E.C.G. machines
- B.P apparatus
- Cardiac monitor
- E.N.T instruments
- Suction machine
- Weighing machine
- Thermometer
- Nebulizer
- Pulse Oximeter
- Knee hammer
- Tracheostomy set
- Suturing set
- Cut-open set
- Thoracentesis set
- Cauterisation set

All the instruments should be properly sterilised and equipment should be checked frequently for any rust, break, or bluntness.

6.9.2 General Rules

On coming across an emergency such as an accident, it is imperative to deal with the situation following a proper procedure. It is important to use one's common sense, be aware of one's limitation and avoiding doing too much.

Assessing the situation

As soon as you have taken control of the situation, it is crucial that to make an accurate assessment of the situation and decide on the priorities of actions. To do this you must consider:

- Whether you or the casualty are in any danger
- If the casualty has any life-threatening conditions
- If any bystanders can help you
- Whether you need to call for assistance

You must minimise the risk of danger to yourself, the casualty, and bystanders, and guard against any further casualties, for e.g. in the case of road accident, instruct bystanders to control traffic, keeping it well away from yourself and the casualty. Watch out for any risks, especially from petrol spillage and switch off the ignition of the vehicles concerned.

Determining priorities

- In order to determine the condition of a casualty, ask them about the incident.
- A reply will help to know about the part affected in the body
- It also tells you many things like if he/she is conscious, their airway is clear, and he/she is able to breathe. In case he/she does not respond to questioning or touch, perform the following check immediately

Airway, breathing, and circulation

- Following the ABC rule frequently, check if the airway is open and that the casualty is breathing properly, if not, then commence artificial ventilation immediately.
- Check pulse for circulation in the body. If absent, commence external chest compression.
- Check for any severe bleeding and control it.

Unconsciousness

Place an unconscious casualty or one whose breathing is noisy in the recovery position and assess the level of responsiveness to it. If there is any sign of spinal injury then:

- Do not turn him/her
- Keep him/her on spinal recovery position

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Unless there is a difficulty in breathing, vomiting is likely to occur.

Actions according to situations

The following actions should be taken in these situations:

- a) **Gas and poisonous fumes:** If possible, cut off the source and ensure adequate ventilation.
- b) **Electrical contact:** Break the contact, if possible and take the necessary precaution against further contact.
- c) **Fire and collapsing buildings:** Move the casualty to safe location immediately, if you can without endangering your own life.
- d) **Shock:** Keep the casualty warm, quiet, and in supine position, until skilled help arrives.
- e) **Other needs:** Unless there is an immediate danger to the casualty from the surroundings, treat all fractures and large wounds before moving a casualty. If the condition is in more danger, support the injured part whilst moving.

Calling for assistance

Once you decide that assistance is required then immediately call for the required service:

- Ambulance
- Police
- Fire brigade
- Gas or electricity boards

Whether you are giving the message yourself or instructing someone to do so, make sure that the following information is passed on

- Your phone number
- The exact location of the incident
- An indication of the type and seriousness of the incident, for e.g. road accident, in which two cars are involved or three people are trapped.
- The number, sex, and approximate age of the casualties involved and if possible, the nature of the injuries.
- Request for special help, if you suspect a heart attack or childbirth.

6.9.3 Examination and Diagnosis

In order to examine and diagnose properly, taking the information of the casualty's history and that of the incident, symptom, signs, and level of responsiveness is crucial. A general examination of the casualty should be carried out quickly to

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discern any imminent threats to their life, to check whether the casualty is in state of conscious or not. There are some rules that you should follow:

- Move the casualty him or her as little as possible to avoid aggravating any injuries.
- Start your examination at the head and work methodically towards his or her feet.
- During the examination, remember to use all your senses—look, feel, listen and smell.
- Always compare one side of the casualty's body with the other as this makes it easier to detect any swelling or irregularities that require first aid.

If at any stage during the examination the casualty's breathing become noisy or it becomes difficult to breath, then place the casualty in the recovery position.

The examination should be performed following this procedure:

- I. Examination of the head:** Recheck breathing, noting down the rate, depth, nature, and any type of odour. Check the inside of the mouth to ensure it contains no foreign matter, such as vomit, blood, food, cough, or loose teeth, which might cause choking.
- II. Examination of lips:** Examine the lips for any signs of burn, crack, or discoloration, which will indicate if there is corrosive poisoning in the mouth. Look inside the lips for blue colour, which might indicate the asphyxia.
- III. Examination of teeth:** Check the teeth carefully to make sure that any recently dislodged tooth, has not fallen down into the back of the throat. Make sure that dentures are firmly set in the mouth and if they are not, then remove them.
- IV. Examination of eyes:** Examine both eyes together comparing the pupils and noting whether they are equal in size or not. Check the white orbit for blood shot eyes.
- V. Examination of nose:** Check the nose for any sign of blood, clear fluid, or mixtures of both, which might be coming from inside the skull.
- VI. Examination of face:** Look at the colour of face—it might be pale or flushed. Feel the temperature of the face to check whether it is too cold or too hot and note down the state of the skin that is, whether it is dry or clammy or even sweaty.
- VII. Examination of ears:** Check the ears carefully for foreign objects and traces of blood and clear cerebrospinal fluid that might indicate skull fracture. Speak into the casualty's ears to test their hearing.
- VIII. Examination of skull:** Gently run the hands over the scalp searching for bleeding, swelling, or any indentation that might indicate a fracture in the skull.

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IX. Examination of the spine: Loosen the clothing around the neck area. Run your fingers gently over the spine from the base of the skull, checking for any type of irregularity in the vertebrae that might indicate a fracture in spine. Check round the neck carefully, to check for any medullation. Check the carotid pulse and note down its rate, strength, and rhythm. Note down the presence or absence of all the movements or feelings in the casualty's limbs.

X. Examination of the lower spine: Pass your hand carefully under the hollow of the back and without moving the casualty or removing their clothing, feel along the spine as high and as low as you can checking for irregularity of the vertebrae or swelling in the lower spine.



First Aid recovery positions

Head

This position is used for all conscious casualties with injuries to the head, excluding nosebleeds and mouth bleeds, it includes;

- Scalp wounds
- Compression injuries to the head
- Stroke
- Fractured skull
- Eye injuries

**W**

This position is used for all conscious casualties with the following;

- Heart attack
- Angina
- Chest Wounds
- Internal chest injuries
- Anaphylaxis
- Vaginal bleeding

**Sitting**

This position is used for all conscious casualties who require airway drainage as long as they are fully conscious including;

- Nose bleeds
- Mouth bleeds
- Fractured jaw (also supporting the jaw)
- Asthma

NOTE; if the casualty becomes dizzy in this position, they should be helped into the W position or safe airway position as necessary.

**Shock**

This position is for conscious casualties suffering from shock, as long as they have no lower limb fractures, conditions include;

- Shock
- Fainting
- Internal bleeding (below the diaphragm)
- Lower limb bleeds.

**NOTE**

Sometimes a casualty may have multiple injuries with conflicting treatment positions. Generally the most serious injury will dictate the treatment position. Just remember the airway must always take priority, then bleeding then breaks.

Fig. 6.2 First Aid Recovery Positions

XI. Examining the trunk:

- Check the chest carefully for unevenness in rib movement or breathing and note down any wounds that are ‘sucking air’
- Check the ribs movement for irregularity or depression that might indicate a fracture in chest.
- Check both the collarbones for irregularity and also the shoulder for a sign of deformity.
- Carefully examine or feel either side of the pelvis looking for the sign of fracture and note down any indication of incontinence in it.

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XII. Examination of the arms: Examine the upper arm bones carefully, then check the bones in the forearm, wrists, hands, and fingers. Check all the parts carefully for any sign of deformity and swelling, which might indicate fractures in the bone. Moreover, check the forearms to see if the casualty is wearing a medical warning bracelet and for injection marks. Needle marks on the forearms might indicate either drug abuse or diabetes.

XIII. Examination of the legs: Check the full legs like hips, knee caps, and thighs, both bones of the lower legs, the ankles, feet, and toes.

Use both of your hands, so that you can examine and compare both sides of the body at the same time.

6.9.4 Level of Responsiveness

There are many stages through which a casualty may pass during progression from consciousness to unconsciousness stage. During progression if the casualty responds well to any stimulants then unconsciousness is only light and of short duration. However, he/she could still be in a very dangerous state. If the response of casualty is poor or absent, then that person is quickly losing the consciousness, posing a greater risk to a person. Recheck the body every ten minutes and note down the casualty’s response to the stimuli of noise, touch, and pain on the face, while pinching the skin, hand, or ankle. Additionally, you should also keep a check on casualty’s breathing, pulse rate, and body temperature. Your findings should be noted down properly.

6.9.5 Aids to Diagnosis

Your diagnosis will be based on the information collected from various sources. By taking the history of the incident into account, ask the casualty for symptoms and pain, and examine him/her for signs, so that an accurate diagnosis of casualty could be made. The following Table 6.1 is summary of how to achieve this.

Table 6.1 Signs and Symptoms**NOTES**

Signs	Symptoms
These are sensations experienced by casualty, about which the information should be gathered from the bystanders or victims themselves.	These are noted by first aider by his or her own senses/knowledge
Pain	Sight
Loss of movement	Respiration
Loss of Sensation	Bleeding
Cold	Wounds
Heat	Colour of face
Thirst	Foreign bodies
Weakness	Swelling
Dizziness	Deformities
Faintness	Reflexes
Nausea	Bruising

External clues

If a casualty is unconscious then, check their pockets, handbag, or brief case for possible clues about them. Appointment cards for a hospital or clinic, or an information card may reveal that the casualty is on steroids or insulin or is liable to epileptic fits, due to which, he/she is unconscious. While lumps of sugar or glucose or candy present may indicate that the casualty is suffering from diabetes or a locket for a wrist or neck containing a strip of paper, describe the casualty's condition.

Treatment

Provide the appropriate treatment for each condition carefully and quickly. It is important to reassure and encourage the casualty constantly. Work calmly and efficiently and pay attention to any remarks or request that the casualty makes. It is essential to not pester the casualty with unnecessary questions as it reflects uncertainty on your part. After giving the necessary treatment to casualty, place him/her in the appropriate position and keep a watchful eye until a doctor arrives. The main aim is to preserve the life, prevent the condition from worsening and promote recovery. To preserve life a first aider should:

- Maintain an open airway by positioning the casualty properly
- Begin resuscitation, if the casualty is not breathing and the heart is not beating, and continue the treatment, until the skilled medical aid arrives
- Control bleeding

In order to prevent the condition from becoming worsening a first aider should:

- Dress wounds
- Provide comfortable support for any fractures or wounds

- Place the casualty in the most comfortable position according to his/her condition

In order to promote recovery a first aider should:

- Relieve the casualty of discomfort and pain.
- Attempt to relieve from the casualty of pain and discomfort.
- Handle the casualty carefully.
- Protect the casualty from the wet and cold

Post-Treatment

Once you have carried out the treatment required by a casualty, they should then receive attention from a qualified doctor or nurse, without any delay. Depending on the severity of the condition of a casualty and availability of skilled help, you should:

- Arrange transport to the hospital by an ambulance or a car for minor injuries
- Hand over the casualty to the care of the doctor or nurse, present at the scene.
- Take the casualty to a nearby shelter or house to await the arrival of any transport and doctor.
- Allow the casualty to go home and advise them to seek medical advice, if necessary.

It is important to note that a casualty should not be sent home, if they had been unconscious or in shock, even for a short time period, as they may need medical aid.

Final report

The casualty should always be accompanied by a brief written report, when they finally leave your care. If there is a need, then you should accompany the casualty yourself and prepare the report. The need to supply complete information about a casualty cannot be emphasised enough. The report should include the following information:

- History of the illness or information about the accident of a person.
- Brief description of the injury or illness.
- The level of responsiveness and any changes in the body
- Any other additional injuries to the body
- Change in the pulse
- Change in the skin colour
- Blood loss sustained
- Any unusual behaviour by the casualty
- Time and information about any treatment, if given

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The first aid is one of the very important procedure for the casualty. It helps to stop the infection from developing and reduce mortality rates.

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Check Your Progress

10. When was the Red Cross Society of India established?
11. List any three contents of first aid kit.
12. What should be done in case of the absence of pulse?
13. Why should a first aider use both hands during examination?
14. List any four signs, which should be checked by a first aider.
15. What are the rules that should be followed by a first aider while checking a casualty?

6.10 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The functions of an emergency department include:
 - a) Delivery of instantaneous and proper lifesaving services to the patients at any time
 - b) Rapid institution of BLS to critically ill at site, en route, and in hospital
2. Triage is the assessment and screening of the patients on the basis of priority for treatment.
3. In sliding rate, patients are charged based on the level of treatment they receive.
4. The health workers in the emergency department are:
 - a) Physicians
 - b) Physician's assistant
 - c) Nurses
 - d) Nursing aides
 - e) Department managers
 - f) Orderlies
 - g) Technicians
 - h) Receptionist
 - i) Clerks
5. The emergency services can be categorised into:
 - a) Major emergency and disaster management
 - b) Referral emergency

- c) Basic emergency
 - d) Stand by emergency
6. The duties of the Department Chief Medical Officer In-charge are:
- a) Checks the punctuality of the other Medical officers and Resident Doctors
 - b) Prepares the duty roster for the doctors
7. The advantages of the decentralisation are:
- a) Direct access to the staff at the clinic for the services
 - b) Reduction in the confusion at the registration desk
8. Cardiac care ambulance provides a network of fully equipped life support ambulances in the metro city. In case of a medical emergency, any nearby resident may access a fully equipped cardiac ambulance, within the shortest possible time.
9. The word, Ambulatory, literally means an injured person who is able to walk.
10. The Red Cross Society of India was established in 1920.
11. The contents of first aid kit are:
- a) Dressing
 - b) Roller bandages
 - c) Cotton wool
12. In case of the absence of pulse, a first aider should commence external chest compression immediately.
13. A first aider should use both of their hands during examination, so that they can examine and compare both sides of the body at the same time.
14. The signs, which should be checked by a first aider are Pain, Cold, Dizziness, and Weakness.
15. The rules that should be followed by a first aider while checking a casualty are:
- a) Move the casualty him or her as little as possible to avoid aggravating any injuries.
 - b) Start your examination at the head and work methodically towards his or her feet.

NOTES**6.11 SUMMARY**

- Medical emergency is a state, in which the patient requires critical and high quality medical care to thwart loss of life or limb and/or to initiate action for the restoration of life.

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- Emergency medical situations usually occur through an unexpected affront to our body or mind, through injuries, infections, obstetric complications, or chemical imbalance.
- No emergency service can work efficiently without its essential components. These components, including staff, care, triage, etc. make the emergency services what it is.
- The models and systems found in the hospital are set up in such a way that priorities are set for the treatment of patients and for ensuring the efficient charging, registration, and record keeping.
- Being an essential part of a hospital, an emergency department often remains over-crowded with patients. Thus, there is good chance of crises happening in this department.
- An ambulance is a transporter for carrying the sick or injured people to the hospital from patients' home or any spot, for treating an illness or injury.
- Ambulatory care is evolving as the real and focal policies and procedures serve as operational guidelines for providing proficient care to patients.
- First Aid is the most important branch of the medical science. It is both an art and a science. First aid is the immediate and temporary care given to the victim of an accident or in case of a sudden illness to a person.
- In order to examine and diagnose properly, taking the information of the casualty's history and that of the incident, symptom, signs, and level of responsiveness is crucial.
- A general examination of the casualty should be carried out quickly to discern any imminent threats to their life, to check whether the casualty is in state of conscious or not.
- The casualty should always be accompanied by a brief written report, when they finally leave a first aider's care.

6.12 KEY WORDS

- **Obstetric Complication:** An acute condition arising from a direct cause of maternal death, such as antepartum or postpartum haemorrhage, obstructed labour, postpartum sepsis, complications of abortion
- **Diabetic Hypoglycemia:** Diabetic hypoglycemia is a low blood glucose level occurring in a person with diabetes mellitus.
- **Haemorrhage:** An escape of blood from a ruptured blood vessel
- **Orthopaedic:** Relating to the branch of medicine dealing with the correction of deformities of bones or muscles
- **Catchment Area:** The area from which a hospital's patients or school's pupils are drawn

- **Nosocomial Infection:** A Hospital-Acquired Infection (HAI) or a nosocomial infection that is acquired in a hospital or other health care facility
- **Medullation:** The formation of a medullary sheath or medulla

Emergency Services

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6.13 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. What are the aims and functions of an emergency department?
2. Write a short note on the priorities of an emergency department.
3. What are the golden rules of first aid?
4. Why is it important to check the level of responsiveness of a casualty?
5. What kind of crises can be faced by an emergency department?

Long Answer Questions

1. Explain the components of emergency services.
2. Describe the actions required by a first aider in different situations.
3. Discuss the complete procedure that should be performed by a first aider during examination.
4. Elucidate the duties performed by staff in emergency services.
5. Discuss the importance of ambulatory services in detail.

6.14 FURTHER READINGS

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UNIT 7 SERVICES OFFERED BY THE EMERGENCY DEPARTMENT

Structure

- 7.0 Introduction
- 7.1 Objectives
- 7.2 Meaning, Importance and Various Services Related to Emergency
 - 7.2.1 Structure
 - 7.2.2 Organisation
 - 7.2.3 Clinical Facilities
 - 7.2.4 Disaster Preparedness
 - 7.2.5 Legal Concerns
 - 7.2.6 Clinical Laboratories: Rules and Regulations
- 7.3 Answers to Check Your Progress Questions
- 7.4 Summary
- 7.5 Key Words
- 7.6 Self Assessment Questions and Exercises
- 7.7 Further Readings

7.0 INTRODUCTION

It can be assertively said that the emergency department and their services are in great demand these days. But the question arises why that is happening. There is a belief amidst the public that for any illness or injury, or whatever kind of a critical condition, the emergency department provide necessary services and treatment, irrespective of the time. People with conditions and cases that normal physicians cannot handle, now approach the emergency departments. Most of the time, all the accident-related cases are brought directly to the emergency department. Consequently, the department's demand is increasing day by day.

However, the word, 'emergency', can often attain different meaning through different perspectives of individuals. It might be and most of the times, a life endangering situation involving physical damage, wound, or any illness, which occurs suddenly and needs instant care and attention. But emergency is not defined properly. For example, for a certain person, his/her inability to contact his family physician at night to treat his constipation of two days may be an emergency. Therefore, the emergency department has to provide an ample range of amenities and facilities for all kinds of situations, be it for a fundamental first aid, a total urbane management of surgical and medical emergencies, a full-scale trauma care, or any kind of general outpatient provisions. Now a days when the outpatient facilities are not available, the treatment is given at the emergency ward, thus it has become an informal walk-in treatment area.

The hospital planners need to identify the changes that should be made in the aspects of ideas, facilities, and work in accordance with the increasing demand of the emergency department. They also need to introduce flexibility in the basic structural designing of the emergency department with respect to the area, personnel, supplies, machines, and equipment to manage cases cautiously as well as competently. If proper materials are not allotted, the department will not function properly and will become a burden even in the best available conditions. Sustaining all day round emergency services is costly because of its high maintenance charges, which remain fixed and its low usage.

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7.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the importance of the emergency services
- Analyse the various services provided by the emergency services
- Discuss the structure and maintenance of the emergency services

7.2 MEANING, IMPORTANCE AND VARIOUS SERVICES RELATED TO EMERGENCY

As discussed in the previous unit, emergency services holds significance for general public. Its level of requirement is reflected in the increasing burden on this department. Thus, it becomes crucial to focus on its aspects like structure, organisation, legal matters, etc. for the smooth functioning of the department.

7.2.1 Structure

Entrance of the emergency department should be separate and away from the hospital's common entrance. The entrance must be visible from distance, so it should be marked and highlighted with sign boards and lights. It should be mapped on the ground floor for easy and quick accessibility. Moreover, it should be in close proximity to admission, medicine, and accounts departments. From different studies, it is found that over 40 percent of the patients require X-ray in the emergency department. However, the usage of portable X-ray units is not acceptable, so choosing a location with close proximity to the radiology department is a better idea. Because of the heavy demand of blood in the emergency department for transfusion, blood bank and other biochemical services department, should be near the emergency ward.

The strategic location of radiology department and other needed laboratories, with respect to the emergency ward, can reduce the operational cost and the time spent in the travel from one point to another. Department should be equipped with elevators for convenience and quick transfer of patients to and fro. The way to the

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casualty ward should be well shielded, to protect patients from weather. Paved entrances and adequate parking space should also be constructed for the vehicles of patients and staffs. The entrance for ambulance should be built in such a way so as to permit the entry of two or more ambulances at a time. Ramps need to be built for the easy access by the patients.

Since, there will be relatives present along with the injured patients and accident victims, handling the movement of people and the traffic should be on priority. The design should facilitate good public relations. Unacceptable provisions, overcrowding, interruptions, and inefficient operation due to defective design will present a poor image of the hospital in front of the public.

7.2.2 Organisation

Since the emergency services range from mild to extreme cases, not all hospitals can provide all the services. First aid help should be made available for emergency situations at every hospital. Having an emergency care and not having necessary facilities is as much of a danger as having an emergency department with all the facilities, but without proper equipment and competent staff available at all hours. Both these aspects should be in perfect balance, so that the department can function properly.

As explained by the ‘American College of Surgeons’ via the committee on trauma, the following are the prerequisites of regimented care for traumatic conditions in an emergency care department:

- ‘A well-organised ambulance service department, which is well resourced with competent professional’
- ‘A well-equipped casualty room for operating individuals’
- ‘A small recovery room’
- ‘An experienced physician, skilled nurse, an attendant on call’
- ‘Skilled and experienced surgeons of various fields to handle any kind of emergency’
- ‘Proper diagnostic and therapeutic services under the supervision of experienced personnel’
- ‘A well organised and documented medical record is made for every patient, and includes records of all injuries, lab testing, medication, and treatment.’

Administrative and public areas

- Reception serves the purpose of maintenance and surveillance of the admissions, entrance area for ambulance, and waiting area. Intercommunication facility is setup in this area, and the major work of coordination and classification is done here during any disaster or emergency condition.

- The receptionist at the reception area should be able to see the happenings of the emergency section, but other patients and their relatives should be kept away from that area.
- There should be enough space for keeping wheelchairs and stretchers, but away from the main walking and traffic area. Stretchers sometimes serve as examination tables in an emergency and should, therefore, be well built and provided with wheel locks.
- Clean toilets, drinking water, vending machines, and public telephones should be provided in hospital waiting areas.
- Proper area should be designated for officials and attendants.
- Night guard's office should be near the emergency department.
- Coffee/Tea and snacks counter should be in close proximity with the emergency department.

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Cardiopulmonary resuscitation is another critical area, where the emergency personnel are busy and active. 'Code Blue' is a hospital term to declare an emergency of crucial importance like a cardiac arrest. A team of skilled personnel are already selected and appointed to instantly respond in such situations. There are circumstances, when the fixed team is already busy, handling a code blue case. During those times, staff outside the department can also be instructed to cope with the situation so that lives can be saved.

7.2.3 Clinical Facilities

In the emergency medical department, the four major useful areas are:

- Trauma and Injury Unit, where primary care for all serious surgical cases are provided
- Examining area for medical purposes
- For cases involving orthopaedic conditions, proper casting and splintage area is built
- Beds for the patients, who are to be kept under observation

The facilities required by the emergency department are:

- All emergency departments must have a Trauma room. This is for attending to the serious cases.
- To serve properly, there are rescue and equipment for life support such as:
 - i. Drugs or medicines
 - ii. Oxygen and other medical gas outlets for vacuum and compressed air
 - iii. Examination table and lights
 - iv. Illuminators used for viewing X-ray reports
 - v. Supply shelves, along with cabinets

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- vi. Treatment rooms with adequate tables, lights, counters for working, drawers, sinks, and store room for medicines
- vii. Nurses on call for the trauma and orthopaedic room.
- viii. Availability of additional space for disaster handling
- ix. Sufficient working area for staff
- x. For certain equipment which need to be easily accessible, there must be a proper storage area.
- xi. Clean rooms
- xii. Washrooms and lavatories
- xiii. Caretaker's room
- xiv. Refreshment area for all the staff and employees
- xv. Safekeeping place for the assets of staffs

7.2.4 Disaster Preparedness

Every hospital has a disaster committee, so that when a case of disaster occurs or they are about to face a disaster, proper safety measures can be taken. Let's take an example of a flood in an area, which is resulting in a lot of emergency cases along with the immediate danger to the building. At that time the emergency department, along with the disaster committee, should coordinate and work towards facing and overcoming the disaster.

The emergency personnel and others, who partake in disaster managing procedures are competent to identify the advanced and relative severity of a patient's state. There is a triage area in hospital, where the patients are first brought and then are sorted according to the degree of their emergency and are finally sent to the proper treatment areas. In a matter of life and death, patients are directly sent to the emergency room or department, extreme casualties are sent to operation room, and the non-life threatening cases to out-patient departments. By doing so, over-crowding can be solved to a certain degree. Priorities are based on the degree to which the patient's life is threatened, as already discussed in previous units.

7.2.5 Legal Concerns

The emergency process handbook must include rules and regulations for legal cases. All the staff associated with the emergency department should be aware of the legal facets of the emergency services. In circumstances, concerning accidents and such, there are a lot of laws objectified with the detection and proper action, which should be taken by the police and other government based agencies. The doctors are obliged to adhere to all those laws and to testify if the need arise.

To avoid any kind of malpractice or medical negligence, the physicians and nurses must be well-aware of the consumer protection forums and Consumer Protection Act. They are responsible for their career and the respect of the hospital. Fool proof documents of all medical records should also be maintained, along with a proper log of patients.

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7.2.6 Clinical Laboratories: Rules and Regulations

The level of healthcare services defines the level of structural and functional availability of a clinical laboratory. Every lab is not provisioned with equipment and services for performing complex procedures and studies. The ethical guidelines for biomedical research on human subjects in the year 2000 was launched by The Indian Council of Medical Research (ICMR), and was revised in 2006, because of the advancement in the potential for clinical research. Other guidelines like the Indian Good Clinical Practices (GCP), were put forward by Central Drugs Standard Control Organisation (CDSCO) in 2002, for guiding the biomedical research in our country.

Indian public health care system being 3-tier, the laboratory services in India are also 3-tier.

Objectives of each laboratory is to recognise the possibility, services, and potential of the services given by it, along with the needed structural requisites, keeping safety measures at the top most level of importance. The basic arrangement services contain:

- Lobby or reception area
- Room for specimen collected
- Lavatories
- Good water supply
- Continuous electricity
- Investigative working area
- Proper cold storage area for samples
- Record keeping room
- Space for glass equipment cleaning and sterilisation
- Competent waste removal system, including biomedical wastes
- Gear for fire-safety
- Adequate arrangements for ventilation and lighting
- Room for meeting and discussions
- Leisure area for staff members
- Services and equipment for communication
- Specimen and samples transportation

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Personnel, training, and development

In order to serve the emergency department efficiently, the clinical laboratory should have trained and efficient staff members.

- There should be a Head of the Laboratory, who will be responsible for all the activities and studies, carried out in the lab.
- The employed staff should be able to work under pressure and perform all the necessary and needed activities, without any difficulty.
- The staff should have a clear cut idea of what they are doing in the lab. All instructions must be given clearly. They should also be ready for any kind of emergencies.
- The skills of the staff should be periodically updated and the authorities should plan a program to achieve this goal.
- An annual assessment of the staffs' performance should also be done, depending upon the rules and regulations of respective laboratories.
- The lab should keep records of all staff members. Records should contain information about:
 - i. Data on qualifications and work experiences
 - ii. Photocopies of certificates with proper state authority registration
 - iii. Photocopy of appointment letter
 - iv. Fully verified health certificate
 - v. Appraisal of performance
 - vi. All awards and certificates of recognition
 - vii. Any disciplinary action taken by the government
 - viii. Reference letter

Safety measures

The people or as we say, staff, working in the laboratories are open to various risks by chemicals, infection causing materials, hazards by fire, etc. The environment will also be at risk due to the presence of hazardous materials in the laboratory. So, safety measures should be given priority. General safety measures according to the ICMR are:

- Laboratory safety procedures and policies should be documented properly.
- People working in laboratory should be well educated about all the safety policies and procedures.
- All hazardous materials used in the lab must be enlisted and accounted for and should be prepared.
- All the lab staff members should follow hygienic practices.

- Sink must be attached with portable eye wash facility.
- The containers having hazardous material should have biohazard symbol or label on them.
- All the specimen collected should be under proper supervision and should also be properly preserved.
- Adhering to the procedures, proper disposal of dangerous and harmful materials must be done.
- All the staff members must be trained in handling emergencies.
- Make sure there are sufficient fire extinguishers present in the lab.
- One must ensure timely checking of all safety equipment.

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Check Your Progress

1. What are the advantages of the strategic location of radiology department and other laboratories, with respect to the emergency ward?
2. Where does the major work of coordination and classification happen?
3. Why did the guidelines like the Indian Good Clinical Practices (GCP), were put forward?
4. Mention any five basic arrangement required in a clinical laboratory.

7.3 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The advantages of the strategic location of radiology department and other laboratories, with respect to the emergency ward are reduction in the operational cost and the time spent in the travel from one point to another.
2. The major work of coordination and classification happens in a reception area of a hospital.
3. The guidelines like the Indian Good Clinical Practices (GCP), were put forward by Central Drugs Standard Control Organisation (CDSCO) in 2002, for guiding the biomedical research in our country.
4. The basic arrangement required in a clinical laboratory are:
 - a. Lobby or reception area
 - b. Room for specimen collected
 - c. Lavatories
 - d. Good water supply
 - e. Continuous electricity

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7.4 SUMMARY

- It can be assertively said that the emergency department and their services are in great demand these days.
- The hospital planners need to identify the changes that should be made in the aspects of ideas, facilities, and work in accordance with the increasing demand of the emergency department.
- The strategic location of radiology department and other needed laboratories, with respect to the emergency ward, can reduce the operational cost and the time spent in the travel from one point to another.
- The design should facilitate good public relations. Unacceptable provisions, overcrowding, interruptions, and inefficient operation due to defective design will present a poor image of the hospital in front of the public.
- Since the emergency services range from mild to extreme cases, not all hospitals can provide all the services. First aid help should be made available for emergency situations at every hospital.
- Every hospital has a disaster committee, so that when a case of disaster occurs or they are about to face a disaster, proper safety measures can be taken.
- The emergency process handbook must include rules and regulations for legal cases. All the staff associated with the emergency department should be aware of the legal facets of the emergency services.
- In order to serve the emergency department efficiently, the clinical laboratory should have trained and efficient staff members.
- The people or as we say, staff, working in the laboratories are open to various risks by chemicals, infection causing materials, hazards by fire, etc. So, safety measures should be given priority.

7.5 KEY WORDS

- **Cardiopulmonary Resuscitation:** An emergency procedure that combines chest compressions often with artificial ventilation in an effort to manually preserve intact brain function until further measures can be taken.
- **Consumer Protection Act:** An act of the Parliament of India enacted in 1986 to protect the interest of consumers in India.

7.6 SELF ASSESSMENT QUESTIONS AND EXERCISES

*Services Offered by the
Emergency Department*

Short Answer Questions

1. Write a short note on the structure of the emergency department.
2. What is 'Code Blue'?
3. Why is it important to be aware of legal aspects for emergency department staff?

Long Answer Questions

1. Discuss the organisation of the emergency department.
2. Describe the various aspects of clinical laboratories in detail.
3. Explain the clinical facilities required by the emergency ward.

7.7 FURTHER READINGS

Llewelyn, Davies R; Macaulay, H.M.C. 1966. *Hospital Planning and Administration*. Geneva: W.H.O, Jaypee Brothers.

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UNIT 8 LABORATORIES FOR INVESTIGATION

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Structure

- 8.0 Introduction
- 8.1 Objectives
- 8.2 Types of Laboratories
- 8.3 General Safety Measures
- 8.4 Rules and Code of Conduct in Case of Emergency
 - 8.4.1 Housekeeping and General Rules for Safety in Lab
 - 8.4.2 Waste Management
 - 8.4.3 Laboratory Workers
- 8.5 Answers to Check Your Progress Questions
- 8.6 Summary
- 8.7 Key Words
- 8.8 Self Assessment Questions and Exercises
- 8.9 Further Readings

8.0 INTRODUCTION

A laboratory is a facility or a building or even a part of a building (e.g. Schools, Universities, Industries) with appropriately controlled conditions and safety standards, equipped with certain scientific tools to conduct vast types of experiments, tests, researches, investigations, manufacturing of various sensitive drugs, medicines and chemicals, and to teach students the science behind numerous reactions and processes.



Fig. 8.1 A Typical Laboratory

Wilhelm Wundt, who was then, a famous doctor and psychologist, was the one to create the world's first ever experimental laboratory in the year 1879, at the University of Leipzig in Germany. However, the earliest scientific laboratory according to the evidence is a home laboratory of Pythagoras of Samos, a Greek philosopher and scientist. This laboratory conducted experiments about the tones of sound and vibration of string, and was thus a more of a physics-oriented laboratory.

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Fig. 8.2 Wilhelm Wundt

In India, The National Physical Laboratory is one of the earliest national laboratories set up in 1900 and Dr. K. S. Krishnan was the first Director of the laboratory.

8.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the types and rules of a laboratory
- Analyse the safety measures required in a laboratory
- Discuss the duties and working environment of staff

8.2 TYPES OF LABORATORIES

Laboratories have specific needs, purposes, and associated risks. Some of those requirements are specific to where it is located, some are with respect to its purpose (e.g. pharmaceutical, chemical), or to an activity (e.g. small volume manufacture of high potent products, work with biological agents). The different kinds of laboratories are:

- I. Laboratories for analysing and quality control:** Here, products and materials are tested against set standards and the absence of impurities in

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the products are ensured. These laboratories form an essential component within the production and the supply chain.

- II. Biosafety laboratories:** The biosafety laboratories are utilised to detect potentially harmful biological agents. This is now achieved through a thoughtful combination of methods, facilities, and equipment. The levels of containment go from BSL1 to the highest level of BSL4.
- III. Clinical and medical laboratories:** These contain appropriate equipment for testing tissue, blood, and other kinds of fluid samples. They are subdivided into various processes such as pathology, serology, histology, virology, bacteriology, and molecular biology
- IV. Incubator laboratories:** These consist of incubators to protect the microbial cultures from the environment. Laboratories conducting microbiological and cell or tissue culture work require control. Parameters of temperature, humidity, and oxygen, and carbon dioxide levels need to be controlled.
- V. Production laboratories:** All kinds of productions for clinical trial and pilot productions, forms a category of their own. Such laboratories can be found in the pharmaceutical, biotech, and the science and technology sectors.
- VI. Research & Development (R&D) laboratories:** These laboratories are especially equipped with material that can aid in research work and development.

8.3 GENERAL SAFETY MEASURES

The following are the rules that must be a part of manual followed by all the laboratories. They include what must be done if an accident or emergency occurs, safety measures, and how they must be handled, and basic common-sense rules. It is important to read all manuals, precautions, signs, symbols, fire alarm system, and follow safety precautions as guided in case of any accidents in the lab. The pathways in the lab must be kept clear and all items not required in the lab must be placed outside or on shelves in the corner. All labs must have smoke detectors and fire extinguishers, in case of fires. One must always work on the desks or area provided to them and place all other belongings either outside the lab or on shelves provided in the laboratory. One must know the location of laboratory's safety aid, which must contain eye washers, safety kits, blankets for covering the injured, stretchers bandages, wheelchairs, first-aids (scissors, savlon, bandage, cotton, ice pads, etc.), etc.

The windows in the lab must always be kept open for ventilation. One must never enter the lab without a lab coat and other appropriate clothing that may be necessary. If one isn't aware of handling of certain equipment, he/she must always ask someone who may be familiar with it (Lab assistant, Lab Teachers, etc.). Lab areas containing biohazard substances, all kinds of radioactive elements, or even lasers must be marked and covered well with warning signs to prevent exposure

to an unauthorised individual. Any flames in the lab must not be used without permission from a qualified supervisor. A gap of 36" must be kept around the fire sprinklers. One must know, where the fire alarms are placed in the lab and how to use it in case of an emergency. One must never drink or eat while working in the lab or even bring any food inside the lab. Laboratory glassware and utensils should never be utilised to eat food even if they are well-washed. While using laboratory equipment, if any damage like cracks is seen on it or if it isn't working properly, it must be recorded and must not be kept for use anymore.

The complete knowledge of lab equipment is necessary before using it. The last person to leave the lab holds responsibility for the final closing of the lab, including turning of lights and fans, ensuring all burners have been turned off, all bottles containing chemicals are closed well and all equipment are well washed and placed back in the correct order.

Never leave an ongoing experiment unattended or until it is fully completed. One must always hold all glassware, solutions, and apparatus up to eye level and never beyond it to prevent it from spilling on the face. Chemical substances must never be smelt or tasted directly by mouth. All labs must have dustbins to discard any waste, whether dry or wet and the staff should avoid littering the floor or working area. One must not pipette or perform any experiments using mouth, as the chemicals may enter the food pipe causing damage to the organs. Lab waste must be disposed of correctly with full knowledge about its disposal.

Any injuries, burns, or accidents must be reported about and addressed immediately, even if it may seem minor. In case of a chemical, splashed onto the eyes or skin, splash cold water onto the area for about 10 minutes. All unsafe conditions in the lab must be reported to the supervisor. The workstation must only contain those items which are to be used at that time.

One should never make too much noise inside the lab, use the phone, or play games inside. A laboratory must be treated like a library and must be kept well organised. All laboratories must document all maintenance, service records that were planned, all equipment utilised, temperature and humidity conditions in the laboratory, any accidents or breakage that might have taken place, etc.

Housekeeping safety rules

Laboratory housekeeping deals with the basic cleanliness, tidiness, hygiene, and health in the laboratory. All work stations must be clean and tidy at all times and also well sanitised and moisture and dust free. The floor and walls must be cleaned from time to time. All fire extinguishers and exits must be easily accessible at all times. Only those equipment and materials required for a particular work must be kept in the working area. Everything else should be stored safely elsewhere. Before using an equipment for an experiment as well as after using, ensure that it's well washed and dried. According to the rule, all light items must be stored on top and the heavier ones at the bottom or stored in a manner that it's easily accessible.

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Dress code safety rules

As expected, lab dress codes are set to avoid wearing those items that may cause any hazards and to establish a uniformity in the lab, so as to prevent accidents or injuries inside the lab, while working with the equipment, substances, etc. Some of the basic dress code are:

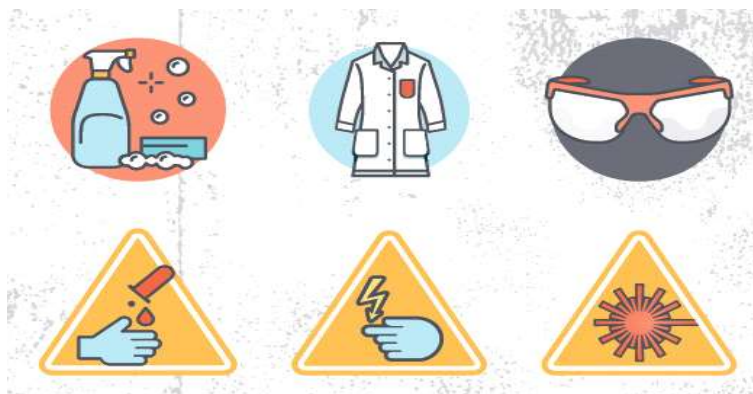


Fig. 8.3 Laboratory Rules

- Never wear any loose, dangling clothes inside the lab
- Hair must always be tied inside the lab and if possible wear head caps
- Any fancy clothing items like jackets, jewellery, boots, etc. must not be worn inside the lab and be removed before entering it.
- One must never enter a lab bare-foot or in improper footwear (Slippers, sandals, heels, etc.). Foot must always be covered inside the lab.
- Never wear any short bottoms inside the lab
- Hands must be well sanitised after working with chemicals and if gloves were used, they must be sanitised as well. These include the basic rules for protection from various hazards as well as basic hygiene rules to follow to avoid any sort of contamination.
- When working with hazardous materials, glassware, acids, heat, etc. one must wear safety glasses.
- Working with long nails inside the lab is not allowed.
- Never touch eyes, mouth, or face after or during handling of any chemicals during an experiment or after performing it, until you have washed your hands. Hand gloves are a must while handling chemical substances.
- Lab coat must be worn inside the lab without fail.
- Before leaving the lab, always wash your hands thoroughly with soap and water.

Chemical safety rules

These rules are a must in the laboratory. These rules help avoid any accidents, spillage or damage to the laboratory. These rules also direct the procedure to follow in case of a chemical hazard, leak, etc. This ensures that the chemical is washed off properly and injuries are avoided.

- Do not allow any solvent, solution, substance, etc. to come into contact with your skin. Always measure and use the exact quantity of chemicals required for an experiment.
- Never leave the chemicals exposed and always put it in an airtight bottle or container. All chemicals should have a label attached to it along with its concentration, the receiving date, expiry date, etc.
- Before using any chemical or acid, read the label twice to ensure it is the right one with the right concentration.
- Discard the unused leftovers and never place it back into the containers.
- Chemicals placed in the lab must not be taken out of the lab or used for other purposes. In case of a chemical spill, it must be cleaned soon before it can cause any harmful effects.
- Transferring, distilling, pipetting of chemicals, etc. must be done with extreme caution.
- Never pour back the chemicals into its container, if used or even unused. Just discard it.
- Water must never be mixed or added directly into concentrated acid.

Electrical safety rules

Laboratories contain electronic equipment which must be well handled in specific manners to prevent any shocks, misuse of electricity, wiring mistakes, high voltages, etc. Electrically powered equipment found in the laboratory consist of vacuum pumps, lasers, power and electrochemical apparatus, x-rays, high voltage devices, generators, LED Devices, Voltmeters, hot plates, mantles, microwave ovens, and ultrasonicators. Shocks are the major hazard in the lab. Before handling any equipment, one must know the proper use of the appliances and its precautions and safety guidelines. All electrical equipment must be well insulated. It is important to wear insulating rubber gloves when working with electrical equipment. Electrical equipment and power supplies must be well isolated from the working area. It is better to avoid use of the plastic equipment in the laboratory, as they do not work under vacuum and certain other conditions.

All power sources to the equipment must be shut down immediately and must be unplugged from the source using safety gloves, if water or any other substance is spilled on it. No failed or damaged electrical appliances must be used. In case of ovens and refrigerators, use the ones, which are specifically designed

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for laboratory, as the domestic ones may not have the appropriate features. Avoid overheating the samples and never work with open oven or fridge doors for too long. Microwave heating presents several potential hazards such as rapid temperature and pressure rise, liquid super-heating, arcing, and microwave leakage. To avoid explosion, do not heat sealed containers in the oven. One should wear insulated gloves, while handling electrical equipment. All high voltage devices with 50 V D.C. must only be handled with permission and authority. Always turn off a high voltage power supply, when you are plugging it. All adjustments, repairs, wiring, etc. of electrical equipment must be performed only by a qualified individual. No voltage experiments must be manipulated in any way. Electrical panels must be installed in such a manner that it's easily accessible. Most electrical instruments emit harmful radiation like photochemical, high intensity light, sound waves, etc. so special precautions must be taken.

Compressed gas cylinders

Always check cylinders, connections, and hoses regularly for leaks. Flammable gas leak detector is an effective option to keep a check on leaks. Cylinders must always be located in well-ventilated areas. The guidelines with regards to gas cylinders in the lab include:

- Paying attention to the right type, quality, and size as per needs.
- The cylinders must be properly transported and stored in a safe area.
- All pipe linings must be attached and connected well, and must be checked regularly.
- An empty cylinder must be returned back to the company, from where it was brought.

Laser safety rules

Although not used in all labs, it is important for labs that use lasers to follow strict guidelines. Everyone working in such labs must be well versed with all hazards associated with it, correct handling of the equipment, and the right apparel and gears to protect themselves from it. Appropriate laser safe goggles must be worn and the upper body must be protected from all types of laser beams (including the eyes, chest, nose, brain). One should never walk through any laser beams as it can be harmful.

Check Your Progress

1. Which is the earliest known laboratory in India?
2. Why must all solutions and apparatus be kept up to eye level?
3. State any three dress codes for laboratory.
4. What kind of hazards are presented by microwave heating?

8.4 RULES AND CODE OF CONDUCT IN CASE OF EMERGENCY

The following general emergency procedures are recommended in the event of a fire, explosion, spill, or any other laboratory accident. All labs must have post numbers to call in during emergencies. All emergencies must be reported in a detailed manner, with how, when, and the nature of the accident, along with the severity of the injury (if any). In such situation, some rules to follow are:

- Never enter or re-enter an unsafe area
- Always help others in the lab and warn them regarding the risk
- Always provide immediate first aid without any delay; small measures including washing under a safety shower and special first-aid should be immediately provided
- Small fires can be put out using a portable extinguisher. Seek fire department in case of large fires
- Medical help must be given immediately in case of any accidents
- Everyone should be informed about the nature of the hazard and how to deal with it. Protection of life and health in a calm environment must be the motto of any laboratory. If possible, one person must remain with the injured person. He/she mustn't be left alone till complete medical help is provided.
- If harmful solutions have been spilled, remove the chemicals, remove the clothing and wash them under running water till proper first aid can be given. If any chemical or acid gets into the eye, wash the eye immediately with cold water for fifteen minutes, and in case of the skin, wash with water showers or place ice packs on it, if it's a minor burn.
- For all experiments conducted, the Standard Operating Procedures (SOPs) are necessary for maintaining a consistent performance and comparison. Guidelines for the SOP's must be given in a format that is clear and concise, and can be clearly understood. It should be followed strictly for accurate and desired results. Quality control (QC) must always be conducted for all products used in the laboratory to detect any minor and major errors and any changes that occur over time to any substances, equipment, etc. thus, ensuring the accuracy or validity of the test results.

8.4.1 Housekeeping and General Rules for Safety in Lab

We have seen certain safety rules related to housekeeping in the previous section, you will learn about certain rules here. Housekeeping generally refers to the management of duties like cleaning, cooking, home maintenance, laundry, etc. Good housekeeping in laboratories may reduce the risk of contamination and protects the integrity of biological experiments. Laboratory personnel are responsible for the cleaning of the benches, floor, equipment, and surrounding

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areas that require specialised technical knowledge. In case of housekeeping, the major role is played by technical knowledge.

General rules

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- Floors should be clean or free from any hazards.
- Never leave any discarded objects, dropped objects, and/or any spilled materials on the floor.
- Materials (tablets, chemicals, floors aisles, and desks), which are not being used at that time, should be kept aside and cleared.
- Exits passageway must be cleaned and there should be at least two passageways to avoid any sort of inconvenience
- Spaces near any shower, eyewashes, fire extinguishers, and electrical controls must be cleaned.
- Drainage problems should be taken care of as sink traps and floor drain traps may be filled with water, preventing the hidden sewer gases from leaving the laboratories.
- After completion of any experiment, the area or room must be cleaned.
- Labs must be cleaned at the end of each day to maintain cleanliness.
- Contamination is very common in the corners and edges of any materials, which may cause incorrect results. So, the edges and top of the benches should be cleaned properly.
- The passageways must be devoid of any obstructions.
- The floors should be free from any sort of liquid to avoid risk of slips and falls.
- The doors and drawers must be closed.
- The chemicals and drugs should be kept safe and packed tightly.
- Bio-hazards and radioactive waste must be disposed of in the specifically labelled container
- All the compressed gas cylinders should be secured properly.
- Appropriate disinfectants should be used
- Cloth covered chairs and stools shouldn't be used in the laboratories.

Storage

- Sharp and pointed tools or materials should be covered properly
- Tools must be placed in a drawer, with labels on them, in order to find it easily.
- Equipment shouldn't be draped over benches
- Clothing should be placed at proper locations

- The equipment, which are not being used very often, should be placed in the storage area.
- Chemical containers shouldn't be placed on the floors.
- Cardboard boxes, equipment boxes, and other materials shouldn't be kept under the shelves, benches, or other cabinets in the laboratory.

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Laboratory safety

The laboratory is a hazardous place to work. Laboratory workers are exposed towards number of hazards such as biological, chemical, physical, and radioactive.

Precaution regarding fire and explosions are:

- Flames should not be left open
- Gas cylinders should be checked for any leakage
- Smoking should be strictly prohibited
- Burning matchsticks should not be thrown in the waste bin
- The location of following equipment should be known by every person in the laboratories:
 - (i) A fire extinguisher
 - (ii) Fire blankets
 - (iii) Sand bucket with scoop

It is very common in a laboratory for chemicals to react in a dangerous manner, if any two incompatible chemicals come into contact with each other. There are some examples which are as follows:

- (a) **Acetic acid:** Incompatible with nitric acid, hydroxyl compounds, permanganates, ethylene glycol, peroxide, and chromium acid.
- (b) **Acetone:** Incompatible with sulphuric acid and nitric acid mixtures.
- (c) **Alkaline metals such as calcium, potassium and sodium:** Incompatible with water and other chlorination hydrocarbons.
- (d) **Chlorine:** Incompatible with ammonia, hydrogen, benzene, and other finely divided metals.
- (e) **Chromic acid:** Incompatible with alcohol, acetic acid, and other liquids, which can cause flames.
- (f) **Copper:** Incompatible with acetylene, azides, and hydrogen peroxide.
- (g) **Cyanides:** Incompatible with acids and alkalis.
- (h) **Flammable liquid:** Incompatible with chromic acid, hydrogen peroxide, nitric acid, ammonium nitrate, and halogens.
- (i) **Hydrogen peroxide:** Incompatible with copper, iron, chromium, and most other metals.

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- (j) **Iodine:** Incompatible with acetylene and ammonia.
- (k) **Oxalic acid:** Incompatible with silver and mercury.
- (l) **Sodium azide:** Incompatible with copper, lead, and other general metals.
- (m) **Sulphuric acid:** Incompatible with permanganates, chlorates, water, and perchlorates.

Precautionary measures against fire

In case of fire, there are some precautionary measures, which should be followed by every lab:

- In case of a small blaze, water, sand, and a fire blanket should be used to put out the fire.
- A fire extinguisher should be used for a larger blaze.
- Water should not be used in case of electrical fire.
- A carbon-dioxide fire extinguisher should be used, after switching off the main electrical switch.
- Sand should be used to smother the fire.
- Water shouldn't be used on a fire caused by organic solvents for e.g. ether, alcohol, petrol, oil, or grease.

Table 8.1 Fire Extinguishers Types and Uses

Type	Used for	Don't use for
Water (H ₂ O)	Wood, paper, and fabric	Alkali metals, flammable liquids, burning metals, and electrical fires.
Carbon dioxide (CO ₂) powder	Electrical fires, flammable liquids and gases	-
Dry powder	Electrical fires, flammable liquids and gases	-
Foam	Flammable liquids	Electrical fires

The common causes of fire in the laboratory are:

- Electrical overloading
- Poor electrical maintenance
- Switched on equipment unnecessarily
- Deterioration of gas tubing
- Naked flames
- Misuse of matches
- Burning matchstick, left in the waste materials
- Explosive chemicals stored in an ordinary refrigerator.

Precautionary measures against infections

There are some precautionary measures, which should be followed by every lab to prevent infections:

- Use push-button pipettes and dispensers in the place of mouth pipettes
- Never blow out by the pipettes, which contain infectious material.
- Protective apparel should be used like disposable gloves, goggles, masks, laboratory coats, etc.
- Whenever gloves are changed, wash hands thoroughly
- A proper needle disposal system should be established to dispose of any needle and sharp objects, after use.
- Protection of face should be a priority because of the presence of some harmful chemicals and other fluids like strong acid or alkalis.
- Hands should be kept away from nose, eyes, and other mucus membranes.
- Clean all surfaces and other reusable devices after use, with disinfectants.
- Discarded tube or any infected material should not be left on the surface of table, shelf, or anywhere in open
- Use face masks and rubber gloves during cleaning of any freezing compartments of the refrigerator
- Personal protective equipment should be worn properly as they can protect one from any infectious materials from passing through the clothes, skin, and undergarments.

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8.4.2 Waste Management

Waste management is necessary for pathology laboratories because of state and federal regulations that control and restrict the disposal of waste generated by all medical facilities. There are some rules, which are to be followed in the disposal of waste in the laboratories:

- Medical wastes should be separated carefully.
- Medical wastes should include infectious wastes such as human blood, body fluids, and discarded microbiological cultures.
- Sharp and broken glasses should be discarded
- Blood collection materials such as syringes, giving and taking sets, and ESR tubes should be disposed of carefully.
- Garbage and trash should include everything except hazardous and medical wastes.
- Corrosive, toxic, carcinogens, ignitable, and other chemicals such as explosive should be included in hazardous wastes.

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The standard waste disposal procedure is:

- Take a puncture proof rigid container and place all the discarded sharps in it.
- Medical and hazardous wastes should be separated from the trash and garbage.
- Discard all the liquid wastes into the sewage system, except those containing heavy metals.
- Waste papers, cartons, thermocol, plastic, and other trash and garbage should be discarded into black-coloured plastic bags.
- Liquid radioactive wastes should be stored for three months before disposal and decontamination.
- All microbiological waste with plates, containers, tubes should be autoclaved before discarding.
- Containers and disposable specimens should be autoclaved in the manure pit.
- Cerebrospinal fluid, decant blood, plasma, serum, etc. should be discarded in any stainless steel kettles.
- Flush urine and stools in toilet and 5% hypochlorite solution should be placed in the reusable containers.

8.4.3 Laboratory Workers

A laboratory worker plays a major role in the safety and conduct of any laboratory. Without their cooperation, it is impossible to have a properly working laboratory. Their responsibilities include:

- The workers play a very major role in finding out the cause of the diseases, for which a physician requires the laboratory test results.
- A laboratory worker should help the patient to recover by finding and providing them accurate results of their tests, requested by the physician.
- The laboratory workers should not offer any personal excuse for delaying and for shortcoming in their performance.
- Any kind of delaying or errors should be reported directly to the superior, without any fear, so that they can help or rectify it as soon as possible.
- Always be honest because wrong results or information may lead to patient's death.
- Professional behaviour and high morals should be maintained by the laboratory workers.
- The information must be kept confidential.
- The results and information about any patient should not be disclosed without taking permission of a physician.

- The physician, who has requested the tests, should be the only one to receive the results.
- The laboratory technician should take care of the equipment carefully.
- It is necessary to keep all the reports ready on time.

The factors that reduce productivity of any laboratory are:

- Poor hygiene
- Short staff
- Poor maintenance of equipment
- Excess of overtime
- Poor communication
- Low salary
- Unsatisfactory work environment
- Poor instrument maintenance
- Low test volume
- Over staff
- Poor inventory
- Inadequate instrument
- Unfair policies of administration
- Unionisation
- Non-testing activities

The factors, which increase the productivity of any laboratory are:

- Better work environment
- Improved facilities
- Improved computerisation
- Adequate staff
- Salary on time, with incentives
- High performance standards
- Employee counselling
- Positive environment by the employees
- Proper facilities
- Maintenance of good hygiene
- Fragrance must be good and in moderation
- Good communication
- Directions must be clearly visible
- The privacy should be maintained by the staff

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- Time-consuming tests should be dropped out from the test menu
- Sent out the non-cost effective tests
- ‘Stat’ abuse should be reduced
- Good connections with other laboratories
- The data should be kept safe.

Check Your Progress

5. How should the lab emergencies be reported?
6. List any five causes of fire in the laboratory.
7. Mention any three factors that reduce productivity of any laboratory.

8.5 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The earliest known laboratory in India is The National Physical Laboratory, set up in 1900.
2. All solutions and apparatus must be kept up to eye level to prevent it from spilling on the face.
3. The dress codes for laboratory are:
 - (a) Never wear any loose, dangling clothes inside the lab
 - (b) Hair must always be tied inside the lab and if possible wear head caps
 - (c) Any fancy clothing items like jackets, jewellery, boots, etc. must not be worn inside the lab and be removed before entering it.
4. Microwave heating presents several potential hazards such as rapid temperature and pressure rise, liquid super-heating, arcing, and microwave leakage.
5. All the lab emergencies must be reported in a detailed manner, with how, when, and the nature of the accident, along with the severity of the injury (if any).
6. The causes of fire in the laboratory are:
 - (a) Electrical overloading
 - (b) Poor electrical maintenance
 - (c) Switched on equipment unnecessarily
 - (d) Deterioration of gas tubing
 - (e) Naked flames

7. The factors that reduce productivity of any laboratory are:

- (a) Poor hygiene
- (b) Short staff
- (c) Poor maintenance of equipment

*Laboratories for
Investigation*

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8.6 SUMMARY

- A laboratory is a facility, with appropriately controlled conditions and safety standards, equipped with certain scientific tools to conduct vast types of experiments, tests, researches, investigations, manufacturing of various sensitive drugs, medicines and chemicals.
- Laboratories have specific needs, purposes, and associated risks. Some of those requirements are specific to where it is located, some are with respect to its purpose, while others to an activity.
- There should be rules that must be a part of manual followed by all the laboratories. They include what must be done if an accident or emergency occurs, safety measures, and how they must be handled, and basic common-sense rules.
- It is important to read all manuals, precautions, signs, symbols, fire alarm system, and follow safety precautions as guided in case of any accidents in the lab. All labs must have post numbers to call in during emergencies.
- Housekeeping generally refers to the management of duties like cleaning, cooking, home maintenance, laundry, etc. Good housekeeping in laboratories may reduce the risk of contamination and protects the integrity of biological experiments.
- Waste management is necessary for pathology laboratories because of state and federal regulations that control and restrict the disposal of waste generated by all medical facilities.
- A laboratory worker plays a major role in the safety and conduct of any laboratory. Without their cooperation, it is impossible to have a properly working laboratory.

8.7 KEY WORDS

- **Serology:** The scientific study of blood serum, especially with regard to the response of the immune system to pathogens or foreign substances
- **Histology:** The study of the microscopic structure of tissues

8.8 SELF ASSESSMENT QUESTIONS AND EXERCISES

NOTES

Short Answer Questions

1. Write a short note on housekeeping safety rules.
2. What are the guidelines regarding gas cylinders in a lab?
3. State any six examples of incompatible chemicals.

Long Answer Questions

1. Discuss the types of laboratories.
2. Describe the chemical safety rules, which should be followed in a laboratory.
3. Analyse the waste management system, which should be followed in a laboratory.

8.9 FURTHER READINGS

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BLOCK - III

LAUNDARY AND LINEN SERVICES

*Safety in the Laboratory
Blood Bank Management*

UNIT 9 SAFETY IN THE LABORATORY BLOOD BANK MANAGEMENT

NOTES

Structure

- 9.0 Introduction
- 9.1 Objectives
- 9.2 Planning and Organisation of Blood Bank
- 9.3 Blood Bank Records
- 9.4 General Policies for Blood Bank Services
- 9.5 Donor
 - 9.5.1 Medical History of the Donor
 - 9.5.2 Physical Examination of the Donor
- 9.6 Donation of Blood
 - 9.6.1 Laboratory Testing of Donor's Blood Samples
- 9.7 Answers to Check Your Progress Questions
- 9.8 Summary
- 9.9 Key Words
- 9.10 Self Assessment Questions and Exercises
- 9.11 Further Readings

9.0 INTRODUCTION

Laundry and Linen services in hospitals require special care and there are certain rules to be followed. But, before you study Laundry and linen services in hospitals. In this unit, you will discuss an important aspect of safety in the laboratory, i.e., blood bank management. Blood bank is a unit, where blood is received from the donors. Donated blood undergoes processing, stored in a hygienic place, and finally issued to a needy patient, on the recommendation of a doctor. Nowadays, voluntary donors are given donor cards, which can be used for one unit of blood, valid for the period of one year. There are some required medical conditions under which only, a donor can donate his/her blood:

- Only healthy and fit donors are selected for donating the blood, after their medical examination and required laboratory procedures.
- Only 300ml of blood is withdrawn from the donor out of approximately 6 liters of blood present in a human body.
- Blood volume depletion is recovered in 48 to 72 hours, while the haemoglobin level in approx. 3 to 4 weeks.

*Self-Instructional
Material*

NOTES

- Blood from a professional donor is mostly low in quality and the chances of transmission of diseases and infections are more than a volunteer donor.

The functions of a blood bank are:

- Collection of blood, its products and their distribution.
- Clinical utilisation of blood and its products.
- Lab procedures such as blood grouping, cross matching, etc.
- Research work
- Promoting training and teaching for safer transfusion and best use of donor blood.

The hospital transfusion services deals with the transfusion of the blood and its products directly to the patient. Hospitals provide services such as compatibility testing, final product preparation like thawing of fresh frozen plasma, resolution of patient's antibody problems, neonatal antibody work ups transfusion, training of medical health care workers, etc.

9.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the planning and organisation of a blood bank
- Recall the features of the blood bank records
- Analyse the procedure involving blood donation
- Discuss the general policies for blood bank

9.2 PLANNING AND ORGANISATION OF BLOOD BANK

A blood bank should be situated in the hospital itself. It should have its own department in the hospital so that it would not be disturbed by the patient's attendants. It should be designed in such a manner that the reception room, donation room and rest room are easily accessible to a person.

Reception room

A reception room should be spacious and ventilated to attend numerous donors. There should be two registration desks, one for the registration of donor and the other one to fill the details and particulars of the respective patients. There should also be one separate cabin for doctor, so that he/she can check whether the donor is healthy enough to donate the blood or not. It should also consist of a laboratory to check the haemoglobin of the donor and various other parameters, which can decide the fitness of the donor. Proper sitting of the donor should be arranged.

Donation room

A donation room should be well ventilated, comfortable, and air conditioned. It should have comfortable beds for the donors, sterilised equipment, and a refrigerator. The restroom should be air conditioned. Donor should be given light refreshment. This room should have comfortable beds or sitting arrangements for the donor. Doctors must keep the donor under observation after donating the blood so that a donor can resume their daily work, without any problems.

Laboratory

A laboratory is one of the most important parts of a blood bank. It should be air conditioned, spacious and should have a refrigerator. It must have experienced lab technicians. It provides test like grouping of recipient blood. A proper labelling of the donated blood is quite important. A blood bank laboratory should have other features like:

- A room for deep freezing refrigerators to preserve plasma and other blood products.
- An immunology lab for testing the Australia antigen, malaria, syphilis, aids, and cytomegalovirus (CMV).
- A working laboratory is for preparation of blood products. It should also be secluded even within the blood bank. Maintenance of sterility is required. It should have restricted entry.
- Other sections such as blood bank store, sterilisation section, and doctor and technical staff premises.

There is a set of pattern for technical purposes, covering all important parts of laboratory procedures. The principle of test provides a statement concerning the reaction involved and clinical importance of the test. All procedures have a standard format:

- Procedure requires preparation of a patient, required specimen, and specific handling conditions.
- Labelling requirements, storage, and preparation for all the reagents.
- Instrument preparation contains start up procedures, calibration, and quality control.
- For acceptable results, calculations include equations and examples
- Result reporting includes reference ranges, acceptable reporting format, and when to refer the results to the consultant.
- Notes of the procedures include sources of error and acceptable alternative procedures.
- Limitation of the procedures contains interfering substances and linearity limits and their effects on the test results.

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- References
- Supplemental material like related charts, worksheets, tables, and appendix.
- Procedure administration includes approval, review, and revision procedures.

9.3 BLOOD BANK RECORDS

The features of blood bank records are:

- It contains donor registers; one for relative donors and other for voluntary donors.
- Receipt register contains the data of blood collected everyday in the blood bank.
- It issues register.
- There are blood stock registers, which contain information about the blood plasma and other products separated from various blood groups.
- Registers for requisition.
- There are registers for blood grouping and cross matching and ABO and RH typing of OPD and ward patients. It also contains the Australia antigen reports and records of all the equipment and chemicals of blood bank.
- It also has attendance register for all the staffs.
- Proper colour labelling of blood should be done carefully; blood group O is labelled with blue colour, blood group A is labelled with yellow colour, blood group B is labelled with pink colour, while the blood group AB is labelled with white colour label.

Documents concerned with blood requisition, donation and issues

- Blood requisition should accompany with requisition form signed by the registrar of the ward. It should contain the proper labelled sample of patient's blood in a plain vial and a blood receipt slip.
- Every donor should have the slip, which is filled by the doctors and signed by the donor in the presence of the doctor.
- Blood can be issued with a transfusion slip and completion of the attached label.

Check Your Progress

1. State the functions of a blood bank.
2. How are the different blood groups classified?

9.4 GENERAL POLICIES FOR BLOOD BANK SERVICES

The following are some of the general policies that must be imposed under the direction of qualified physicians while delivering blood bank services:

- I. Quality management program comprises of:
 - A program of quality assessment and improvement to ensure that policies and procedures are maintained and followed.
 - Medical director gives the review on detailed procedures manuals annually.
 - Quality control for reagents, equipment, and methods.
 - Review the programs that monitors all categories of transfusions, including criteria for evaluating orders, usage, administration policies, and compliance with these policies.
 - All tests are required to be conducted according to the standards, in an accredited laboratory.
 - Containers and anticoagulants used for preservation must strictly meet set standards.
- II. Safety programs incorporate:
 - Programs, which are used to minimise risk for the employees, donors, volunteers, and patients.
 - Procedure for biological, chemical, and radiation safety and monitoring.
 - Handling and discarding of blood and its components in a manner that recognises the potential for exposure to infectious reagents.

NOTES

9.5 DONOR

Not just anyone can become a donor. A donor has to go through various processes in order to be qualified to donate blood. The procedure involves:

- a) There are some reasons for donating the blood, i.e.
 - Volunteer donation
 - Donation to replace product used
 - Donation to cover family and friends in some type of insurance program.
 - Paid donation.
- b) Donors who do not receive any kind of payment for donating the blood are considered as volunteer donors.

NOTES

- c) Donor screening is done prior to donating the blood. It has two purposes, i.e. to protect the donors who should not be donating for health reasons and other to protect the recipients.
- d) The process of donor screening involves registration of the donor, medical history, and physical examination.
- e) Data collected from each donor includes:
 - Donor's full name.
 - Permanent address and business cum professional plus telephone number.
 - Gender of the donor.
 - Date of birth or age of the donor, which has to be around 17. No upper age limits exists for donation.
 - Date of last donation as the donor must wait for 56 days for the donation of the blood again.
 - Written informed consent of the donor to proceed for the donation.
 - A source of identification like driving licence number, identity card, passport, etc.
 - Donor's occupation and where he or she will return to work after donation. Some heavy work is considered risky after donating the blood as the plasma volume lost during the donation returns back to normal levels within 24 hours, but red cells volume returns back to pre-donation levels only after 56 days.
 - Race or ethnic group of the donor, which can be useful in selecting donors to screen for specific antigen negative products in the future.
 - Information on credit or replacement group, if the blood bank participates in such programs.

9.5.1 Medical History of the Donor

When it comes to medical history, some of the points to consider are:

- Different names should not be used for blood donation.
- Amount of given blood, plasma, or platelets in the past 8 weeks.
- Did he/she ever experience chest pain, heart disease, or a lung disease
- Did he/she ever have blood disease, bleeding disorder, or cancer.
- Did he/she ever have the history of jaundice
- Did he/she ever receive growth hormones
- Any history of drugs

- Is he/she feeling healthy on the day of donation
- Did a donor ever have malaria in past three years or had taken antimalarial drugs
- Did a donor receive blood or organ in a last year
- Has a donor got a tattoo on his ear or skin, piercing, or accidental needle prick during last twelve months
- In the past year did a donor ever come into a close contact with the patient suffering from hepatitis or aids
- Pregnant donors are deferred from blood donation until 6 weeks after delivery or termination of pregnancy.
- Consumptions of teratogen drugs for acne make the donor unsuitable

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9.5.2 Physical Examination of the Donor

Physical examination of the donor includes:

- General examination of the donor.
- Determination of the haemoglobin level, which should be 12.5g/dl or more.
- Pulse rate should be regular and in between 50 to 100 beats per minute.
- Blood pressure should not exceed 180/100mm Hg.
- Temperature should not exceed 37.5 degree celsius
- Potential donor must weigh at least 50 kg (110lbs) to avoid risk of hypovolemic shock from blood donation.
- Amount of blood to be safely drawn from the donor may be calculated under the following equation:

$$\frac{\text{Amount of blood to be withdrawn}}{100} \times 14 = \text{Amount of anticoagulant}$$

Vein puncture site of the donor

Some considerations regarding the location of vein puncture site include:

- Inspection of the area situated in front of the elbow (antecubital area)
- Both arms should be checked.
- Locate and select a particular vein to be punctured.
- Multiple small scars along veins is a proof that the donor is using intravenous drugs and thus, should be permanently deferred.
- In case, dermatitis is seen in antecubital area, then it should not be selected for puncturing the veins.

NOTES

Use of tourniquet in blood collection

A sphygmomanometer placed on the upper arm and inflated to 50mm Hg is the most suitable method of obstructing venous return. It is important to make sure that arterial flow is not impaired. Radial artery palpitation at the wrist is a common way of ensuring that tourniquet is not too tight. One of the dangerous situation, called air embolism, is the sudden releasing of the pressure in the cuff, when glass bottles are being used for collection of blood. There is a condition in which deliberate obstruction of arterial inflow happens followed by release of the cuff pressure. It happens when the insertion of the needle in a very large vein may make this big vein to go into the spasm causing slowing or stopping of flow of blood.

Standard of practice for screening of blood donor

- Whole process of screening of the donor takes place in 15 minutes.
- Donor should be explained the procedure of blood donation completely.
- Medical history and physical examination of the donor helps ensure the safe blood for the patients.
- Now, the donor is ready to move into the donation room for donating the blood.

Preservation of blood in vitro and containers

Nowadays plastic containers are designed, instead of glass bottles, to allow the maximal product stability and cell viability. All containers are sterilised and used only once. There are some anticoagulants that are used for preservation and storage of blood, i.e.

- Acid-citrate dextrose (ACD)
- Heparin
- CPD
- CPDA-1
- CPD plus AS1 and AS2. AS stands for additive solutions which consist of saline, dextrose, mannitol, and adenine
- Citrate phosphate double dextrose (CP2D) plus AS. This solution consist of saline, dextrose, and adenine.

Check Your Progress

3. Which anticoagulants are used for preservation and storage of blood?
4. What is the equation for the amount of blood that can be safely drawn from a donor?

9.6 DONATION OF BLOOD

*Safety in the Laboratory
Blood Bank Management*

Following the qualification of a suitable donor, the procedure for the donation of blood begins:

- I. Preparation of an area of about one and a half inches in all directions, which includes,
 - Scrub vigorously with fifteen percent aqueous soap or detergent solution for thirty seconds to clean the dirt, fat, skin cells, and other debris.
 - Remove soap and froth with 10% acetone in 70% isopropyl alcohol, and allow it to dry.
 - Apply tincture of iodine and allow it to dry.
 - Remove the iodine with 10% acetone in 70% isopropyl alcohol, moving away from the venepuncture site outwards in spiral. Allow the solution to dry.
 - Cover site with dry sterilised gauze, if venepuncture is not done immediately.
- II. Do not repalpate the vein after the venepuncture site has been prepared.
- III. Anti-coagulated bag should be inserted several times to ensure satisfactory mixing of anticoagulant with the blood.
- IV. Allowable quantity of blood is collected in the anti-coagulated bag (450+₋50).
- V. After collecting the blood, about 30ml of blood sample is sent to the laboratory for testing process.
- VI. In order to reduce possible errors, rechecking should be done of the numbers on all the items, when the blood leaves the donors' bedsides.
- VII. Gloves should be used by workers, who have cuts and scars or any ulcers.
- VIII. Once the bottle is filled with blood up to the measured mark, do the following:
 - To stop the flow of blood, clamp the tube of blood taking set with artery forceps.
 - Release the pressure in arm to 0mm Hg.
 - Put a piece of dry gauze over the needle and take it out of the donor's arm.
 - Put a cotton piece over the site of the puncture.
 - Take the collecting set out of the bottle and store the bottle after collecting blood sample in 2 pilot tubes for test.
 - Put an adhesive plaster over this site of venepuncture after bleeding stops.

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IX. Needle must be discarded in a puncture proof container after the collection of blood.

X. There are some steps which should be followed after the donation procedures is completed, i.e.

- Donor should be requested to rest for 10 minutes before getting up. One of the staff members should help the donor to get up.
- Offer the donor a snack or drink before leaving.
- Attend to the donor, if he is suffering from light headedness, nervousness, uneasiness, or their colour turns slightly pale.
- Donor should be requested to drink water or more fluids during next 4 hours after donation.
- Raise the arm and the pressure should be applied on the bandage. In case of the bleeding, wait for some time until the bleeding stops.
- Ask the donor to report to blood transfusion centre, in case the donor becomes ill in 3 to 4 days after donation and conduct investigations. Do not use this donated blood till investigations stops.

XI. The actual donation time for a unit of whole blood is 8 to 10 minutes.

XII. The donor is eligible for next donation after 8 weeks of previous donation. During these 8 weeks, the donor's iron content gets replenished.

Autologous donation

It is the donation of the blood for donor's own use. Getting one's own blood is the safest route, because it contains no foreign antigens and thus, there is no chance of any infections happening, other than what patient already have. Autologous donation is usually performed:

- To prevent transfusion transmitted diseases.
- To get rare blood type.
- To protect patients, with a record of severe transfusion reactions.
- To allow transfusion availability to those patients, whose religious belief do not permit blood transfusion.

Drawbacks of autologous donation are:

- Phlebotomy induced anaemia
- Hypovolemia
- Expensive autologous products
- There are some adverse reactions that can occur at the time of donation.

Criteria for autologous donation are:

- There is no upper or lower age limit
- Haemoglobin 11 g/DL
- There is no weight limit
- Frequency of donation should not exceed more than 3 days
- No bacteraemia
- Patients can donate 2 to 3 unit of blood over a period of 7 days. The oldest unit is transfused to the patient and 2 more units of blood are withdrawn.

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Specialised Donations

Under specialised donations comes the process of Apheresis. In this process, whole blood is removed from a donor or patient, then that blood is processed to separate selected elements. The un-harvested portion is returned back to the patient. There are three general types of apheresis, i.e.

- Cytapheresis:** The removal of blood cells from blood, especially prior to returning the resulting blood to a patient
- Plasmapheresis:** A process that filters the blood and removes harmful antibodies
- Bone marrow processing

Complication in apheresis:

- Hypotensive transfusion reaction:** HTR is a rare condition involving low blood pressure associated with administration of blood products.
- Mechanical haemolytic anaemia:** A mechanically induced damage to red blood cells; Red blood cells, though flexible, may succumb to physical shear and compression under constant pressure.
- Decrease in the number of T and B cells, which may persist for eight months after apheresis.

Preparation of segments from the blood unit

- Segments are sealed portions of the integral blood donor tubing that leads from collecting bag to the needle during phlebotomy.
- Segments having small aliquots of donor blood can be used for pre-transfusion testing.
- Segments are made by dividing the integral tubing and is sealed by heat treatment or by adding metal clasp.
- At the time of bag manufacturing, each segment has a unique number assigned. This number is permanently attached to the blood bag.

- The donor sample is used to confirm ABO type of the unit and the Rh type, to perform compatibility testing and to provide material for additional tests.

Adverse donor reaction

NOTES

It is obvious that the donation process is safe, but sometimes adverse reaction may occur like:

- Hematoma:** Development of hematoma at venepuncture site. Apply pressure for 5 to 10 minutes to reduce the chances of hematoma. Ice can also applied to this site to reduce the size of hematoma.
- Nausea and vomiting may occur.
- Serious cardiac problem like cardiac arrest
- Syncope/Fainting:** There are some donors who are liable to faint for a few hours, even with the donation of 400ml of blood, if they get up suddenly. It may occur due to weakness and dizziness, which may cause loss of consciousness and involuntary passing of urine or stools. Climbing ladder or taking flight should be avoided for twelve hours after the donation of blood.
- Tetany:** It rarely occurs with donors. It is caused due to hyperventilation. It happens mostly in anxiety prone people. Rebreathing from the cylinder can help in providing relief.
- Air embolism:** Blood is usually taken into a glass bottle carefully according to gravity or its flow is assisted by having the vacuum in the bottle. When the blood stops flowing, the pressure in the vein gets decreased by releasing the sphygmomanometer cuff. The pressure in the bottle is becomes higher than the pressure in the vein and the air bubbles pass from the bottle into the donor's arm. This process is called air embolism.
- Puncture of artery:** A rapid flow of blood may occur, if the needle is removed, followed by an extensive bruising. It has the possibility of forming arteriovenous fistula.

9.6.1 Laboratory Testing of Donor's Blood Samples

The laboratory testing is very important for safe blood transfusion. Following tests must be done on each donor's sample before every donation:

- ABO grouping:** Both red cells typing and serum typing should be performed. Rh typing is done using anti D reagent. If the sample is found to be D positive, the unit is labelled Rh positive. The sample that are negative for both D are considered as a Rh negative.

b) Antibody screening: It should be done in those donors, who have a history of transfusion and pregnancy. Following are the methods to detect antibodies:

- Anti-globulin test
- Low ionic strength solution
- Room temperature antibody screening test
- Detection of significant IgG antibody reactive at 37 degree celsius test.
- If the donor carries significant antibodies, then red cell product should have minimal quantity of plasma.
- Wash red cells with saline before transfusion or storage.
- Plasma is removed and red cells are frozen.

c) Test for syphilis: If screening test is reactive, specific test for syphilis spirochaetes are performed.

d) Cytomegalovirus screening: It is done to provide blood that is free from cytomegalovirus for specific patients who would have problem.

e) Sick cell trait test: This trait does not cause problem when transfused in adult patients. However, infants may not survive.

Laboratory tests cannot solely provide 100 % safety of blood sample because:

- New infectious agents continue to emerge and produce new disease. If disease is found out to be transmissible by blood transfusion, then the new test should be added in the above mentioned tests.
- Every disease has a phase. When a person gets exposed to the disease, it doesn't produce any antigen or antibody sufficient enough for in vitro detection. It is considered as a window period.
- Adverse reaction from the transfusion cannot be excluded in spite of all possible precautions. It can only be minimised.

NOTES

Check Your Progress

5. How is Rh typing performed?
6. Define air embolism.

9.7 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The functions of a blood bank are:
 - a) Collection of blood, its products and their distribution.
 - b) Clinical utilisation of blood and its products.

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- c) Lab procedures such as blood grouping, cross matching, etc.
 - d) Research work
 - e) Promoting training and teaching for safer transfusion and best use of donor blood.
2. Different blood groups are classified as: Blood group O is labelled with blue colour; blood group A is labelled with yellow colour; blood group B is labelled with pink colour; while the blood group AB is labelled with white colour label.
3. The anticoagulants, which are used for preservation and storage of blood are:
- a) Acid–citrate dextrose (ACD)
 - b) Heparin
 - c) CPD
 - d) CPDA-1
 - e) CPD plus AS1 and AS2. AS stands for additive solutions which consist of saline, dextrose, mannitol, and adenine
 - f) Citrate phosphate double dextrose (CP2D) plus AS. This solution consist of saline, dextrose, and adenine.
4. The amount of blood that can be safely drawn from a donor may be calculated under the following equation:
- $$\frac{\text{Amount of blood to be withdrawn}}{100} \times 14 = \text{Amount of anticoagulant}$$
5. Rh typing is performed using anti D reagent. If the sample is found to be D positive, the unit is labelled Rh positive. The sample that are negative for both D are considered as an Rh negative.
6. When the blood stops flowing, the pressure in the vein gets decreased by releasing the sphygmomanometer cuff. The pressure in the bottle is becomes higher than the pressure in the vein and the air bubbles pass from the bottle into the donor's arm. This process is called air embolism.

9.8 SUMMARY

- Blood bank is a unit, where blood is received from the donors. Donated blood undergoes processing, stored in a hygienic place, and finally issued to a needy patient, on the recommendation of a doctor.

- A blood bank should be situated in the hospital itself. It should have its own department in the hospital so that it would not be disturbed by the patient's attendants.
- A blood bank should be designed in such a manner that the reception room, donation room and rest room are easily accessible to a person.
- General policies must be imposed under the direction of qualified physicians.
- Not just anyone can become a donor. A donor has to go through various processes in order to be qualified to donate blood.
- The laboratory testing is very important for safe blood transfusion.

NOTES

9.9 KEY WORDS

- **Immunology Lab:** Focuses on identifying antibodies, investigating immune system, and determining organ, tissue, and fluid compatibility for transplantation
- **Cytomegalovirus (CMV):** A common virus that can infect almost anyone. Once infected, your body retains the virus for life.
- **Australia Antigen:** HBsAg is the surface antigen of the hepatitis B virus (HBV).
- **Hypovolemic Shock:** An emergency condition in which severe blood or fluid loss makes the heart unable to pump enough blood to the body
- **Venous Return:** The rate of blood flow back to the heart

9.10 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. Write a short note on the organisation of a blood bank.
2. What is included in the data collected from each donor?
3. What are the constituent of the blood bank records?

Long Answer Questions

1. Discuss the points, which should be considered with respect to medical history of a donor.
2. Explain the procedure a donor has to go through in order to be qualified.
3. Describe autologous and specialised donation in detail.

9.11 FURTHER READINGS

NOTES

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UNIT 10 LINEN AND LAUNDRY SERVICES

Linen and Laundry Services

NOTES

Structure

- 10.0 Introduction
- 10.1 Objectives
- 10.2 Meaning and Importance of Linen and Laundry Services
 - 10.2.1 Laundry Management
- 10.3 Location and Design of Laundry Department
- 10.4 Linen Classification
 - 10.4.1 Linen Items for Laundering
 - 10.4.2 Types of the Laundry Services
- 10.5 Equipment of Laundry
 - 10.5.1 Selecting the Right Equipment
 - 10.5.2 Equipment List
- 10.6 Requirement of Linen in a Hospital
- 10.7 Processing Quality of Laundry
 - 10.7.1 Assessment of Laundry Service
- 10.8 Answers to Check Your Progress Questions
- 10.9 Summary
- 10.10 Key Words
- 10.11 Self Assessment Questions and Exercises
- 10.12 Further Readings

10.0 INTRODUCTION

The laundry and linen service of a hospital is a very essential service, which helps prevent any infection from happening and impacts a hospital in a positive way. If a laundry section is clean and dirt-free, it affects the patient's health in a good way while a dirty one, becomes another cause of an infection for a patient. Moreover, it is not good for the image of a hospital. Hence, it is very important for each and every hospital to keep their linen and laundry standards high, so that patients, doctors, and all the workers feel satisfied. It is no wonder that laundry and linen service has become a crucial part of hospital management.

10.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the meaning of linen and laundry service
- Analyse the importance of linen and laundry service
- Discuss the types of linen and laundry service

10.2 MEANING AND IMPORTANCE OF LINEN AND LAUNDRY SERVICES

NOTES

The laundry service include tasks like sorting, drying, ironing, folding, washing, mending and delivery. In modern society, every patient expects their linen of being clean and healthy. With regards to developing or under-developed nations, this service becomes even more crucial as high population demands high level of caution in terms of risk towards infections. A proper laundry service is a core of better health conditions for a patient.

Regular changing and supplying of bed linen and all other linen items, which are required by a patient, affects patients' health in a positive manner, providing positive reviews to the hospital. Sometimes, in an operation theatre, there is a poor supply of the linen items, due to which, operation time gets delayed. Linen items have also found to be not washed properly in many hospitals. The main purpose of the linen and laundry is to provide all the needs of the hospital at correct time and correct place. All these can only be possible, if it was covered under the hospital management.

It is also very important for the staff, who are going to interact with the patients, to keep their entire dress clean to avoid exposing the patients to any germs. As it is a very important department of the hospital, so the management of the hospital should be such that it can fulfil the needs of the patients in a satisfied manner by:

Linen and laundry service has many functions such as:

- Collecting the infected and dirty linen from the ward and other sections
- Sorting out the linen
- Removing the blood clots and body fluid
- Decontaminating the contaminated linen
- Cleaning or washing of linen, followed by drying
- Replacing or repairing any damaged item as soon as possible
- Pressing of the linen
- Folding of the pressed linen
- Storing of the pressed item
- Distributing the well-processed items to their respective user department.
- Back-up options are kept for emergency
- Maintaining the registers

Its service with regards to a patient can't be emphasised anymore. A good quality laundry and linen service for patient includes:

- The utmost care of a patient's comfort
- The linen items should be changed according to a patient's need.
- All the items should be clean, have good fragrance, and be pleasant to the patient.
- All the items must have perfect size according to a patient.

The quality of linen and laundry service depends upon the facilities and delivery services available in a hospital. The patients, generally, correlate a hospital's overall status with its laundry service. Now a days, more and more people are becoming health conscious, so it is imperative for these people that a hospital maintains hygiene. The linen is not dirty merely in terms of dirt and soil, but it is harmful in terms of the infectious microbes. This makes it necessary to not only wash it, but decontaminate it. This can only be done by advanced equipment and method of washing. A hospital must be strict about such cases as it is related to the health of a person and more importantly the reputation of a hospital.

10.2.1 Laundry Management

The manager of the laundry department is responsible for imposing all the principles and rules of laundry. Laundry workers work under them. In some places, there are more than one well-qualified manager. Some of their responsibilities are:

- They must instruct the workers, whenever required.
- They should maintain the norms, as much as they can.
- Keep the room of linen storage clean.
- Check out the proper connection of electricity, water, etc.
- Check out the workers' hygiene
- They must be aware of the availability or the scarcity of the equipment.
- To check out the area of washing
- Availability of linen items should also be considered by the manager

Although any particular training is not required, but the workers need to be well educated about the contamination and infection, and the norms of the laundry so that they won't make any mistakes.

10.3 LOCATION AND DESIGN OF LAUNDRY DEPARTMENT

It is very important to have a good location for the laundry department. Laundry department must be located at a place, where there is proper ventilation and natural light. It should never be on any floor, other than the ground floor, where it should be connected to the power as it uses energy, water, and heat. The department

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should always be strategically located so as to be accessible from every route, saving time and effort of the workers. It should be in the backside area so not to obstruct other services.

Generally, the biggest mistake on the part of the hospital management is giving less importance to the laundry department, resulting in an improperly designed laundry section. It is important to have strict norms and principles in place, so as to have an effective laundry service department. Since, it contains both the soiled and infected item so there should be a division between these two sections, either by line or any other barrier method, in order to keep them separated. While designing the laundry section, the item or material, which is being used for barrier method must be kept solid. Sometimes, during the washing process the airborne bacteria of infectious side, can settle down on the cleaner side. Thus, the two sections must be designed in such a manner that they remain well-segregated. Other considerations include:

- The bag in which used laundry items have to be kept should be of two different colours for soiled and infectious items.
- Both bags should be closed or covered, especially the infectious one
- Their place of storage should also be different.
- Workers should be well familiar with the norms
- Workers should be advised by the management to wash their hands properly

While designing the laundry department, careful planning is required to meet all the needs of the laundry services, which requires:

- Electricity
- Heat
- Water
- Air (compressed)
- Steam
- Temperature

Among the above-mentioned requirements, water is the most important requirement, so the laundry department should be designed in such a way that water is properly supplied, in a sufficient amount. The requirement of electricity is also there, so power should also be constantly and adequately supplied to the laundry section. Steam room should be as near as possible. It is very important to keep the temperature at correct level. In order to maintain an efficient laundry service department, design and location are crucial elements.

Linen room (central)

Under the supervision of linen supervisor, all the work of the linen service in a hospital is controlled by central linen room. This room supplies all the linen items to the hospital. This room must be designed close to the washing section. Sometimes this room is located at the side of the entrance of the laundry department. The main function of this room is the smooth supply of the linen items to the hospital, wherever and whenever it is required. Basically, the backup plans are prepared in this room, in case of any damage of any equipment or items (including sewing). As hospital service is open for 24 hours, so the linen department should be large enough to supply all the needs continuously. For the storage of linen items, shelves or racks should be built in the central room of laundry. For the storage of smaller items, other sections should be built to avoid any inconvenience. All the big equipment as we had discussed earlier, obviously need more space so the central room should be large.

Tailoring room

It is the part of the laundry department, which is most interior and must be under the supervision of the laundry supervisor. This room is particularly used for the damaged items, which require repair. All the equipment which is required for tailoring, for e.g. a sewing machine, must be present in the room in adequate number. All the items which are going to be repaired must be mentioned in a book or recorded. This helps supervisor to know the efficiency of the machine or other items.

Check Your Progress

1. What are the two most important requirements of the laundry department?
2. Why is laundry service crucial for developing or under-developed nations?
3. List any three responsibilities of a manager.

10.4 LINEN CLASSIFICATION

The laundry and linen services can be classified on the basis of various factors. Following are the classification of linen, which are used in a hospital:

- a) Common purpose linen:** These are the linen, which are not used by patients, either directly or indirectly. It consists of all the curtains, table cloth, and all other items of the hospital.
- b) Linen of patient care:** All the clothes, which are worn or used by the patient fall under linen of patient care, including shirt, pants, coats, etc.

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c) **Linen for bed:** These are the linen, which are used by the patient indirectly. It generally includes items such as pillow cover, bed-sheet, blankets and its cover, etc.

d) **Linen of operation theatre or labour room:** It includes all those items, which are used by the doctors or surgeons. It is not used by the patient directly or indirectly. It consists of items like cap, towel, gowns, apron, along with shirt, pants, top, etc.

10.4.1 Linen Items for Laundering

By keeping the purpose of laundry in mind, the linen of hospital can be classified into:

- a) **Used (dirty) linen:** These are the linen which are commonly used.
- b) **Contaminated linen:** These are the linen which are contaminated with body fluid or blood.
- c) **Odious linen:** These are the linen, which are not really infected, but have foul smell.
- d) **Infectious linen:** These are the linen which are declared as infectious and contaminated by the doctors.
- e) **Infectious linen with foul smell:** These are the linen with infection and foul smell; these are generally the clothes, which are left for six hours after use.

Depending upon the above classification, the linen items are washed according to their categories of infection.

- Used linen and the contaminated linen should be washed separately
- Odious linen should be washed separately from the used and contaminated linen
- Infectious linen and the infectious linen with foul smell should undergo processing or different method of washing or pressing, before they are washed with common laundry.

Since washing is the most common stage, at which an infection generally spreads and the linen items get contaminated, so it is very important to follow the methods of washing as per the classification of the linen items.

10.4.2 Types of the Laundry Services

The laundry system itself are of different types; they include:

- I. **Laundry of rental system:** These are generally for the hospitals, which are small in size and financially, not so strong to have laundry of their own. In such cases, hospitals generally take laundry on rent from broker or other contractor. The hospitals consisting of up to 60 beds, generally adopt the rental system.

II. Laundry of plant system: These are for the hospitals, which are bigger in size and are able to afford the laundry plant, which is the best method for cleaning and washing the linen items of the hospital. It also helps to overcome the loss of linen. The hospital consisting of up to 500 beds, generally come under the plant system.

III. Laundry of contract system: It is a type of system in which hospitals have their own linen but they don't have their own laundry system. For the purpose of laundry, contractor are hired to collect all the linen and supply it to the hospital after laundering it. Even the hospitals, with their own laundry system, are adopting this system. The biggest demerit of this system is that sometimes the linen get mixed i.e. clean linen with the infectious linen, resulting in contamination.

IV. Cooperative system: It is a system which is used to define the laundry which work for many hospital. As it is also for the hospital of small size and other nursing homes.

The demerit of this system is also the contamination or the infection.

And also if there is problem in laundry, it creates problem in hospitals too as their work is affected.

Check Your Progress

4. What is the biggest demerit of contract system of laundry service?
5. Which type of hospital generally opt for rental system?

10.5 EQUIPMENT OF LAUNDRY

The equipment in laundry should be arranged in such a manner so that the work of laundry should not get affected. All the machines of washing should be kept close to the laundry department. In order to maintain the standard of the work, all the equipment needed by the department must be available. In the earlier days, the standard of laundry department was not well developed but now a days, it is quite improved due to the availability of the required equipment. Machines, which are automatic and require less effort are much preferred. Other important features of laundry equipment are as follows:

- Increase in production.
- Water, heat, and electricity are better utilised through such machines.
- Machines reduce the effort required by a worker
- Automatic dispenser formula
- In such machines, operation is also controlled automatically.

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- Such machines also help in combining the materials of washing.
- Rinsing
- Extraction
- Folding

10.5.1 Selecting the Right Equipment

Choosing or selecting an equipment is a very important task, as one needs to consider the requirements of the hospital. Considerations such as the size of an equipment should be kept in mind, while selecting. During selection of the laundry equipment, the factors which should be kept in mind are:

- Must be affordable
- Must come under Indian standard of bureau
- Must be easy to maintain and in case of damage, the parts are easily available
- Must work efficiently even in extreme condition
- Must consume less power, water, heat, steam, etc. at the normal rate and should not allow the wastage of other materials of washing
- Must not be labour and time consuming
- Must be easy to use

10.5.2 Equipment List

Some of the equipment, which are very important and generally found in the laundry are as follows:

- Washing machine with double door
- Washer-extractor
- Machine which combine the three main methods of laundry extraction, washing, and rinsing
- Machine for drying
- Pressing machine
- Compressor
- Scale
- Sewing machine
- Trolley for dry linen
- Trolley for wet linen
- Trolley for hanging items

- Different trolley for distribution
- Hand press or iron
- Machine having discharge facilities

To achieve optimum utilisation of the laundry department, proper planning of equipment is required like selection of equipment, fixed time of loading and unloading, etc. The planning must be done under the supervision of laundry manager. Some of the points, which should be kept in mind are as follows:

- Time of loading or unloading of the soiled material must be fixed and machine must be able to work without any problem, for e.g. if the washer extractor consists of 60 kg load then it must have a dryer, which has the capacity of 40 kg.
- A machine should be operated according to its sequence, so that no pressure is put on the machines.
- There will be reduction in the output of machine, under following conditions-
 - i. Increase or decrease in the time of cycling
 - ii. Loads incompatible to the weight of the machine
 - iii. Overloading, increasing the time of cycling
 - iv. Under-loading, leading to waste of effort and time

Check Your Progress

6. Mention any three features of laundry machines.
7. Why is proper planning required?

10.6 REQUIREMENT OF LINEN IN A HOSPITAL

In case of hospital laundry, linen is not measured in terms of size, but in weight. In western countries, the requirement of linen in hospital is different, it varies from 4 to 8 kg each day, depending upon the size of a bed. While in our country, the linen, generally, weighs around 4 kg. It mostly varies depending upon the requirement of the hospital. Following are the different types of linen and their weight in percentage from the total linen used:

- a) Sheets:** 80 per cent.
- b) Towels and other items of linen room:** 13 per cent
- c) All the work of hand:** 7 per cent

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Table 10.1 Weight of Different Linen Items

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Linen Items	Weight of the Items
Shirt of surgeons	250 gm
Pants of surgeon	250 gm
Gown use during surgery	550 gm
Head cover	16 gm
Face cover (mask)	16 gm
Head wear of ladies	30 gm
Shirt of patient	260 gm
Pant of patient	260 gm
Petticoat of patient	210 gm
Jacket of patient	200 gm
Pillow cover	130 gm
Bed sheet	800 gm
Cover for trolley	400 gm
Blanket for patient	Approx. 1000 gm
Blanket for baby	Approx. 400 gm
Towel for OT	450 gm
Towel for bath	Approx. 800 gm

The above Table 10.1 displays the weight of the fabrics, which varies depending on the kind of fabric. The weight can also vary with the size, but such cases are rarely observed in laundry of the hospital. Such cases are generally observed in the western countries.

Principles for per bed linen

In case of the bed linen, there must be three pairs of bed linen for each bed.

- One is for use
- One should be ready, whenever required
- One in the laundry for the process of processing or washing
- The rest two should be available for any emergency

Following are the norms for the number of linen:

- Bed sheets:** 3 pair for each bed
- Cover for pillow:** 3 pair for each bed
- Pants and shirts for man:** 3 or 4 for each bed

- d) Petticoat and blouse for women:** 3 or 4 for each bed
- e) Towel for patient:** 2 or 3 for each bed
- f) Blanket for patient:** 2 per bed

Change of linen is required on daily basis or at least on the alternate days. In such cases, the need for linen gets automatically increased. It is very important to have linen items, according to the norms of the hospitals, so that there would not be any scarcity of the linen items, either for the patients, staff members, or doctors. Sometimes, the requirement of linen is on the emergency basis, so availability must be in such a manner, that there won't be any scarcity either on daily basis, alternate basis, or even in emergency situations.

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10.7 PROCESSING QUALITY OF LAUNDRY

Following points should be followed by the laundry for maintaining the processing quality:

- All the policies or the procedure should be written on the manual irrespective of the activity.
- The method of washing must be written in the manual in the sequential manner.
- Washing method should differ according to the fabric.
- The work of laundry including the cleaning to pressing should be completed in least amount of time.
- The time should be fixed for collecting the linen (dirty) items and their (clean linen) supply to the required department.
- For the workers, the hours for working should be fixed.
- The linen items should be selected according to its quality or standard. It must be of good quality and material.
- All the clean linen items including the linen items of doctors, staff, and patient should not be present in only one section of the hospital; it must be present on each floor.
- The linen items must be purchased under the supervision of laundry manager, or after a discussion with them.
- For the process of distribution, an efficient plan should be made. The time of distribution should not be affected.
- Separation of the linen items must be done properly based on whether it is dirty or infected.
- It is very important to keep a record of number of linen items, which are going to be cleaned.

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- The linen items requiring repair should be kept separately from the other linen items, including clean and infected linen.
- The items, which had been repaired, must be checked regularly, to avoid the chance of damage again.
- There must be a regular meeting between higher authorities and the manager of laundry department for maintaining the linen.
- The quality of water supplied to the laundry, must be checked by an expert regularly, so to avoid any infection.
- Records of expenditure on each linen item must be available in the department.
- Damaged equipment, along with their cost of maintenance and efficiency should be recorded.

Different methods should be applied and the authorities must visit regularly to ensure the smooth working of laundry department.

10.7.1 Assessment of Laundry Service

The steps which are involved in the assessment of laundry service are as follows:

- I. Feedback from patient:** It is very important to receive any complaint or review regarding the laundry services, which include:
 - Whether linen items are washed properly or not
 - Whether there is an adequate supply of linen items or not
 - Whether the linen requirement is fulfilled by the laundry or not
- II. Complaints:** Any complaints by any of the staff members or doctors should be considered by the authorities.
- III. Cost in the repair of the damaged linen:** It is very important method of the assessment, as it helps to manage the laundry department economically.
- IV. Loss of the linen items:** Sometimes, linen item gets misplaced, stolen, or damaged. In such cases, the supervisor or other authorities should be informed immediately in order to assess the situation.

Check Your Progress

8. What percentage of the total weight is taken by the towels and other linen items in total linen uses?
9. What are the issues of linen regarding which feedback should be received?

10.8 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The laundry service is crucial for developing or under-developed nations as high population demands high level of caution in terms of risk towards infections.
2. The two most important requirements of the laundry department are water and electricity.
3. The responsibilities of a manager are:
 - a) Whenever required they must instruct the workers.
 - b) As far as possible, they should maintain the norms.
 - c) To keep the room of linen storage clean.
4. The biggest demerit of contract system of laundry service is that sometimes the linen get mixed i.e. clean linen with the infectious linen, resulting in contamination.
5. The hospitals consisting of up to 60 beds, generally opt for the rental system.
6. The features of laundry machines are:
 - a) Rinsing
 - b) Extraction
 - c) Folding
7. A proper planning of equipment is required to achieve optimum utilisation of the laundry department.
8. Thirteen per cent of the total weight is taken by the towels and other linen items in total linen uses.
9. The issues of linen regarding which feedback should be received are:
 - a) Whether linen items are washed properly or not
 - b) Whether there is an adequate supply of linen items or not
 - c) Whether the linen requirement is fulfilled by the laundry or not

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10.9 SUMMARY

- The laundry and linen service of a hospital is a very essential service, which helps prevent any infection from happening and impacts a hospital in a positive way.
- The laundry service include tasks like sorting, drying, ironing, folding, washing, mending and delivery.

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- It is very important to have a good location for the laundry department. The department should always be strategically located so as to be accessible from every route, saving time and effort of the workers.
- Generally, the biggest mistake on the part of the hospital management is giving less importance to the laundry department, resulting in an improperly designed laundry section.
- It is important to have strict norms and principles in place, so as to have an effective laundry service department. The manager of the laundry department is responsible for imposing all the principles and rules of laundry.
- The equipment in laundry should be arranged in such a manner so that the work of laundry should not get affected.
- In case of hospital laundry, linen is not measured in terms of size, but in weight.

10.10 KEY WORDS

- **Pressing:** The process of taking clean clothes and pressing the wrinkles out as well as reshaping the garment to its original form
- **Washer-Extractor:** The machines using high speed spin between wash and rinse cycles, 'extracting' the wash water from fabrics using centrifugal force

10.11 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. Write a short note on the design of a laundry service department.
2. How is laundry generally classified in a hospital?
3. What are the factors, which should be kept in mind, while selecting laundry equipment?

Long Answer Questions

1. Discuss the importance of linen and laundry service.
2. Describe and classify laundry service on the basis of purpose and types.
3. Evaluate the processing quality of laundry department.

10.12 FURTHER READINGS

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*Linen and Laundry
Services*

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UNIT 11 LAUNDRY ARRANGEMENT IN THE HOSPITAL SETUP

NOTES

Structure

- 11.0 Introduction
- 11.1 Objectives
- 11.2 Laundry Arrangements and Services
 - 11.2.1 Objective and Advantages of Effective Laundry Services
 - 11.2.2 Classification of Hospital Linen
 - 11.2.3 Types of Laundry Services
 - 11.2.4 Workflow and Staff of Hospital Laundry
 - 11.2.5 Work Schedule of Hospital Laundry
 - 11.2.6 Washing Materials and Arrangements
- 11.3 Answers to Check Your Progress Questions
- 11.4 Summary
- 11.5 Key Words
- 11.6 Self Assessment Questions and Exercises
- 11.7 Further Readings

11.0 INTRODUCTION

The word 'laundry' finds its origin in the word 'launderer', or 'laundress'; which essentially means a washer man. Hence, the word laundry defines the activities that pertain to the cleaning and/or washing of clothes and linen.

When hospitals are talked of, the first thoughts that come to mind are those of doctors and nurses, owing to their high contextual visibility, their services and the bonds that are created with patients during the processes of treatments. However, equal importance and emphasis is must be given the not-so-popular departments in a hospital, like kitchen, laundry and housekeeping. These are also referred to as supportive services. Large amount of space is allocated to these services, also they account for major chunks during the planning and construction of the hospital. One of the most important services is that of laundry and linen, which also accounts for a department which has the most complaints in almost every hospital.

11.1 OBJECTIVES

After going through this unit, you will be able to:

- Discuss in the laundry services and arrangements
- Analyse the objective of maintenance of the laundry services
- Discuss the functions of laundry services of a hospital
- Understand the types of laundry services
- Recognize the equipments that are used in a typical hospital laundry

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11.2 LAUNDRY ARRANGEMENTS AND SERVICES

Emphasis on the patient's hygiene and personal needs is nearly as important as the prescribed medication in a hospital. Nursing and providing clean and safe food is also vital. The comfort and the safety of the patient against infections is affected by the quality of the linen and hence this quality is imperative. In the provision of providing high quality medical care, clean and superior quality linen is important, while also maintaining a clean and aesthetic environment. Staff attending to each patient must also maintain exceptionally high levels of personal hygiene and cleanliness. Neat uniforms and properly maintained employees contribute towards the maintenance of a good image of the hospital.

Any hospital relies strongly on its laundry service; especially in today's times, when the patients and kin expect the change of linen on a daily basis. This also increases the demand of the number of linens. Also, it maintains the overall life of linen by maximizing its resting period and minimizing the wear and tear. The laundry in a hospital can be maintained and run in two ways. It can both be developed and run in-house, or it can be contracted/outsourced. In case of in-house laundry services, the equipment for drying and washing must be proper and well installed. In case of outsourcing, there must be enough provision of area for the reception of soiled linen, storage, processing and clean linen area. Sorting areas are also required. Other equipment for transport like trolleys, staff, etc. must also be taken care of.

Principle of Washing

There is exertion of periodic and mechanical forces on the fibres, which causes small flows in and out of the pores in the fibres. The process of diffusion ensures that soap reaches the dirt through the outer layer of the fibre. The dirt particles within the fibre are loosened by the convection flow through the fibres with moving layers of soap along pore walls.

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11.2.1 Objective and Advantages of Effective Laundry Services

Hospital laundry has a main objective of providing of adequate quantities of linen for Operation Theatre, OPDs and other departments. Provision of clean linen for medical personnel is equally important. Also, the quality of linen must be matched as per the requirement.

Advantages

Having an effective laundry services in a hospital carries the following advantages:

- Since laundry is done regularly, preferably daily, the risk of hospital associated infections (HAI) goes down.
- The provision of better healthcare services, which is the main agenda of a hospital, is completed more efficiently.
- Better psychological satisfaction of the patients and improved aesthetical environment.
- With a good laundry service comes a more positive image of the hospital as laundry is an important public relations venture.
- Better satisfaction of the customer ensures higher profit derivation of the hospital, which is the primary aim of any business establishment.

Functions

The main functions of the laundry services of a hospital include:

- Collection of used or infected linen from all places in the hospital.
- Processing of the used/infected linen using laundry equipment. These processes are divided into disinfecting, sorting, washing, conditioning, ironing and folding.
- The condemnation and replacement of damaged articles.
- Putting all items together for sterilization.
- Distribution of the cleaned linen to the respective departments.
- Maintenance of the stored linen.

Quality of Linen

The linen used in hospitals must meet the following requirements:

- Non-irritant
- Microbial-repellent
- Good aesthetics
- Not highly flammable
- Easy to clean and disinfect
- Poor static conduction (generation of static electricity on clothing)

- High tensile strength (ability of a material to withstand a pulling force)
- Have a fluffy and soft surface
- Quick efficiency for drying
- Repels moisture

Laundry Arrangement in the Hospital Setup

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Importance of Clean Medical Linen

Clean medical linen carries importance due to various factors. Since hospitals are an on-the-edge environment, there is hardly any scope of error pertaining to departments like laundry, food, etc. Therefore, the importance of clean laundry is high and can be found as under:

- Prevention of transmission of germs from one patient to another, or from attendants to patients
- Hospital linen comforts the patients and makes them feel at ease
- Vitalities owing to sanitation fronts
- Aesthetic values
- Dirty or smelly and stained linens give off a bad impression as well as cause discomfort to the patient and waiver their families' trust from the hospital

11.2.2 Classification of Hospital Linen

Classification of hospital linen can be done on many fronts. Lets discuss there one by one.

Classification of Linen on the Basis of use

- **General purpose linen:** It includes simple use items like curtains, drapes, tablecloths and similar items that are commonly used. These clothes are the type that aren't used for patient care.
- **Ward linen:** These are ward items used for patient care, namely bedspreads, blankets, pillowcases.
- **Patient linen:** This linen is used exclusively and extensively by the patient and plays a major role in decision of time of recovery and hence cannot be meddled with. They include gowns, pyjamas, coats, shirts, etc.
- **Procedure Linen (OT, OPD, Labour Room linen):** Items like pyjamas, kurtas, gowns, coats, shirts, etc. won by OT professionals and surgeons, and also masks, caps, surgical gowns, OT towels and trolley covers. These are required in labour room and procedure rooms.

Classification of linen on the basis of degree of contamination

- **Clean Linen:** Cleaned, washed and processed linen makes up this category.
- **Soiled:** Linen used by patients.
- **Infected/fouled:** Linen which is exposed to bio-hazardous conditions like fluids, blood, etc.

Self-Instructional Material

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Classification of linen based on its purpose

- **Linen for housekeeping:** curtains, drapes, tablecloths, etc.
- **Patient linen:** bedspreads, pillowcases, pyjamas, shirts, coats, gowns, etc.
- **Staff linen:** doctor and medical staff aprons, gowns, caps, masks, etc.

Classification based on the colour of linen

- **OT Linen:** This should be green in colour, including doctor's gown.
- **Ward linen:** ward linen must be white in colour.
- **Patient linen:** this linen should be blue in colour.

Location of a Hospital Laundry

The laundry must be situated such that it receives ample sunlight as well as ventilation. The ideal location of hospital laundry is considered to be on the ground floor, in an isolated building near to the power plant, as the laundry is one of the departments that uses the most power, water and steam. Since the running costs of laundry are variable subject to the incoming of clothes to be washed, it is imperative to plan the laundry such that the cost doesn't vary much during this given situation. This can be done by minimizing the space of horizontal and vertical distance between the washing area and other departments.

The optimum location is one which is convenient to the user units as well as close to the service elevator(s). However, it also depends on the type of healthcare establishment and may also find a proper house in the basement that has a proper drainage system. The better possible place for the laundry department is closer to the Central Sterile Service Department or CSSD or to the dietary department since the laundry requires steam which is easily available at both these places due to the boiler.

Layout of a Hospital Laundry

As compared to other methods, a well-organized and mechanized laundry is safe as well as convenient and dependable. It is expected of the laundry's designing to facilitate these given processes which are used during laundering:

- Reception of the soiled laundry.
- Sorting: The soiled laundry may be categorized on the basis of material, colour, type of cloth, etc.
- Sluicing: Rising or preliminary washing of the clothes
- Washing: This is the main process which accounts for the cleaning of the soiled clothes
- Hydro extraction: This makes the clothes ready for drying by warding off the extra water left in the clothes after washing.

- Drying
- Pressing: Ironing the clothes for a better acceptability and comfort
- Packing, storage and distribution of cleaned clothes/laundry

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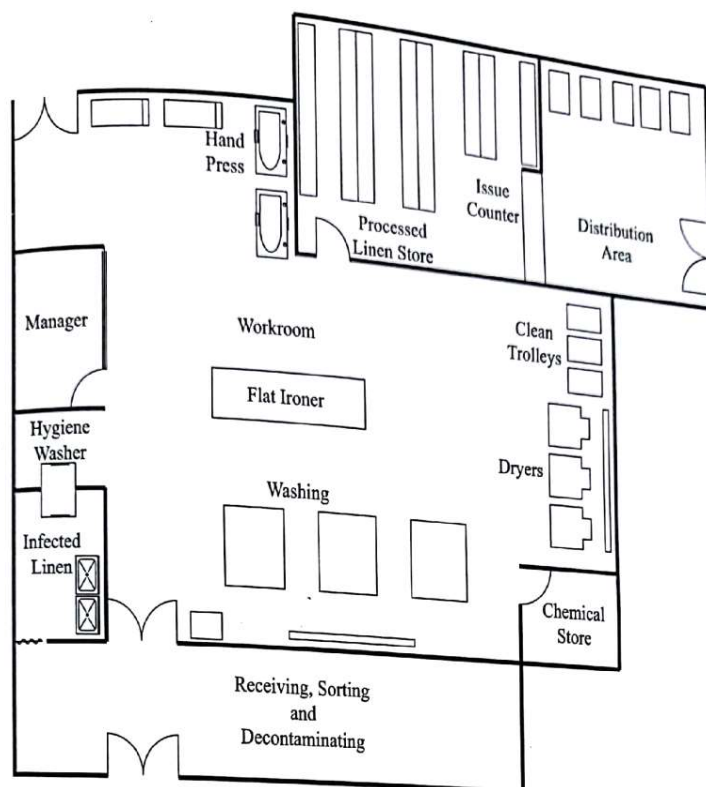


Fig. 11.1 A Simple and Planned Layout of a Hospital Laundry

- Separation of rooms is important. There must be a different room for holding the soiled/dirty clothes before they are ready to be taken for processing.
- The planning should be such that the linen to be washed should be 4kg per bed per day with a 10-20 per cent chance of infection. Per operation, around 8kg of linen load to be processed can be expected, as is the case with deliveries too.
- Separation of infected clothes from soiled or cleaned ones is vital. There must be separate partitions for the handling of infected and soiled linen. Also, it should be a mandate to sluice and disinfect the infected linen before further processing it.
- Physical and well maintained separation of cleaned and soiled laundry is a must. Separate entry and exits for the laundry help in maintaining this.
- Damage and defection of machines must also be kept in mind, which ask for a higher head room for repair and service. A head room of 4.5m is recommended.

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- Since there is a lot of water work involved, the floor must also be facilitative of the same. It must be smooth, clear and water repellent.
- Surface of walls should also be washable and smooth.
- Infection attracting surfaces must be demolished/discouraged, for example, dust collecting projections.
- Width of the door must be facilitative of trolley and machinery movement. Recommended width is a minimum of 1500mm.
- Proper ventilation is also a must in the laundry, as infections are in plenty. An air exchange rate of 10 per hour is appropriate.
- Proper exhaust pipes for steam are required and hence a proximal closeness to the boiler room is also called for.
- One of the most important components and resources needed in a laundry is water. A water supply of 10 litres of cold water and 15 litres of hot water per kilogram of linen is required.
- Utilization of a solar resource whenever possible should be encouraged, like for thermal systems.
- Water softening plants may also be needed in case of hard water supply.
- General safety provisions like those for fire safety are also required, such as installation of fire alarms, sprinklers, water hoses and fire extinguishers.
- Mobile storages of clean and soiled laundry are required and may bring to use trolleys. This must be separate for both kinds of linen.
- Installation of equipment should be such that it must be able to handle the workload of seven working days per week.
- Also, a laundry system and arrangement must also have separate spaces for the following setups:
 - Sewing room
 - Laundry manager's office
 - Cleaning facilities and restrooms for the workers.
 - Drainage system should be proper and must contain lint collection and sedimentation chambers.
 - During the planning and designing of the laundry, future needs for expansion and increased workloads must be kept in mind.

Size

For the maintenance and proper utilization and usage of linen, the general principle followed is that for every one piece of linen in use, four other pieces are already being processed or have been processed and are stored. Therefore, the inventory

maintained consists of daily use items multiplied by five in number. Here's how each one of them is in use at any particular time:

- A pillowcase being used at the patient's bed.
- A clean pillowcase in the closet at the nursing station.
- A dirty/soiled pillowcase in the laundry basket/with the dirty clothes.
- A pillowcase being cleaned in the laundry.
- A clean pillowcase in the linen store.

Capacity

An estimation of the machinery required and the probable requirements pertaining to equipment and alike can be made by the compilation of a list of the quantity and types of clothes to be washed per week. For planning, however, this may not be possible. Hence, the following method can be used for the estimation:

- **American Guidelines:** Laundry is averaged at 15 pounds (6.8kilograms) per bed for a day and an additional 25 pounds (11.33 kilograms) for operations and/or deliveries.
- **British Guidelines:** Here, the estimation is made at a weekly basis, wherein each bed is averaged at giving out laundry at a rate of 60 articles per week, each article standing at 0.39 kilogram individually.
- **Indian Standards:** The basics of Indian standards lie with a simple thumb rule, which says a bed produces 3-5 kilograms of laundry to be processed per day.

A categorization of the soiled laundry is also made. Soiled laundry is categorized as under:

- Normal used and soiled linen:** This estimates a total of 80-90 per cent of all linen that comes in for processing and this type has been used for regular day to day activity.
- Infected or fouled linen:** This type of soiled laundry, although only 10-20 per cent of the total, is of concern as it is infectious and may also be biologically hazardous. Therefore, it commands for special treatment and processing. All napkins used for new-born's and infants are treated as infected.

For the estimation or calculation of the actual workload, linen for all seven working days is laundered on six working days of the week for the whole week. This implies that the laundry must carry the capacity to launder a week's linen within six working days.

Soiled and infected linen is comprised of large flat clothes (sheets, etc.), smaller flats (pillowcases, napkins, etc.), tumble work (bathing towels, bed-sheets, top-sheets and blankets, etc.), ironing-work (garments, etc.), nursing and paediatric clothes, and isolation clothes.

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Designing of the Laundry Department

There are mainly 3 steps in the processing of laundry that decide the work sequence or workflow: receiving, processing and dispatching. It is imperative for the designing of the laundry department to be in accordance with these three steps for the laundry to have a smooth workflow and effective working.

There are similarities in the ways of working of laundries in hospitals and other institutions like hotels. However, what sets hospital laundries apart is the kind of clothes they receive for processing, i.e. hospital laundries receive specialised items and tasks, such as infections, bio-hazardous linen, etc. The most important of these specialities of hospital laundries are disinfection and control of infection due to the larger availability of fouled/infected linen. These factors contribute to laundries requiring special designing to cater to the removal of these infections (bacterial) and also processes like asepsis.

It is majorly a failure on the part of the hospital planners and the administration in foreseeing the layout and following the principles of central sterilization and the supply department. A strict separation must be maintained between normal soiled linen and the infected or fouled linen. This may be reached by installing a few equipment and better designing: installation of double door, pass through washing machines through walls which separates the soiled area and the clean area. Linen is put in on the soiled side and it is taken out on the cleaned side.

However, this physical separation of the soiled and clean areas relies heavily on the designing of laundry and also on infection control. In conventional means, the steps are carried out in the same room as other laundry institutions like hotel laundries. However, this opposes the recently established concept, which separates the cleaned and soiled functioning throughout the hospital. While washing the linen, a huge amount of bacteria is released into the air in the laundry. This bacterium then settles down on the clean linen, thereby infecting them. This re-infected linen is then distributed to the patients. This increases the risk of infections in patients which can only be prevented by the separation of processing areas.

It is vital to understand that this separation must start right from the time of soiling or infection. One method to ensure this is the separation of laundry bags distinguishable by colour. There must be enough space to close the bags and the bags must properly be sealed and should temporarily be deposited in well-ventilated holding area, where it waits till being collected for laundry processing.

Types of Laundry Services Designs

There are three types of designs mainly used for laundry services:

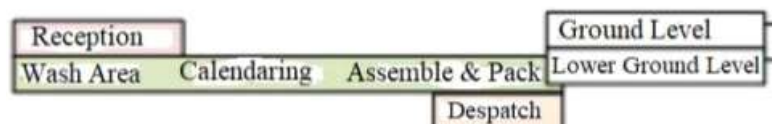
- i) Straight: There is a straight aligned workflow in a line consisting of a reception area that receives dirty clothes, a process area and an output area.



- ii) U-flow: Here, the arrangement of the three areas given above is in a U-shaped area.



- iii) Gravitational: This particular design type is used for vertical multi-floored buildings.



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Structural and Engineering requirements

- A large amount of water is consumed by mechanical laundry. 25 litres of hot water and 15 litres of cold water is used for processing every kilogram of linen.
- A requirement of 100 litres a day per bed is essential for smooth functioning of laundry services as a thumb rule.
- 300 lux lighting in the working area is needed. The reflecting values of paint surfaces required are estimated between 0.74-0.81.
- In case of hard water supply, it is essential that a water softening plant be set up in the premises.
- An adequate number and positioning of 15A sockets is required. Backup-supplied 400watt and 3-phase connection is needed.
- In order to minimize the distribution losses, collocation of a steam plant is required. This collocated plant must provide steam at 170-180°C temperature and at 100-125 Psi (Pounds per square inch)

11.2.3 Types of Laundry Services

- In plant system laundry services are further of two types:
 - a) **General System:** Here, the hospital has its own laundry system. It is considered the best system for laundry management in the hospital. These laundries are more suitable for bigger hospitals.
Advantages:
 - o Loss and damage of linen is reduced
 - o Reduced cross infection
 - b) **Barrier System:** This is better for sites demanding higher hygiene levels and this is done by creating barriers/separations between soiled and fouled linen, which reduces the scope of cross infection. The best

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machine for a larger hospital is the tunnel-washer. Soiled linen is placed in one end of a long tunnel and are moved through the washer by the use of an Archimedes' screw. Separation of loads into batches is done by this screw. Entry of water and chemicals into the machine is from the other end. Hence, every time the laundry goes into the next batch, it is cleaner and cleaner water is used.

Smaller laundries or establishments make use of pony barrier washers. The machine contains two doors, one each on the clean and dirty sides. Loading of laundry is done on the dirty side, followed by the closing of the doors. Post this, the program is set up and initiated by pressing the start button. Upon the conclusion of one cycle of laundry cleaning, the workers on the clean door's side remove the processed laundry. There are minimal or no germs that have proceeded to this part of the machine.

- **Off-site System:** This is subdivided into three types
 - a) **Rental system:** Hospitals that run on a smaller magnitude find their comfort in this type of laundry services, mainly because it isn't considered economical for them to have their own laundry. Here, hospitals hire the clean and laundered linen from contractors. Hospitals with capacities of up to 50 beds may use this type of laundry system.
 - b) **Contractual system:** In this system, hospital is the owner of the linen, but with zero capabilities of laundering it. A contractor is hired. They collect the linen and deliver it to the hospital post cleaning. At times, the provision of space and facility for laundering may be done by the hospital itself.
 - c) **Co-operative system:** This laundry system is the most economic. The reason for this is that here a single laundry caters to multiple hospitals here. Again, this is more suitable to small scale hospitals. However, chances of cross infections are higher due to more workload and different sources of infection. In cases of strike, however, more than just one hospital's functioning is affective.

Equipment Used in Hospital Laundry

Since laundries in hospitals are important and specialised departments, there are some equipment that are used in a typical hospital laundry:

- Double-door washing machine
- Hydroextractor
- Sluicing machine
- Flat work ironer or rotary iron
- Tumble dryer
- Utility press

- Mushroom press
- Table trolley
- Hand iron
- Linen hamper
- Wet linen trolley
- Dry linen trolley
- Distribution trolley
- Hanger trolley
- Air compressor
- Platform scale
- Motorized sewing machine

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11.2.4 Workflow and Staff of Hospital Laundry

There are basically two ways of the working of a hospital laundry which is also based on the two types of clothes that come in, i.e. infected linen and normally soiled linen. Basic principles of both remain the same, with minor changes due to infections in fouled clothes.

- *Soiled linen:* The first step is storage after the collection from different departments. Then there is classification on the basis of types and sizes of clothes (large flats, small flats, speciality items, etc.). Post this, washing and extraction is done; after which there is conditioning of clothes. After conditioning, general processes like drying, ironing and calendaring are carried out. The assembling and packing is then done before dispatching of laundered articles back to the departments for usage.



Fig. 11.2 Laundry Cycle

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- *Fouled/infected linen:* The major difference between the treatment of fouled and soiled linen ideally begins at the sorting and collection. It is then taken to a specially constructed central disinfection area. Here, the storage, sluicing and washing of the infected or fouled linen are carried out. After this, the extraction is done and the clothes are conditioned. The processes after this are common to both soiled and infected linen, namely ironing, drying and calendaring. The assembling and packing is then followed by distribution and dispatch to concerned departments.

Area of Disinfection

As already discussed, there must be separations for the treatment of soiled and fouled linen. Fouled and infected linen must be categorized to one section of the reception area. There, it is stored until washing, if only in bags. Next, it is loaded in washing machines. It is important to separate this area from the area of cleaned laundry. This separation is best done by the use of double-door and pass-through washing machines. Ideally, sluicing machines are used for sluicing and disinfecting before washing fouled clothes.

However, it is not necessary to separate the reception points of fouled and normally soiled linen. Post unloading on the cleaner side of the laundry, both infected and soiled linen are treated normally for finishing.

Utilities

In the preliminary stages of planning and designing of a hospital laundry, careful studies and predictions of the utilities and services must be made. The vital and basic requirements of the laundry are water, power and air compression. There is a lot of water consumption in the laundry. Sufficient sources to meet this requirement must be present. Also, there must be enough planning for the regular and timely discharge of effluents. Needs for adequate power must also be fulfilled at all times. Whole of the normal power must be kept standby. Steam and temperatures are also needed in adequate quantities. The most important thing with the provision of steam is that the laundry must not be at a far off distance from the source of steam. This minimizes the line losses of steam and also provides ample heat to the work ironers and presses. Compressed air is also needed for the operation of these flat work ironers and presses.

Projections must be made for the maximum possible demands of the utilities. The most accurate way for this calculation is the usage of different types of equipment simultaneously and also keeping in mind the scope of future expansion and increase of workload along with.

Selection of Equipment

As with everything else, the laundry services and systems have also revolutionised in the recent years. The development of labour saving machinery is widely seen,

with the likes of automatic washing machines. These have resulted in the better economy of the operating personnel and operating time. Some common features nowadays are automatic formula dispensers, automatic controls, sorting and counting devices; and machines that have combined washing, rinsing and extraction. There is also flat work folding machines for automatic folding now.

Selection of proper size and kinds of equipment is important in order to balance the economics and thus the production. Therefore, the laundry equipment must properly be selected. These are the factors that must be kept in mind:

- Maintenance of continuity of workflow
- Economical consumption of consumables like water, power, steam, etc. and of washing materials like detergent, etc.
- Work efficiency under normal working conditions.
- Ease of maintenance and the easy availability of spare parts
- Design reliability and compliance with Bureau of Indian Standards
- Reasonable capital cost

Staffing Pattern

The most important component of any organisation is its staff; and laundries are no different. There are mainly four types of workers needed to work in a laundry. These can be classified as supervisor(s), operational workers, skilled workers, semi-skilled workers and unskilled workers. Mandatorily, training in infection control and hygiene are a must for workers.

Calculation of Staff in Laundry

There are basic parameters for the staff calculation that is needed to work in the laundry. For every thirty beds, one man must be employed for work. Also, for 60-70 kg of linen, one washer-man is needed. Excluding managers, approximately 7 people are needed for a 50-75 bed hospital. For a 100-175 bed setup, a workforce of 12 people is adequate. For a larger establishment of having 200-300 beds, 15 people are required to do the laundering jobs.

However, as the size of the hospital setups increases, a 500 bed hospital needs a laundry manager, a supervisor, a couple of store keepers, 6 orderlies, a tailor, one sweeper and 17 operators.

11.2.5 Work Schedule of Hospital Laundry

The laundry requires a proper working schedule for its proper functioning and for avoiding unnecessary and unplanned shabby work. Hence, weekly and daily working schedules are made for laundry.

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Daily schedule

For the daily working, all equipments in the laundry must be properly cleaned. This prevents the spreading of the infections. Also, since equipment helps in the cleaning of fouled or soiled clothes, they themselves get dirty in the process. This calls for the proper cleaning, which makes it imperative for cleaning to be made a daily task in the laundry department.

Post cleaning, daily work procedures of laundry collection, categorisation and processing are carried out as usual. The cleaned laundry is then distributed as per schedule.

Another important task to be completed each day is the examination of the working of the equipment in the laundry. In some large setups, there may be checklists that persons need to tick off for proper functioning of certain elements and equipments.

Weekly schedule

On weekends (i.e., on Friday, Saturday, or Sunday) detergent and other consumables are bought and stores are replenished for the next working week.

On every Sunday, doctors' coats and useable items are cleaned. Weekends are chosen for this as doctors generally have a weekly off on Sundays.

Further, working of the laundry is also divided as per monthly works.

Monthly schedule

Some tasks also need to be done on neither a weekly nor a daily basis. This work affects the long-term functioning of the laundry system and hence needs to be done only once a month.

These tasks include checking of the linen stocks. It so happens that during the use of linen for various purposes in the hospital, wear and tear occurs and certain articles may need to be replaced. This can be done once a month.

Next, for any organisation to function, calculation of expenses is important. Laundry expenses need calculation on a monthly basis. This helps in minimizing extra losses and cost cutting.

Entries need to be made for damaged clothes. Contaminated and faded fabrics must also be noted in the register.

11.2.6 Washing Materials and Arrangements

There are three variables that affect the conditions needed to wash and process linen. These variables vary in accordance to the magnitude of soiling of the clothing articles. These variables are temperature, water level and time taken for processing. Lesser soiled linen requires low temperature, lesser water and lesser time. Contrarily, higher amount of all three are needed for more soiled articles. There

are electric heating elements in the machines that heat up the water. White clothes require a higher temperature of water (60-90°C). Whites are always washed separately from coloured. Coloured clothes are washed at cooler temperatures of 30-40 degree centigrade. Detergent functions better at the optimum temperatures. During washing, the clothes are irritated in detergent which softens the dust. However, both the detergent and dirt remain on the clothes.

Dry cleaning

Dry cleaning is essentially laundering, but it contains a chemical solvent rather than water. It is generally used for those fabrics that are harmed by water, or those that may get damaged in washing machines.

However, contrary to the popular belief, it is not really dry as there are other solvents used to soak the clothes. Tetrachloroethane, trichloroethane and petroleum spirits are some such solvents.

Detergent enzymes

Detergent formulation finds an integral part of its origin in detergent enzymes. Industrially, detergent enzymes' market share is pretty high as compared to the shares of other enzymatic applications. In order to be used for detergent activity, an enzyme must have the following characteristics:

- A higher temperature range of action; about 20-50°C, and even more
- Its optimum pH for functioning should be around alkaline or higher
- Its compatibility must be towards detergent
- Its specificity must be towards numerous proteins

Some prominent detergent enzymes are proteases, amylases, cellulases and other miscellaneous enzymes such as peroxidases and pullulanase. Reduction of the phosphate content in these detergents has been in trend. It can be replaced by sodium carbonate plus proteases.

As the study suggests, proper laundry management in a hospital is imperative for the proper functioning and customer satisfaction of a hospital.

Check Your Progress

1. What are the main functions of a laundry department in a hospital?
2. What do you mean by patient linen and procedure linen?
3. List the basic and vital requirements of a laundry.
4. Name the common features frequently seen in automatic washing machines.
5. List the essential laundering job profiles in a hospital setup.

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11.3 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

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1. The main functions include collection of used and infected linen, processing of infected linen, replacement of damaged clothes, washing, distribution of clean linen and maintenance of stock of linen.
2. Patient linen, is used exclusively by the patient. It includes gown, pyjamas, coat, shirts, etc. Procedure linen, it consists of linen used in OT, labour room, etc. It includes surgeons coat, mask, shirts, OT towels, trolley covers.
3. The vital and basic requirements of the laundry are water, power and air compression. There is a lot of water consumption in the laundry. Sufficient sources to meet this requirement must be present.
4. Some common features in automatic washing machines nowadays are automatic formula dispensers, automatic controls, sorting and counting devices; and machines that have combined washing, rinsing and extraction. There is also flat work folding machines for automatic folding now.
5. The essential laundering job profiles in a hospital setup include a laundry manager, a supervisor, a couple of store keepers, orderlies, a tailor, sweeper and operators.

11.4 SUMMARY

- The word 'laundry' finds its origin in the word 'launderer', or 'laundress'; which essentially means a washer man. Hence, the word laundry defines the activities that pertain to the cleaning and/or washing of clothes and linen.
- Any hospital relies strongly on its laundry service; especially in today's times, when the patients and kin expect the change of linen on a daily basis. This also increases the demand of the number of linens. Also, it maintains the overall life of linen by maximizing its resting period and minimizing the wear and tear.
- The laundry in a hospital can be maintained and run in two ways. It can both be developed and run in-house, or it can be contracted/outsourced. In case of in-house laundry services, the equipment for drying and washing must be proper and well installed.
- In case of outsourcing the laundry services, there must be enough provision of area for the reception of soiled linen, storage, processing and clean linen area. Sorting areas are also required. Other equipment for transport like trolleys, staff, etc. must also be taken care of.

- Hospital laundry has a main objective which lies in the provision of adequate quantities of linen for Operation Theatre, OPDs and other departments.
- Clean linen is of great importance particularly in hospitals as it prevents transmission of germs from one patient to another or from attendants to patients, comforts patient, and builds image.
- General purpose linen are the one which are not used for patient care and includes curtain, tablecloths, drapes. Ward linen, are the ones which are used for patient care like blanket, pillow cover, bedspreads.
- The laundry must be situated such that it receives ample sunlight as well as ventilation. The ideal location of hospital laundry is considered to be on the ground floor, in an isolated building near to the power plant, as the laundry is one of the departments that uses the most power, water and steam.
- The better possible place for the laundry department is closer to the Central Sterile Service Department or CSSD or to the dietary department since the laundry requires steam which is easily available at both these places due to the boiler.
- As compared to other methods, a well-organized and mechanized laundry is safe as well as convenient and dependable.
- An estimation of the machinery required and the probable requirements pertaining to equipment and alike can be made by the compilation of a list of the quantity and types of clothes to be washed per week.
- Design is made in such a way that it facilitates the reception of soiled linen, sorting (categorised on the basis of material, colour, type of cloth.), Sluicing (preliminary washing of clothes), washing, hydro extraction (making clothes ready for drying), drying, pressing and finally packing.
- For proper functioning and maintenance of laundry, a general principle of one piece of used linen is used. For one piece of linen, four other pieces are already being processed or have been processed or stored.
- There are mainly two kinds of laundry services, which are named in-plant system and off-site system. The in-plant system is more suitable for larger hospitals and has two sub-types: General System and Barrier System. Off-site laundry system is found to be more suitable for smaller hospitals. This system has 3 sub-types: rental system, contractual system and co-operative system. Each has its own advantages and disadvantages.
- Some common laundry equipment include double door washing machine, hydroextractor, sluicing machine, flat work ironer, tumble dryer, utility press, mushroom press, table trolley, hand iron, linen hamper, wet and dry linen trolleys, distribution trolley, hanger trolley, air compressor, platform scale and a motorised sewing machine.

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- Hospital laundry segregations are made on the basis of fouled linen and soiled linen separated. The basic laundry cycle remains the same; i.e. collection of dirty laundry, sorting dirty laundry, pre-treating stains, washing, rinsing and drying, repairing and ironing, folding and stacking and finally delivering the linen to the respective rooms.
- There are three basic requirements of a hospital laundry, constituted as compressed air, water and power. The water consumption in a laundry is exceptionally high.
- Laundry mustn't be far from the source of steam, so this affects the basic placing and planning of the laundry right from scratch, since this placing minimizes the losses of steam and also provides proper heat for the functioning of ironers and presses.
- Four types of workers are needed in the laundry service, supervisor, operational worker, skilled workers, semi skilled workers and unskilled workers.
- For weekly schedule, on every weekend detergents and other consumables are brought and stored for further use. On every Sunday, doctor's coats items are cleaned. In monthly schedule, work is carried out once in a month. This work affects the long term functioning of the laundry. It includes checking of linen stocks, calculation of expenses. Washing conditions are affected by the three variables, these vary in accordance to the magnitude of soiling of the linen.
- Coloured clothes are washed at cooler temperature. Detergent functioning is better at optimum temperature. Other type of linen cleaning is dry cleaning. Dry cleaning contains a solvent rather than water. It is used for those fabric which are damaged by water. Tetrachloroethane, trichloroethane and petroleum spirits are used as solvents.
- There are different enzymes used in the process of cleaning. Detergent formulation is an integral part of detergent enzymes. Detergent enzymes have high shares in the industry than that of any other enzymes.

11.5 KEY WORDS

- **Sluicing:** It refers to wash or rinse freely with a stream or shower of water
- **Archimedes' Screw:** It refers to a machine used for transferring water from a low-lying body of water into irrigation ditches.
- **Barrier Washer:** It refers to a machine which is equipped with two opposing doors one for the introduction of dirty linen and the other for the unloading of cleaned laundry, so that avoiding the recontamination.

11.6 SELF ASSESSMENT QUESTIONS AND EXERCISES

*Laundry Arrangement in
the Hospital Setup*

Short Answer Questions

1. Mention the main functions of the laundry services of a hospital.
2. Comment on the quality of the linen used in hospitals.
3. Briefly discuss the categorization of soiled laundry.
4. What are the types of laundry services designs?
5. List a few equipments used in hospital laundry.
6. Write a short note on working schedule of hospital laundry.

Long Answer Questions

1. Discuss the advantages of effective laundry services in a hospital.
2. How the hospital linen is classified? Discuss.
3. Describe the layout of a hospital laundry.
4. Elaborate on the different types of laundry services.
5. Discuss the workflow and staff of an hospital laundry.

11.7 FURTHER READINGS

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BLOCK - IV
GENERAL KITCHEN EQUIPMENT STORE

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UNIT 12 HOSPITAL DIETARY SERVICES

Structure

- 12.0 Introduction
- 12.1 Objectives
- 12.2 Importance and Functions of Hospital Dietary Services
- 12.3 Structure
- 12.4 Procedure
- 12.5 Organisation
- 12.6 Dietary Record
- 12.7 Menu Planning
 - 12.7.1 Menu Structure
 - 12.7.2 Standard Hospital Diets
 - 12.7.3 Therapeutic Diets in Hospitals
- 12.8 Quality Control
- 12.9 Answers to Check Your Progress Questions
- 12.10 Summary
- 12.11 Key Words
- 12.12 Self Assessment Questions and Exercises
- 12.13 Further Readings

12.0 INTRODUCTION

Diet is one of the most important aspects of human life that promotes good health and development. If taken correctly, it can prove to be the best method to gain resistance to infections and get rid of any diseases. Dietary service is one of the most crucial parts of the hospital services, as it helps in the recovery of patient's health by keeping in mind the need of the patients as per their physiological condition. It involves counselling the patients about the different foods and diets. The food service department is mostly managed by a person, who is educated in the field of food or by a professional dietician.

12.1 OBJECTIVES

After going through this unit, you will be able to:

- Understand the importance of hospital dietary services
- Analyse the functions of hospital dietary services
- Discuss the organisation of hospital dietary department

12.2 IMPORTANCE AND FUNCTIONS OF HOSPITAL DIETARY SERVICES

Hospital dietary service goals can be best achieved by the cooperation, mutual understanding, coordination, and the dedication of all who are concerned with the patient care. The functions of dietary service department are as follow:

- Delivering the best quality of food to the patients at a reasonable cost
- Establishing policies, plan layouts, and equipment requirements
- Purchasing raw materials that are required to make food, checking their quality, quantity, storage, and food preparation.
- Educating and counselling of both in-patients and out-patients on special dietetic problems before discharge.
- Providing accurate nourishment to the patients as per their disease, clinical condition, anthropometric measurements, food allergies, preferences, and the instruction of the doctor.
- Proper planning of the menu as per the patient's preferences as there should be a variety of food choices available to them.
- Training of dietetic interns
- Educating kitchen's staff about the maintenance of hygiene, when handling the food.
- Maintaining of all day records of the diets/meals.

Dietary services in hospitals are divided into three components:

- I. Dietary counselling** includes the interaction of patient with the dietician. Dietician visits every indoor patient and examines the nutritional requirement of the patient and plans diets accordingly. Counselling of the patient is done to inform them about the food that should be taken and that should be avoided.
- II. Dietary education** includes the training of hospital and kitchen staff regarding the cooking of food to maintain the quality of food, cleanliness of kitchen and prevent food contamination. It may also include training of dietetic interns to provide them the work experience.
- III. Diet planning and management** plays an important part in food service department. It includes everything from the infrastructure of kitchen to diet planning. The infrastructure of the kitchen is planned by keeping in mind the elements like location of the kitchen, which generally is the ground floor for easy delivery of the food. The layout of the kitchen is divided into various sections like the receiving area from where all the raw materials and other preferable food items are received from the vendors. Quality and quantity

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of all the items are also checked in this area, followed by the sanitisation. There is a storage area, near the kitchen, where food is stored, according to the conditions required by them in terms of temperature, moisture, sunlight, etc. Different sections are provided for different types of works like chopping section, bakery section, chapatti section, boiling section, liquid section, food serving section, etc.

Nutrition needs of the hospital patients

After admission to the hospital, the nutritional assessment of the patient is performed. Screening is done to check the risk of malnutrition at the time of admission and during the stay as well. A nutritional care plan is made, implemented, and evaluated. The following things are considered during a medical/nursing assessment.

- Food preferences are into account.
- Food related allergies are kept in mind for patient facing any allergies.
- Cultural/religious/ethnic preferences, like some people prefer no onion and garlic in food as per their religious beliefs, are considered.
- Social/environmental related choices must be considered.
- Inability to eat or drink as per the specific patient is kept in mind, for example, people having difficulty in chewing food are provided with soft diet.
- Equipment support is provided to be used when having food.

12.3 STRUCTURE

The facilities that are required by the hospital dietary department are planned, by keeping in mind the factors like number of beds, number of meals, type of food, number of workers, and other additional needs. The hospital kitchen generally is located on the ground floor near to the lifts for easy delivery food to different sections of the hospitals. The storage area should also be near the hospital kitchen. The layout of the hospital kitchen should be designed in a functional manner so that every section of the kitchen corresponds with the flow of work. The layout must follow the fixed guidelines of activities that has to be performed in every section like room service, storage area, food preparation area, cooking area, sanitisation area, bakery section, staff room, office area, and dietician room.

Areas of kitchen

The different sections of kitchen include:

- I. Receiving Section:** In this section, all the raw material like fruits, vegetables, eggs, meat, etc. are received and evaluated in terms of quality and quantity. This section provides all the required equipment for measuring and quality checking.

II. Storage: Storage of different food items requires different conditions. Storage area should be near the receiving area. There are two kinds of storage; dry storage and refrigerated storage. Dry storage is used for staple items and cold storage is used for perishable items. Generally, hospitals buy various items like dry fruits, pulses, spices, etc. in large quantities to meet the sudden demand, if required.

III. Food preparation area: This area is also divided into several sections like chopping area, boiling area, liquid section, etc. Every section should be well equipped. The most important thing is that the size of this area should be large. The points that should be considered while preparing food are:

- Well oriented physical layout corresponding to the sequence of the work should be adopted.
- Only standardised recipes should be preferred.
- Proper cooking techniques should be followed to preserve the original flavour and to enhance nutritional value.
- Less time should be consumed in preparation of food.
- Well managed and supervised work should be ensured.

IV. Serving room: In this section, food trays, received from the preparation area, are prepared. Food trays are then set as per the diet prescribed by dietician. A kitchen dietician checks every food tray. Trays are then put into the tray carts or food trolleys to deliver food to patients. This section also requires a particular set of equipment like refrigerators, table tops, cupboards, cutlery, dishes, napkins, and other things.

V. Food delivery: Food trolleys are now equipped with an electric outlet to maintain food temperature. There are different ways of delivery in different hospitals like many small hospitals serve in separate hot food containers, while some serve in ordinary trays.

VI. Dishwashing area: This work is now less cumbersome because of the electric machines that are available. At one end the used dishes are loaded, while on the other clean dishes are taken. Separate baskets may be used for different types of dishes. Drainage of water should be checked often.

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Check Your Progress

1. Who is responsible for the management of the food service department?
2. Why should a hospital kitchen be located near to the lifts?
3. Mention any two considerations during a medical/nursing assessment.

12.4 PROCEDURE

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To maintain the quality of the whole process, the following procedure and practices should be followed:

- There should be a systematic menu planning for nourishment of the patient, according to the preferences of the patients.
- There should be a different arrangement for the preparation of therapeutic diets such as diabetic, low cholesterol, gluten free, soft diet, normal diets and others as advised by the doctors and dieticians.
- The services should not only be provided for patients but also to the staff separately.
- Dieticians should visit every patient daily to examine the nutritional status of the person, his/her dietary intake, give dietary advices and counsel them.
- Dietary counselling of out-patients should also be conducted.
- Information about the patients having food allergies should be transferred to the kitchen to prepare food accordingly.
- Food trays should always be covered with a lid or a foil to prevent contamination.
- The temperature of the food should be checked before taking food out from the kitchen.
- The cooked food should not be held for too long. It should be served in proper time.
- Supervision of the kitchen staff should be done frequently, especially at the time of serving food.
- The meal charges should be told to patients at initial stage of admission to the hospital.

General rules for the maintenance of food hygiene are:

- At the time of delivery, the temperature of the frozen food should be below -18 degree C and chilled food should be below +3 degree C.
- Unwanted food should be discarded after one hour and should be covered.
- Reheating of food should be avoided, if possible.
- Food should be distributed at a correct temperature like hot food at 63°C and cold food at 10°C.
- Cooking should be done at the day of consumption only.
- Storage conditions should be clean, dry, and pest free with a set temperature.

12.5 ORGANISATION

Historically, the chief of dietary service department is a dietician. However, in larger hospitals the people with a professional degree in management, food service, or hotel management are appointed, along with the dietician for supervision. In hospitals, with less capacity of patients, the dietician serves as a supervisor as well as the manager of the dietary service department. This service has two main functions: one is the administration and other is the food production for the patients.

Administrative service includes purchasing, menu planning, storage, etc. which take a lot of time and labour. This service also includes training students and dietetic interns. The large number of workers in this department are unskilled. They demand less salary, which lead to the poor work quality and extra efforts of the seniors to train them. Most hospitals make compulsory medical and physical examinations for all the workers in the food service department to ensure that they do not harm patient by infecting them. Therapeutic nutrition demands a well-educated, experienced dietician. Most of the diets are modified, if directed by a doctor in terms of its salt content, oil content, consistency, etc. There should be a proper marking of every type of diet. The staff of a hospital dietary department include:

- a) **Chief dietician:** The role of chief dietician is to make indents, quality control, and provide guidelines. The record of supervision is kept and stock is verified by chief dietician. Menu planning, diet counselling, and training are also done under his/her supervision. In food service, her/his role is to check uniforms, working hours, and proper food serving timings. Personal hygiene of staff, hygiene in cooking of food and the respective area are also evaluated by them. The therapeutic diets' preparation is also one of their major tasks.
- b) **Steward:** Their work is to supervise overall work and control the kitchen. They collect all raw materials from the storage area for cooking food. They check the distribution of food in wards. They manage general cleanliness of the kitchen.
- c) **Store keeper:** His/her main role is to keep the records of all items, whether dry or wet. They maintain the storage conditions required by perishable items and keep a check on any pest infestations in food items. They issue food items to the stewards.
- d) **Head cook:** They evaluate the cooking of food and supervise kitchen hygiene and food quality. They also keep a check on pilferage and manage kitchen discipline.
- e) **Assistant cook:** Their work is to distribute food to different sections. They assist in cooking like in cutting, peeling, grinding, etc.
- f) **Cooks:** They cook all the food.

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12.6 DIETARY RECORD

Dietary record is kept by the nursing staff and the floor dietician. This record contains information about the patient's medical record, dietary assessment, dietetic care, food habits, and dietary treatment. Effective recording helps the health care team in providing appropriate care to the patients. The dietician records the following factors with regards to the patient's modified diets:

- a) **Confirm the diet order:** Within the 24 hour of admission, it is important to take the dietary preference of the patient into account (except patient on the fluid diet).
- b) **Diet history:** A record of medical history, food allergy, life style, socio-economic status are taken into consideration. Then, assessment is done to evaluate patient's nutrition care plan.
- c) **Nutrition care plan:** Type of diet, calories, sodium, and potassium is noted. There is also a record of patient's dietary intake, fluid intake, and medication. Any changes in the treatment are also noted.
- d) **Nutritional care discharge plan:** This includes brief information about the food to be included and excluded from the diet, at home. This is for the convenience of the patients and their families. A copy is kept in the department for the reference. It also includes a plan of patient's continued nutritional care and the date of any future visit.
- e) **Dietetic counselling:** Any advice given by physician should be considered by the patient and the family members. A written consultation by dietician includes patient's assessment, dietary history, past medical record, previous dietetic care, and recommendation for normal or modified diet. Counselling should be recorded in patient's medical record.

Check Your Progress

4. What does administrative service include?
5. What does the profession of steward include?

12.7 MENU PLANNING

In nutrition care, processed food is as important as medical therapy. Food, which is provided to the patients, relies on the catering service of the hospital. Catering service should focus on the food preference of the patients like their cultural preferences and their needs. Attention should also be paid to the appearance, colour, texture, taste, and selection of food. Other than these considerations, menu should be nutritionally balanced. For an effective menu planning, sufficient

information and input must be collected from various groups in the hospital. For nutrition care, clinical standards should be set for food. Planning should be done by a group of people responsible for implementing local protocols designed to provide food and fluid to the patients. This group should include senior dietician, senior doctors, senior nurses and senior members of the catering staff. Planning of menu should include:

- Menu preparation should be done by using standard recipes.
- Menu should meet the food and fluid requirements of the patient.
- Menu should be planned according to the condition and appetite of the patient.
- Meal timings should be planned.
- Nutrition content of the menu should be analysed.

12.7.1 Menu Structure

Every hospital has a different menu structure. Menu is planned considering the dietary needs of the population, age group, and disease. It includes different courses of meal. There are also in between meal service and snacks. It is planned in such a way that it allows enough time between meals and snacks. Under a menu structure, portion size of the meal is also planned. It also considers the amount of calories, proteins, fats, vitamins, and minerals obtained from a meal. The meals are divided according to the time of day:

- a) **Morning:** In a hospital menu, there is a provision for serving tea or coffee.
- b) **Breakfast:** According to the condition of the patient, breakfast is served. Breakfast includes cereal (including wholegrain varieties, porridge, oats, milk, bread, beverage, etc.). Breakfast aims to provide sufficient calories and protein to meet the nutrient needs. It should provide 200 kcal.
- c) **Mid-Morning:** At this time light snacks such as fruit and beverage are served, which provide approximately 150 kcal.
- d) **Lunch and dinner:** This meal is capable of providing 500-600 kcal. This meal contains cereals, pulses, fruits, vegetables, and fat. This meal is larger than other meals.
- e) **In between meal snacks:** In between meals, snacks are served to meet additional nutritional requirement of the patients. Snacks also add flexibility and variety to the menu. There is pre-planned list of snacks from which patient can choose, according to their taste. Snacks should be served one hour prior to the next meal. It should include at least one fruit. Snacks can be served three times per day.

In-ward or services are essential as they ease supply and storage of food, whenever required. If any patient refuses to eat any item, then that food can be

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stored and used for future demand. This service is of importance, when the hospital kitchen are closed like at night. Beverages, juices, tea bags, sugar sachets, hot water, cold water, etc. can be easily supplied when needed. Policy for ward supply about stock rotation, storage, food handling and hygiene should be developed at local level for quality control.

12.7.2 Standard Hospital Diets

For hospitalised patients, dietary modification of normal diet is performed. This modification is done based on the condition of the patient, willingness to eat, along with some dietary restrictions. Length of stay in hospital is short and meeting optimum nutritional requirement is important. Due to therapeutic restriction, surgery, stress, and illness, adequate intake is often compromised, which may lead to slower recovery. Therefore, it is important to plan diet, which can meet all the nutrient needs even with the compromised intake of food and fluids. There are different types of standard diets in the hospital that can be categorised as:

Normal diet

Normal diet is the diet with no restrictions. This type of diet is given to patients whose condition does not require any special instructions. Patients who can tolerate regular food are served this type of diet. All types of food can be included in the diet like whole grains, juices, fruits, milk products, chapatti, eggs, tea/coffee, fish, etc. But consumption of red meat, pork, alcohol, processed foods are avoided. It is a well balanced diet and provides adequate nutrition. It provides 1800-2000 kcal and 40-45 g of protein.

Clear liquid diet

This type of diet is given to patients who are suffering from acute illness or have underwent some kind of surgery. In this diet patients are advised to avoid intake of whole foods. This diet is free of residue, non-gas forming, non-stimulating and non-irritating. This type of diet cannot meet the nutritional requirement and should not be continued for a long duration. At first, the amount of fluid intake is restricted to 30-60 ml per hour and then is gradually increased according to the patient's tolerance. This diet gives 300 kcal and no protein. Clear liquid diet is suggested to the patients suffering from any of these conditions:

- Acute infectious condition
- Acute inflammation of intestinal tract
- Colon or rectum surgery

Full-fluid diet

This is an intermediate diet between clear fluid and soft diet. This diet is given to patients suffering from acute gastritis, acute infections, and diarrhoea. This diet is

also suggested to the patients, who can tolerate milk and can eat solid or semi-solid foods. This is adequate in terms of the nutrition and can be continued for a long time. This diet is given in 2-4 hours interval. It provides body with 1200 kcal and 35 g protein.

Soft diet

This type of diet is often given to patients, who have difficulty in chewing. It is the most frequently used hospital diet. It may also be suggested to patients suffering from acute illness, acute infection, or post-surgery. It is made up of easy to digest foods which do not contain harsh fibre, are low in fat, and are not highly seasoned. Soft diet provides 1500 kcal and 35-40 g protein.

12.7.3 Therapeutic Diets in Hospitals

A therapeutic diet is the diet that limits the intake of some foods or particular nutrients. It is a part of the medical therapy. This is normally planned by the dietician, depending upon the physiological conditions of the patient. It is a modified form of a regular diet. Reasons for prescribing therapeutic diets are:

- Maintenance of nutritional status
- Increase or decrease in the calories given in food
- Provision of greater amount of nutrients like protein
- Decrease in the amount of some nutrients like potassium
- Provision of texture modifications in case of problems with chewing
- Exclude allergic food from diet if any or any food intolerance like gluten or lactose

Some common therapeutic diets are:

- a) **Renal diet:** This is for people, who are suffering from kidney diseases like CKD. This diet depends on certain parameters of the patient like serum potassium level, serum sodium level, etc. This diet is usually restricted to potassium, fluid, and protein.
- b) **Low fat/low cholesterol diet:** This diet is usually less in total fat and contains 250-300 mg cholesterol. This diet is given to the patients suffering from diseases of liver, gall bladder, or pancreas.
- c) **High fibre diet:** This diet is usually prescribed in case of diseases related to GI tract, heart diseases, and other metabolic diseases. This diet contains high fibre content as it is rich in fruits, whole breads, legumes, vegetables.
- d) **No salt diet:** In this food, addition of salt is restricted while cooking.
- e) **Diabetic diet:** This diet is restricted to calories, carbohydrates, and fat. Food with low glycemic index are included in this diet. The most common caloric levels that are prescribed to the patients are 1200, 1600, 1800, and 2000.

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NOTES**Check Your Progress**

6. Who should be responsible for menu planning?
7. Which type of patients are given clear liquid diet?

12.8 QUALITY CONTROL

Providing safe and nutritious food to the patients in the hospital, is a major responsibility of catering staff and hospital staff. Staff must be trained and supervised to ensure proper implementation of standards of quality for food safety and hygiene. Quality control should include hazard analysis, inspection, and microbiological analysis.

Hazard analysis

For safe catering, practical methods need to be designed, which should be based on risk assessment and involve looking at each operation step by step. Hazard analysis system ensures that products produced are chemically, microbiologically, and physically safe. For hazard analysis, hospital should follow HACCP guidelines, which will help in quality control.

Inspection

Health authorities should conduct daily inspections of kitchens and all food handling areas, including catering staff, supervisory staff, and infection control doctor. This inspection should monitor sources of micro-organisms, areas prone to growth of microorganisms, conditions favouring their multiplication, temperature, and the routes of cross-contamination. Based on this inspection, a risk assessment report must be prepared and should be submitted to the hospital chief executive and control of infection committee.

Microbial analysis

Microbial analysis is a planned procedure, which includes assessment of food, which are frozen to investigate suspected food-borne illness. This inspection is carried out by catering manager and medical microbiologist. Food samples are collected every day and sent to laboratory to determine its quality.

Pest control

Places, which are warm, moist has food and water, which are used by rodents, ants, and cockroaches for shelter. In hospital, these places can be hospital kitchen, boiler rooms, ducts and drains, water supplies, structural defects, etc. The presence of pests is not recognised until their outbreak. To prevent the growth of pests, authorities should take care of cleaning, having proper food storage and prompt

waste disposal system. Also, the structural defects must be repaired. Regular inspection should be carried out by trained pest control officer. It is better to carry out inspection at night.

Quality control is not only limited to the food. It is equally important to maintain hygiene of equipment, in which, food is cooked, storage area of food, kitchen and raw food items. There are few things which should be regularly followed to maintain hygiene and quality of the food.

Quality of raw material

Before serving food to the patients, raw food items should be checked for the quality. At the receiving area their quality is checked and then they are transported to the kitchen for production. Every hospital follows certain guidelines such as:

- Policies are made for checking raw material.
- Raw material should be obtained from government or co-operative stores.
- Raw material should be fresh.
- It should be clean, free from dust, and pests.
- Kitchen advisory committee should check the quality.

Quality control of food

It is important to maintain the quality of food prepared. In hospital setting, it is even more important as the contamination can cause infections. Only standardised recipes should be used for cooking. Hospital should also conduct regular health check-up of the kitchen staff. Hospital kitchen staff should be well trained. To check the quality of the cooked food, surprise inspection should often be conducted by senior officials.

Quality control of the area

It is important to maintain hygiene of the area, where food is cooked and stored. For quality control of the kitchen and storage area following policies are followed:

- Regular cleaning of the kitchen area.
- Cold room temperature and deep freezers should be monitored regularly.
- All the equipment should be checked and calibrated.
- Proper waste disposal systems should be there.
- There should be proper ventilation and smoke exhaust.

Quality control of ward kitchen

In every ward a separate kitchen is there. It is used to store small amount of food and water. As it is present in the ward, it is used by nursing staff, patient attendants, and food handlers. It can be a source of infectious agents. To prevent contamination nursing staff should be trained, a written cleaning schedule and waste disposal

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policies should be prepared. There should be regular cleaning of refrigerators and microwaves.

Time for the quality evaluation

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- At the time of inspection when pests/rodents are observed
- Inaccurate quantities of food serving
- Wastage, theft, or pilferage of food
- The delivery of wrong food items to the patient
- When there are incidents of food poisoning
- Feedback received from patient, visitors, and staff in terms of quality, quantity, freshness/staleness, taste, etc.

Check Your Progress

8. Which places in the hospital are prone to pests?
9. Mention any three guidelines for maintaining the quality of raw materials.

12.9 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. A person, who is educated in the field of food or a professional dietician is responsible for the management of the food service department.
2. A hospital kitchen be located near the lifts for easy delivery of food to different sections of the hospitals
3. The considerations during a medical/nursing assessment are:
 - a) Food preferences should be taken into account.
 - b) Food related allergies are asked to avoid risk to patient
4. Administrative service includes purchasing, menu planning, storage, etc. which take a lot of time and labour. This service also includes training students and dietetic interns.
5. The profession of steward include supervising overall work and controlling the kitchen. They collect all raw materials from the storage area for cooking food. They check the distribution of food in wards. They manage general cleanliness of the kitchen.
6. A group including senior dietician, senior doctors, senior nurses and senior members of the catering staff should be responsible for menu planning.

7. Clear liquid diet is suggested to the patients suffering from any of these conditions:
 - a) Acute infectious condition
 - b) Acute inflammation of intestinal tract
 - c) Colon or rectum surgery
8. Places, which are warm and moist, having food and water, are prone to pests like hospital kitchen, boiler rooms, ducts and drains, water supplies, structural defects, etc.
9. The guidelines for maintaining the quality of raw materials are:
 - a) Policies are made for checking raw material.
 - b) Raw material should be obtained from government or co-operative stores.
 - c) Raw material should be fresh.

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12.10 SUMMARY

- Dietary service is one of the most crucial parts of the hospital services, which helps to support the recovery of health by keeping in mind the need of the patients as per their physiological condition. It involves counselling the patients about the different foods and diets.
- The facilities that are required by the hospital dietary department are planned, by keeping in mind the factors like number of beds, number of meals, type of food, number of workers, and other additional needs.
- Historically, the chief of dietary service department is a dietician. However, in larger hospitals the people with a professional degree in management, food service, or hotel management are appointed, along with the dietician for supervision.
- Dietary record is kept by the nursing staff and the floor dietician. This record contains information about the patient's medical record, dietary assessment, dietetic care, food habits, and dietary treatment.
- For an effective menu planning, sufficient information and input must be collected from various groups in the hospital. For nutrition care, clinical standards should be set for food. Planning should be done by a group of people responsible for implementing local protocols designed to provide food and fluid to the patients.
- Providing safe and nutritious food to the patients in the hospital, is a major responsibility of catering staff and hospital staff. Staff must be trained and supervised to ensure proper implementation of standards of quality for food safety and hygiene.

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12.11 KEY WORDS

- **Dietetics:** The scientific study of the food that people eat and its effects on health
- **Anthropometric Measurements:** Systematic measurements of the size, shape and composition of the human body

12.12 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. What are the functions of hospital dietary department?
2. What are the general rules for the maintenance of food hygiene?
3. What is the role of chief dietician?

Long Answer Questions

1. Discuss the components of dietary services.
2. Describe the various areas of the kitchen.
3. Evaluate the importance of dietary records.

12.13 FURTHER READINGS

- Llewelyn, Davies R; Macaulay, H.M.C. 1966. *Hospital Planning and Administration*. Geneva: W.H.O, Jaypee Brothers.
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UNIT 13 KITCHEN: EQUIPMENT, STORE AND VISUAL ARRANGEMENT

*Kitchen: Equipment,
Store and Visual
Arrangement*

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Structure

- 13.0 Introduction
- 13.1 Objectives
- 13.2 Introduction to Equipment, Store and Visual Arrangement
 - 13.2.1 Kitchen Spaces and Location
 - 13.2.2 Structural Features
- 13.3 Answers to Check Your Progress Questions
- 13.4 Summary
- 13.5 Key Words
- 13.6 Self Assessment Questions and Exercises
- 13.7 Further Readings

13.0 INTRODUCTION

Hospital kitchens are designed in such a way that they provide maximum productivity and meet the complete nutritional needs of patients. When we use technically advanced equipment for cutting vegetables and fruits, it reduces the chances of contamination and maintain the hygiene which is necessary for patients.

Most of the patients depend on the food prepared in hospitals because it provides good nutrition to the body and also balances its nutritional status. Over the years the food service in hospitals has changed drastically and nowadays patients also have the option of choosing food of their choice which is also healthy and beneficial for their speedy recovery.

When we eat a well-balanced diet it provides nourishment to our body and also helps in its growth. Hospitals, these days, have advanced machines for chopping vegetables into small pieces and cooking. These machines avoid contamination and are hygienic.

13.1 OBJECTIVES

After going through this unit, you will be able to:

- Describe the various equipment used in kitchen hospitals
- Explain the process of visual arrangement of food
- Discuss the features of day store
- Analyse the different sections of kitchen and food

13.2 INTRODUCTION TO EQUIPMENT, STORE AND VISUAL ARRANGEMENT

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The food service department in hospitals is also known as the catering department. This department has the responsibility of providing food to the patient according to their body's need and ability. Type of diet which is given to patient is decided by the dietician and the dietician takes care for the quality of food and food service. In hospitals it is common that visitors or family members come to meet the patients, so it is very important to maintain hygiene in hospitals.

Some patients do not accept the service provided by hospitals because they might have issues about food hygiene. Therefore it is imperative to provide proper hygienic food to patients to maintain the image of the hospital. The functions of the food department in hospitals are:

- Maintain the quality of food
- Avoid contamination and dirt
- Maintain the patients' health
- Provide food according to their health

13.2.1 Kitchen Spaces and Location

Kitchens are always designed according to their use in terms of work and nature. Kitchens can be square, L-shaped, U-shaped, rectangular and others.

- **Square kitchen:** These kitchens are not so common because of space utility issues. It is hard to use the space in centers. In square kitchens the pipes and drains are in the centre of the kitchen under the floor. If any leakage occurs in pipes it causes situations like floods in kitchens due to which the kitchen becomes unhygienic and unsafe. It is difficult to repair the leakage in pipes under the floor.
- **Rectangular kitchen:** These kitchens are very commonly used for the whole day activity. They provide large space for cooking so that different types of food can be prepared.
- **U-shaped kitchens:** These kitchens are most useful because they require less walking. In these kitchens work flow is very smooth and the sink is designed at the end of wall.
- **L-shaped kitchens:** This type of kitchen is useful for small canteens and tea and coffee shops.
- **Parallel kitchen:** These types of kitchens are useful for self-service canteens. In mid-day meal schools these types of kitchens are useful because there are no dining facilities.

Location of kitchen

In the past kitchen were designed in the basement area and there was less light and hygiene. As the air could not move in and out of the kitchen it caused bad stench. Basement kitchen shows harmful effect on food quality and it also effects the production of food. New rules and laws have been imposed now and according to them there is no allowance to design the kitchen in the basement. In most hospitals kitchen is located on the ground floor and there is light and easy allowance for air to pass in or out of the kitchen. These kitchens are safe, hygienic and can easily supply the food for the patient in the ward and the room.

Cooking area

In the cooking area only trained staff is allowed to enter. The trained staff knows how to cook food and maintain hygiene in the area. In a hospital kitchen there are five types of areas or demarcations.

- Meat, poultry and fish
- Vegetables
- Sauces and soups
- Therapeutic diets food
- Pastry and hot sweets

a. Meat poultry and fish cooking section

Firstly we put out meat and fish from refrigerators and then wash and weigh them and also give labels to them. After washing and weighing the quantity of food, the food is placed into the equipment for cooking.

For cooking ovens, high pressure steamers, boiling pans and grills are available.

b. Vegetables cooking section

Firstly the vegetables are taken from the refrigerators and then washed. For the cooking of vegetables convection ovens, high pressure steamers, bratt pans, fryers and boiling pans are used.

c. Sauce and cooking section

Sauces and gravies are also served to the patients. For preparation of sauces boiling pans, chilling machine and combination cooking are used.

d. Diet cooking section

As the name indicates special diet is prepared in this section. In a hospital some patients are under special diet and these special diets are known as modified diets. When a balance diet is modified according to the patient's health and the disease, the diet is known as therapeutic diet and this diet helps in the treatment of

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the disease. Convection ovens, boiling table and pans are the normal equipment used for the preparation of the diet.

e. Pastry and hot sweet cooking area

In this section convection oven and pastry ovens are present because these equipment help in the preparation of pastry.

13.2.2 Structural Features

It is important to design the kitchen according to its service and machines. If we plan a kitchen without the knowledge of equipment and its service, it creates problem in food distribution, hygiene and sometimes it also causes the situation of risk for the kitchen staff.

Gas connection, electricity and drainage are also the main features of the kitchen design. A kitchen should be designed according to all these factors. Lightning is also one of the main features of kitchen design.

Let's study some of the important structural features in a hospital kitchen.

1. Drainage

Hygiene and cleanliness of a kitchen depend on its drainage system. During the designing of a kitchen it is important to take care of the drainage system.

During washing of vegetables some leaves and the covering of fruits gets collected in the drainage system due to which the flow of water stops and it causes unhygienic water contamination. Sometimes fat also gets collected in drains and it prevents the flow of water.

Blocking of water flow causes problems like contamination of food and smell in kitchen. Contamination of food causes disease and it makes the patients' condition worst.

Size of drains must be 10 to 15 cm in diameter. To avoid the problems like blocking of water in drains and backward flow it is necessary to use grease traps in the drains.

2. Water supply

Water is an important component of any kitchen as it is used in washing of vegetables to the cooking of food and dish washing. So the supply of hot and cold water is necessary.

It is important to remember that the supply of both hot and cold water should be from the dirt free or uncontaminated source. Always avoid the use of contaminated water. In every three months it is necessary to check the quality of water.

Water tanks are used for storage of water. It is important to cover the water tanks properly to avoid water contamination. Cleaning of water tanks every month is recommended to avoid problems of contamination.

3. Floors

The floor of kitchen must be smooth and clean. During selection of material for the floor of kitchen always avoid slippery tiles because the use of tiles on floors cause a raise pattern and they require joints which makes the floor difficult to clean. The formation of raise pattern in floors causes noise when the food trolleys are passed from them.

So always make a uniform floor to avoid such discrepancies. Avoid floor material which easily becomes dirty and is difficult to clean. Always use waterproof floors and dark colours for floor because it is very normal to get stains on kitchen floor. Avoid the use of carpets.

4. Walls

Walls of a kitchen must be joints free, easy to clean, smooth and should never absorb moisture. Always make kitchen walls with hard plaster and also use emulsion with it. Use of emulsion makes it easier to remove stains.

Ordinary emulsions are always hard to wash and during the removal of stains from the walls they leave behind blurred or smeared marks on the wall surface. If the steam equipment is placed near the walls it is important to give the plastic finish to the wall. Never give tiles finish to the walls. Plastic finishes are bad absorbent and the steam produced from steam equipment does not affect the surface of walls. Enamel and oil paints are also used for the walls. Always avoid the formation of interior ledges because they easily collect dust and make the walls dirty. Walls should be waterproof and durable.

5. Windows

If there is any open window it should be pest proof. For the reduction of solar glare, the tinted glasses may be used. For stopping the slopping of birds exterior sills are used. Avoid interior sills.

6. Door

Insects and pests easily enter through the doors so always make external doors pest and insect proof to avoid them. Always make doors wide because there is a continuous passage of trolleys and equipment. The height of doors should always be according to the equipment and trolleys to avoid damage. Security devices are also associated with the doors to avoid the entry of people. There are sliding doors for chilled food stores and cold doors.

7. Ceilings

Ceilings should be clean and heat resistant and not affected by steam or gases and should be easy to maintain. Always use smooth, sealed and joint less ceilings for the cooking area. It is always important to spend money on long lasting things.

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8. Work surfaces

Work surfaces must be smooth. Always use stainless steel as work surfaces but it has cost implications. Laminated plastics and ceramic tiles can be used as work surface to reduce costs. Laminated plastics are low in cost and they are easy to maintain. Hardwoods are highly costly and difficult to maintain and they easily get stain marks and becomes dirty.

Marbles look well and maintain hygienic quality so they are good for food preparation work. Knife marks may easily transfer on all types of work surfaces so always use separate wooden surface in any work area where we use these sharp equipment for the cuttings of vegetables.

9. Lighting

Lights in kitchen should be designed in such a way that they work as the best source and provide light in the whole kitchen. Above the food preparation area and work tables light fitting must be present. There are many type of lights available in the market like:

- **Fluorescent:** Lights which have high temperature are not recommended for kitchens. CFL bulbs are also suitable if they provide sufficient lighting. They are high in cost but their maintenance is easy and they are more economical to use than other lightings.

They give six times more light than other lights with same electricity consumption. Lighting which does not give good colour effects are not advisable for use in kitchens, dining room or storage areas because they cover the colour of food. Tubes which have better colour and low light input are best for kitchens.

- **Mercury lamps:** These lamps are good for kitchen because they have ranges from 80 to 400 watt capacity. These are same as white fluorescent tube lamps in many features like colour, appearance, and life. These lights are good for eyes and they can also be easily fixed in ceilings. To protect the lamp from moisture and dirt, seal is used and it is easy to clean the seal.

For the steam canopies and rustproof materials additional lightning equipment is used. It is necessary for the equipment to avoid getting moisture and heat.

During the fitting of lighting it is important to arrange the lights properly. To avoid shadow falling on the work surface, always arrange the lights in parallel series and not in right angles. Water proof and easy to clean light fittings are used.

10. Ventilation

There is no special method of ventilation but the combination of window and exhaust fans is good. During cooking condensation starts. In this process there is the formation of droplets of moisture on equipment, food and surfaces. Due to the formation of moisture droplets on food surfaces it leads to contamination of food.

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Mold and bacteria can easily contaminate the food. Proper ventilation provides fresh air in kitchen due to which the bad smell from kitchen is driven out.

It also drives out extra heat from the kitchen and maintains the temperature. So during the designing of the kitchen it is necessary to provide attention on planning of ventilation and lightning in food preparation. Windows and exhaust fans are used as the equipment of ventilation.

It is necessary to cover the windows from the fly proof shutters to avoid the entry of insect, flies and dust. Vapour extractor hoods are also used for ventilation. Vapour extractor hoods are used for the removal of fumes, vapours and odour present over the cooking equipment.

There is a risk of fire in kitchen due to condensation of fat but the use of these hoods with the grease trap filters reduce the cases of fire.

Exhaust fans are used for the outlet of air from the kitchen. Fans are designed near the ceilings in the walls. Kitchen should be bright and happy as possible. Always use cool colours for eye like blue, beige and cream as these colours prevent the feeling of heat.

Always use light colours for good visibility. Selection of equipment with natural colours also comfort the eyes.

If the environment of kitchen is nice it helps to bring best skill and creativity from the staff.

Layout of Kitchens

Layout means the arrangement of equipment and services like drainage water and fuel supply. For the planning of layout different type of knowledge is required.

- Architecture
- Construction and design engineering
- Operation
- Management or administration of kitchen

Work centers in kitchen layout

To design work centers in kitchen always remember four concepts:

- **Smooth work flow:** It is important to manage a smooth work flow in kitchen. To manage smooth work flow all the cooking activities should be performed in sequence. There are four main work sections in a kitchen:
 - o cooking centre
 - o holding, service or dishing out centre
 - o washing area
 - o food preparation centre (all type of foods)

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- **Space between work tables and sinks:** It is important to provide enough space between the equipment. If there is less space between equipment and work tables of sinks it becomes difficult to work.

To avoid congestion it is important to balance the use of equipment. At one time only one person uses single equipment. If a lot of people come into the kitchen at the same time it causes congestion and contamination.

- **Comfortable work environment:** If the kitchen area is comfortable then the staff can do the work easily. Factors that affect the work environment are:

1. Temperature and humidity of the environment
2. Structural features
3. Hygiene and sanitation

Human equipment store

Equipment can be classified according to their:

- weight or size
- order of use
- mode of operation

Food preparation equipment

- Dough making machine
- Peeling machine handling
- Boiler for tea and coffee
- Chopper
- Espresso coffee machine
- Mixer
- Grinder
- Beater
- Bread slicer
- Wet grinder
- Digital catering thermometer
- Cooking equipment
- Ovens
- Fryer
- Griddle plate
- Salamander
- Mobile cooking rotisserie

- Rice milk cooker
- Fryer
- Double fryer
- Gas cum griddle plate
- Boiling pans
- Stock pots
- Gas cooking plate
- Chapatti plate and puffer
- Microwave oven
- Automatic egg boiler
- Burner cooking range
- Egg boiler for six
- Sandwich toaster

Holding service and clearing equipment

- Bain-marie holding and service counter
- Bulk food trolley
- Hot food service trolley
- Trolley
- Tea service trolley
- Tray carrier
- Collection trolley
- Clearing trolley
- Cutlery carrier
- Meat delivery container

Washing and storage equipment

- Pot washing units
- Mobile sinks with drains
- Wall mounting sink with drain board
- Dishwashing machine
- Pot rack
- Storage rack
- Two sink washing
- Three sink unit

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- Mobile work table
- Storage drums
- Glass washing machine

A lot of other equipment are used in kitchen.

- Boling tops
- Atmospheric steamers
- Pasta boiler
- Insulated food storage
- Food dicer
- Hand mixer
- Water filter system
- Mop sink
- Chest freezers
- Dishwashers
- Floor trough
- Chiller tank
- Water meter
- Tray dolly
- Heat lamp
- Beverages table
- Induction charger
- Packing tables with scales
- Basket
- Printer
- Scanner
- Hand sink with soap towel
- Pump fill station
- Food dicer
- Evaporator coil
- Exhaust ventilator
- Bakers table
- Double door upright fridge
- Chopping board
- Stock pot

- Wok
- Stir fry pan
- Sauce pan
- Roasting pan
- Grill pan
- Frying pan

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Service trolleys

Trolleys are used to bring food and beverages from the kitchen to the ward and room for the patients. These are the tables which are attached with the wheels. Some of the common types include:

- fruit trolley
- salad trolley
- bulk food trolley
- hot food trolley
- collection trolley
- clearing trolley
- tea service trolley

With the help of service trolleys it is easy to deliver food and beverages for the patients.

Flatware

These are the cutlery and silverware. These include spoons and forks and knife. With the help of flatware it is easy to eat food because they easily mix and cut and hold food of different sizes easily. These are made up of base metal such as steel. Some of the common types include:

- soup spoons
- fish knives and forks
- large knives and forks
- fruit knives and forks
- coffee spoons
- tea spoons
- service spoons and fork
- steak knife
- grape fruit knife
- cheese knife

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Hollowware

This equipment is used to bring food to the patient from the kitchen. These hollowwares are made up of metal or ceramic containers. Some of the common types include:

- Soup tureens - used to bring soup in large quantity from the kitchen
- Soup bowl – used to give soup to patients
- Tray – use to serve glassware and hollowware
- Round flat tray with lid – used to bring vegetables
- Water jug - to bring water
- Sauce boats – to presence sauce on tables

Special tableware

- Sugar tongs

Linen

The different types of linen used are:

- Table cloth
- Slip cloth
- Napkins
- Buffet cloths
- Burners
- Copper pans
- Punch bowl
- Cake stand
- Pudding cups
- Wooden salad bowls
- Melon bowls
- Bread baskets

Knives

A knife is made up of sharp blade and handle which is used to cut the vegetables into the small mixes. Blades are made up of steel and alloys and the handle is made up of wood which is associated with plastic sometimes. Some of the common types include:

- Chef's knife
- Carving knife
- Paring knife

- Utility knife
- Boning knife

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Filet knife

- Cleaver
- Steak knife
- Mincing knife
- Oyster knife
- Peeling knife

Cooking utensils

Different utensils are made for different type of cooking but pans and pots are used for almost every cooking.

• Chakla & belan

Chakla is made up of wood and marbles. This is round in shape. Belan is made up of wood and these are rolling wood which is used to roll the dough into breads and chapattis.

• Baking trays

These trays are made up of aluminium and steel and are available in large and small sizes.

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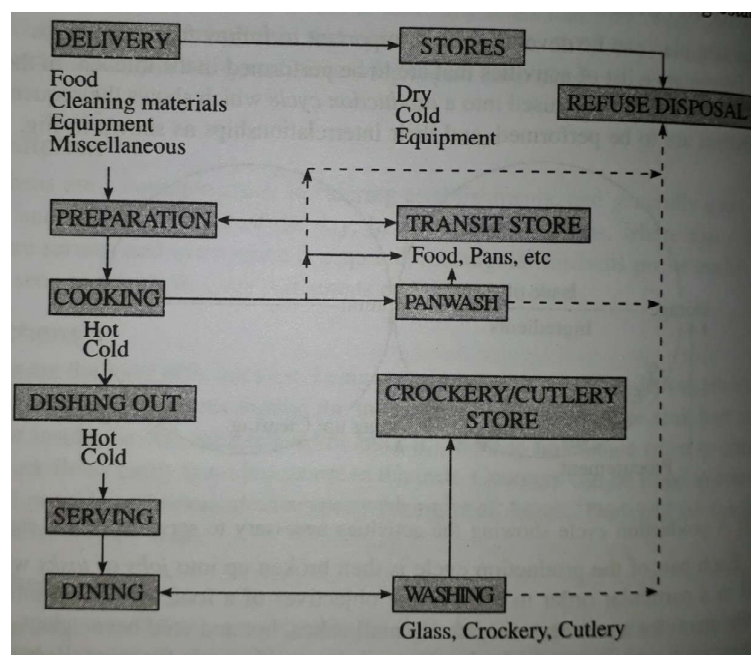


Fig 13.1 Catering Management an Integrated Approach

Source: Catering Management an Integrated Approach- Mohini Sethi, Surjeet Malhan

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Check Your Progress

1. What is the other name of food service department in hospitals?
2. State the functions of the food department in hospitals.
3. Why are square kitchens not so common?
4. What are the main features of kitchen design?

13.3 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The food service department in hospitals is also known as the catering department.
2. The functions of the food department in hospitals are:
 - Maintain the quality of food
 - Avoid contamination and dirt
 - Maintain the patients' health
 - Provide food according to their health
3. Square kitchens are not so common because of space utility issues. It is hard to use the space in centres. In square kitchens the pipes and drains are in the centre of the kitchen under the floor.
4. Service, machine, gas connection, electricity and drainage are some of the main features of the kitchen design.

13.4 SUMMARY

- The food service department in hospitals is also known as the catering department. This department has the responsibility of providing food to the patient according to their body's need and ability.
- Some patients do not accept the service provided by hospitals because they might have issues about food hygiene. Therefore it is imperative to provide proper hygienic food to patients to maintain the image of the hospital.
- Kitchens are always designed according to their use in terms of work and nature. Kitchens can be square, L-shaped, U-shaped, rectangular and others.
- In the past, kitchen were designed in the basement area and there was less light and hygiene. As the air could not move in and out of the kitchen it caused bad stench. Basement kitchen showed harmful effect on food quality and it also effected the production of food.

- It is important to design the kitchen according to its service and machines. If we plan a kitchen without the knowledge of equipment and its service, it creates problem in food distribution, hygiene and sometimes it also causes the situation of risk for the kitchen staff.
- Hygiene and cleanliness of a kitchen depend on its drainage system. During the designing of a kitchen it is important to take care of the drainage system.
- Water is an important component of any kitchen as it is used in washing of vegetables to the cooking of food and dish washing. So the supply of hot and cold water is necessary.
- The floor of kitchen must be smooth and clean. During selection of material for the floor of kitchen always avoid slippery tiles because the use of tiles on floors cause a raise pattern and they require joints which makes the floor difficult to clean
- Walls of a kitchen must be joints free, easy to clean, smooth and should never absorb moisture. Always make kitchen walls with hard plaster and also use emulsion with it. Use of emulsion makes it easier to remove stains.
- If there is any open window it should be pest proof. For the reduction of solar glare, the tinted glasses may be used. For stopping the slopping of birds exterior sills are used. Avoid interior sills.
- Work surfaces must be smooth. Always use stainless steel as work surfaces but it has cost implications. Laminated plastics and ceramic tiles can be used as work surface to reduce costs. Laminated plastics are low in cost and they are easy to maintain.
- Lights in kitchen should be designed in such a way that they work as the best source and provide light in the whole kitchen. Above the food preparation area and work tables light fitting must be present.
- There is no special method of ventilation but the combination of window and exhaust fans is good. During cooking condensation starts. In this process there is the formation of droplets of moisture on equipment, food and surfaces.
- Layout means the arrangement of equipment and services like drainage water and fuel supply. For the planning of layout different type of knowledge is required.
- Trolleys are used to bring food and beverages from the kitchen to the ward and room for the patients. These are the tables which are attached with the wheels.
- Hollowware is used to bring food to the patient from the kitchen. These hollowwares are made up of metal or ceramic containers.

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13.5 KEY WORDS

- **Hollowware:** Hollowware is metal tableware such as sugar bowls, creamers, coffee pots, teapots, soup tureens, hot food covers, water jugs, platters, butter pat plates, and other items that accompany dishware on a table.
- **Mercury-vapor lamp:** It is a gas discharge lamp that uses an electric arc through vaporized mercury to produce light. The arc discharge is generally confined to a small fused quartz arc tube mounted within a larger borosilicate glass bulb.

13.6 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. Write a short note on the location of kitchen.
2. What are the five types of demarcations in a hospital kitchen?
3. State the two types of lighting sources used in the kitchen.
4. Why is proper ventilation required in hospital kitchen?

Long Answer questions

1. Describe the process of proper utilisation of kitchen space.
2. Explain the structural features of a hospital kitchen.
3. Discuss the four concepts of work centres in kitchen layouts.
4. Discuss the different types of food preparation equipment.

13.7 FURTHER READINGS

- Llewelyn, Davies R; Macaulay, H.M.C. 1966. *Hospital Planning and Administration*. Geneva: W.H.O, Jaypee Brothers.
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UNIT 14 KITCHEN SERVICE AND FOOD DISTRIBUTION

*Kitchen Service and
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Structure

- 14.0 Introduction
- 14.1 Objectives
- 14.2 General Kitchen
 - 14.2.1 Special Diet Kitchen
 - 14.2.2 Food Distribution in Hospitals
- 14.3 Answers to Check Your Progress Questions
- 14.4 Summary
- 14.5 Key Words
- 14.6 Self Assessment Questions and Exercises
- 14.7 Further Readings

14.0 INTRODUCTION

A food service and kitchen department in a hospital is one of the most important support service. Patients have various food habits. They may also require therapeutically modified diets based on their condition and treatment. The policy decision to have an inhouse food service department or outsource is to be taken by the management of the hospital. Hospital kitchens have the dual responsibility of providing a good quality of meal and also creating nutritious menus for patients and visitors. Hospitals kitchen need to be fully scalable and offer healthy food to busy professionals, patients, and visiting friends and family. For this purpose, the kitchen should be hygienic and compliant with all safety standards at all times. This unit will discuss kitchen and food distribution services in hospitals.

14.1 OBJECTIVES

After going through this unit, you will be able to:

- Discuss the layout and functions of the hospital kitchen
- Describe food distribution in hospitals

14.2 GENERAL KITCHEN

For commercial kitchen health care centres, the following are required:

- CAD design
- Project management
- Catering equipment

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- Safety and health
- Hygiene standards
- Electricals and ventilation work

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Hospital kitchen layout

It includes space for preparation, cooking, washing, and service station. The back of the kitchen contains multiple areas dedicated to preparing:

- Items which are cold and hot
- Desserts
- Bakery goods
- Dietary items such as bland diets and liquid items
- Food for the patients with ailments such as diabetes, heart diseases

Today, hospitals are taking steps towards holistic living arrangements. This includes pod style living with multiple bedrooms situated around a central living, dining, and kitchen area. The satellite kitchen should be located in the paediatric or oncology wings. In this way hospitals create a secure space that has all the hallmark of a commercial kitchen with a much smaller footprint.

Functions

The kitchen unit of hospitals have the following functions:

- a. Providing diet to patients considering ailment, nutritional requirement and food habit.
- b. Counselling to ambulatory patients regarding food.
- c. Devising policy of in-service training programmes

Location of kitchen unit of the hospital

- a. The unit should be located on the ground floor.
- b. Storage area should be close to the unloading dock.
- c. It should be easily accessible to the vertical transportation system to facilitate delivery of food to the patients.
- d. The preparation rooms and the cafeteria dining room should have convenient access to the patients, attendants, visitors and staff.

Hospital Kitchen Facilities

A hospital kitchen area should have the following facilities:

- (a) **Receiving area:** Material and supplies should be equipped with scales to weigh. It should have easy access for taking supplies from the loading platforms.

- (b) **Storage area:** An area for the storage of crockery, cutlery, utensils as well as a dry storage area for staples and refrigerated storage for perishables. For meat products, large hospitals have walk-in coolers and refrigerators with varying degree of temperature.
- (c) **Preparation area:** Sorting, peeling, slicing, chopping, mincing, and kneading is done here. It should be located between the storage and cooking area. The preparation area requires double sink with draining boards, worktops, peelers and grinders.
- (d) **Production / cooking:** It requires cooking, baking, and frying equipment. The area should be located between the preparation area and service area.
- (e) **Service area:** Food that is prepared is received and assembled in this area into food trays. Some other equipment that are required in this area include refrigerators, table tops, and cupboards for storing trays, prepared foods, cutlery and other items.
- (f) **Food distribution:** It should be provided with space for storage, loading, distribution receiving and sanitizing of the food service carts. Trolleys should also be provided in the area.
- (g) **Dishwashing area:** Automatic dishwashing machines, storage racks, stainless steel sinks should be installed in the area. This area should also provide space for receiving, scraping, rinsing, sorting of soiled tableware.

Floors and walls

The floor should be impervious to slippage, while the walls and ceilings should be smooth, impervious and easily cleanable.

Garbage disposal

Steps should be made for regular wet and dry garbage storage removal and disposal in accordance with waste management guidelines and rules.

Offices

An office should be provided in the kitchen area for the manager, dietician and supervisor. It also includes staff change rooms, lockers for workers, toilets and rest rooms for staff.

Other consideration

- Ventilation and lighting
- LPG should be available
- Adequate steam supply
- Fire protection device

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- Hot and cold-water outlets
- Between kitchen and outside air lock should be provided

Kitchen Stewarding

This department is responsible for storage, maintenance, cleanliness of cutlery, crockery, hollowware, chinaware and glassware. They are also responsible for the cleanliness of kitchen and the washing of pots and pans. Gas connection service and coal supply are also provided by them. They are also responsible for pest control activity of a kitchen. They remain an integral part of the food service operations. Employees in the kitchen stewarding department include the executive kitchen steward, who is primarily responsible for planning, organising, directing and controlling the stewarding activity. He also ensures that the kitchen and restaurants smoothly get their needs. Other employees include the kitchen steward, utility workers, dish washers, and porters.

Function of Dietary Services

The following are the functions of dietary services:

- Patient gains adequate nutrition related to their clinical condition, age, weight, allergies
- Production and preparation of food as per the request of staff and other members
- Hygiene and proper storage of foods
- Patient diet counselling
- Resolving food item issues related to storage, preservation and hygiene
- Keeping the account system and inventory system up to date
- Maintaining the data of diets meals day wise.

By implementing the programmes of continuous quality improvement, the quality of dietary services can be improved.

For evaluating the quality of the food services, the following criteria are used:

- Complaints received from patients and staff
- Consumers outward effects and instances of food poisoning
- Serving wrong diets to patients
- Food wastage
- Non-serving or serving of food that is in inadequate quantity
- Pests and rodents' control

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14.2.1 Special Diet Kitchen

An integral part of the hospital kitchen is the special diet kitchen. A qualified dietician supervises the special diet kitchen. Students and interns carry out the actual preparation of special diet kitchen during their training. Special diet kitchen supplies their menu from the main kitchen and transports the trays through the same tray carts. Their menu is a special and modified of the basic menu of the hospital kitchen. The special diet kitchen area should be located in the main kitchen or close to it. The special diet kitchen also requires pots, pans, vessels; they are just like the main kitchen but on a much smaller scale. The special diet kitchen also requires scales for weighed diets.

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14.2.2 Food Distribution in Hospitals

As a hospital is an organization dedicated to health and healing, it should provide for healthy, nourishing foods and an environment that helps them strengthen and heal. In this, the importance of good nutrition and diet cannot be ignored, especially when it comes to recovery from illness. Thus, one can say that good food and nutrition in a hospital is necessary.

According to some studies, it has being observed that malnourished patients stay 2-3 days longer in hospital than nourished patients. Malnourished patients are at higher risk of readmission in hospitals within 30 days, therefore, good food is important in the treatment of patient and is a part of the total care.



Fig. 14.1 Example of Hospital Food

Food Service Department

In hospitals, the food service department is one of the major departments and is headed by a specialist who is either a professional (food service) manager or a chief dietician. The function of the food service department basically is to provide

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the best possible food at a cost consistent with the policy of the hospital. The department checks the regular supply, quantity, quality, storage and distribution of food. The function of the department is also to establish standards for planning the menu, and preparing and serving food. The department also provides counselling and education to the patients and their families on any specific dietetic problem, and also plans and implements patient therapy prior to their discharge from the hospital. The department also cooperates with the medical staff, and gives instructions to nurses, medical and dental students and interns about principles of nutrition and diet therapy.

Food Delivery

Nowadays, electrical food trolleys are being used that can be plugged into an electrical outlet in order to keep the food warm & hot. Bulk thermal containers are used for liquids, a tray truck which is also called an airline truck with separate heated and refrigerated sections are used for hot and cold food respectively.



Fig. 14.2 *Electrical Food Trolley*

Generally, many hospitals distribute food to individual patients in an individual hot food container carried in an open food cart. The patients are served on trays at the patient's floor. The hot bulk cart contains hot food in bulk that is used to serve hot food. Smaller hospitals may serve the same in an ordinary tiffin carrier. Beverages like coffee and tea are poured into cups in the patient rooms. The system should be crosschecked in case the serving of food takes more than normal 45 minutes. Food containers are used throughout the system from food preparation to re-heating. Hot food after cooking are portioned into shallow food containers. The containers then are loaded onto mobile racks which holds between 18 and 30 containers. These racks of food containers are usually wheeled from the cooking area to the portioning room into chillers or freezers.

Ward Pantry

The basic function of the pantry includes the preparation of snacks, beverage, storage of crockery, refrigerated storage of food, and minimal washing-up. The pantry space will be required for holding chilled meal service equipment as well as to hold trolleys to check the temperatures in any other meal service activities. Generally, the design of the trolley signifies the amount of space required. Easy

access to the ward pantry by staff, patients, and visitors needs to be maintained which is to be ensured by the project team. The trolleys may require additional electrical power points. Ventilation in the pantry should be sufficient as the trolleys and reheating units may produce additional heat within the pantry.

Prior to serving food to patients in the delivery trolleys on arrival at each ward, and again after reheating, one should check and record the temperature of the chilled food. A small worktop for writing, and storage space for disinfectant wipes and disposal gloves should be provided. The tray service crockery cutlery used for main meals is returned to the central wash-up. A hand-wash basin and a double sink and drainer should be provided in the pantry. A room adjacent to the pantry is needed by the staff responsible for reheating and serving frozen food to patients as well as for storing uniforms, for changing and for hand washing prior to serving the meals. Alternately, in hospitals where there is inadequate space in the ward pantry for storing and parking the trolleys, an area retrofitted for reheating/registration and serving several wards is known as a hospital satellite kitchen.

Types of Food Service

Let us now discuss the types of food services in hospitals.

(a) Cafeteria food service

Where a frozen food service is introduced into a day hospital, in addition to it, a refrigerator capable of holding the containers of food between 0°C and 3°C should be provided, fitted with an external recording thermometer. A space is required to park the insulated chilled meal delivery trolley. And an oven or reheating unit for heating the frozen meals.

The meals are delivered in the trolley packed in multi portion size trays which should be unloaded and placed in the frozen food refrigerator. Prior to the meal service time, the trays of food should be loaded into the oven or reheating unit. The shelves should be capable of holding the weight and quantity of containers of food required. After re-heating, they must be placed into the heated serving counter cut outs ready for serving and held at a core food temperature of above 63°C.

(b) Family food service

The bulk containers of frozen meals are delivered to the unit and, on arrival, the containers of food to be served hot. After reheating the containers, the food is placed in the cafeteria type serving counter and the main course item put onto the prewash heated plates and served to the patients. The accompanying vegetables and sauces in their dishes are put onto the table for patients to serve themselves.

(c) Staff and Visitors Food Service System

The food for staff is delivered to the end kitchen from the CFPU in multi-portion containers or in disposable containers. The mobile insulated box with the containers

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of food is wheeled from bulk chilled food store and taken to the staff food reheating and cooking area. The food should be held in a frozen food refrigerator capable of holding peak midday meal requirements, at a temperature btw 1°C and 3°C.

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(d) Coffee Lounge & Restaurant

The entrance lobby should be open with easy access to toilet facilities. The food can be reheated in the adjacent staff food service room. A hatch and pass through hot cupboard should be provided to ensure that the reheated food is transferred to the counter without delay.

A separate room is provided in the end kitchen for providing cold food. Food delivered may include salads, sandwiches, rolls, and cold sweets. If disposable container is used, it is essential to ensure that existing server equipment is able to hold them for service. The frozen food display counters and self-service units used for holding this food must be capable of maintaining the food as required by food safety guidelines.

A trolley is used to deliver the reheated staff food and also space is required for parking these trolleys. A trolley is also used to take the empty and unserved containers of food to the end kitchen central wash-up for washing and disposal. All cold food and reheated cooked foods unserved after meal service are treated as food waste and are disposed of accordingly.

(e) Food store

Food store should be situated adjacent to the main counter and also requires ventilation, as bread, cereals and biscuits will be stored here. Storage units are also required for condiments, sauces, jams etc. The wall should be tiled and the work top of stainless steel. Dairy items should be stored in a refrigerator which should be in the main room and not within the food stored.

Check Your Progress

1. Where should the kitchen unit of the hospital be located?
2. List the criteria for evaluating food services.

14.3 ANSWERS TO CHECK YOUR PROGRESS QUESTIONS

1. The kitchen unit of the hospital should be located on the ground floor.
2. For evaluating the quality of the food services, the following criteria are used:
 - Complaints received from patients and staff
 - Consumers outward effects and instances of food poisoning

- Serving wrong diets to patients
- Food wastage
- Non-serving or serving of food that is in inadequate quantity
- Pests and rodents' control

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14.4 SUMMARY

- The hospital kitchen unit includes space for preparation, cooking, washing, and service station.
- Today, hospitals are taking steps towards holistic living arrangements. This includes pod style living with multiple bedrooms situated around a central living, dining, and kitchen area.
- Steps should be made for regular wet and dry garbage storage removal and disposal in accordance with waste management guidelines and rules.
- Kitchen stewarding department is responsible for storage, maintenance, cleanliness of cutlery, crockery, hollowware, chinaware and glassware.
- An integral part of the hospital kitchen is the special diet kitchen. A qualified dietician supervises the special diet kitchen.
- In hospitals, the food service department is one of the major departments and is headed by a specialist who is either a professional (food service) manager or a chief dietician.
- The basic function of the pantry includes the preparation of snacks, beverage, storage of crockery, refrigerated storage of food, and minimal washing- up.

14.5 KEY WORDS

- **Catering:** It is the business or activity of providing food and drink at events, for organizations, etc.
- **Oncology:** It refers to the department for the study and treatment of tumours.
- **Ambulatory Patient:** A type of patient who has a procedure where he or she is able to walk in for the procedure and then is able to walk out.

14.6 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short Answer Questions

1. What is the layout of the hospital kitchen?
2. What are the functions of the kitchen unit of the hospital?

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Long Answer Questions

1. Describe the facilities in a hospital kitchen.
2. Examine food distribution in hospitals.

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14.7 FURTHER READINGS

Llewelyn, Davies R; Macaulay, H.M.C. 1966. *Hospital Planning and Administration*. Geneva: W.H.O, Jaypee Brothers.

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